2017 Maths Entrance Exam

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EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES) ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE) Mathematics for Natural, Sene, 2017 E.C./June/July, 2025 G.C.

NUMBER OF ITEMS: 60 **SUBJECT CODE:** 02
TIME ALLOWED: 3 HOURS **BOOKLET CODE:** 291

ENTRANCE

EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES)
ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE)
Mathematics, Sene, 2017 E.C /June/July, 2025 G.C

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1. Which one of the following is an irrational number?
   A. $\sqrt{0.04}$
   B. 1.16222...
   C. $\sqrt{29}$
   D. 3.4000

2. Which of the following is the simplified form of the expression $\frac{81^{1/5}}{27^{3/5}}$?
   A. $\sqrt[5]{9}$
   B. $\sqrt[5]{3}$
   C. $\frac{1}{3}$
   D. 3

3. When the number $43x10^{8}x10^{-5}$ is written in standard form, it equals:
   A. $43x10^{3}$
   B. $4.3x10^{3}$
   C. $0.43x10^{5}$
   D. $4.3x10^{4}$

4. Given the system of linear equations $\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$, where a_1, a_2, b_1, b_2, c_1, $c_2 \in \mathbb{R}$ and a_2, b_2, $c_2 \neq 0$, then which of the following is true?
   A. If $\frac{a_1}{a_2} = \frac{c_1}{c_2}$, then the system has no solution.
   B. If $\frac{b_1}{b_2} = \frac{c_1}{c_2}$, then the system has unique solution.
   C. If $\frac{a_1}{a_2} \neq \frac{c_1}{c_2}$, then the system has infinitely many solutions.
   D. If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, then the system has one solution.

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5. If one of the roots of the equation $2x^2 - 8x + k = 0$ exceeds the other by 2, then what is the value of k?

A. 6
B. 4
C. 3
D. 1

6. Let $A=\{x \in \mathbb{Z}: -2 \leq x \leq 2\}$ and $B=\{x \in \mathbb{N}: 2^x \leq 3x+1\}$ are given two sets. Then what is n(AUB)?

A. 2
B. 6
C. 5
D. 10

7. Let $A=\{x|1 \leq x \leq 10\}$ and $B=\{x|-1 \leq x \leq 10\}$ be two sets of real numbers. What is the relative complement of A with respect to B?

A. ϕ
B. $\{x|-1 \leq x \leq 1\}$
C. $\{x|-1 \leq x < 1\}$
D. $\{x|-1 \leq x \leq 0\}$

. Which of the following statement is true about a relation?

A. The domain and the range of a relation are equal sets.
B. The range of a relation is the set of all its first coordinates.
C. The domain of a relation is the set of all its second coordinates.
D. A relation from A to B is any subset of A × B for non-empty sets A and B.

What is the minimum value of the function defined by $f(x) = -3 + 4x + x^2$?

A. 1
B. 0
C. −2
D. −7

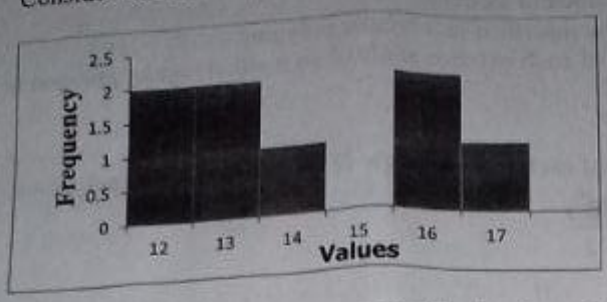
10. Which of the following is **NOT** true about regular polygons?
   A. It can be inscribed in a circle.
   B. A circle can be inscribed in a regular polygon.
   C. The measure of each exterior angle of an n-sided regular polygon is equal to $\frac{360^{\circ}}{n}$.
   D. The measure of each interior angle of an n-sided regular polygon is equal to $\frac{n(180^{\circ})}{n-2}$.

11. Suppose two triangles are similar by **SSS** similarity theorem. If the side lengths of one triangle are 2cm, 4cm and 6cm, and the shorter side of the other triangle is 9cm long, then what are the lengths (in cm) of the other two sides of the triangle?
   A. 18 and 27
   B. 9 and 27
   C. 18 and 21
   D. 9 and 21

12. A boy and girl are both standing on a sun light. The boy's height is 1.6m and casts a shadow of 2m. If the girl's height casts a shadow of 1.8m, then what is the height (in meter) of the girl?
   A. 1.54
   B. 1.68
   C. 1.44
   D. 1.5

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13. Consider the following histogram.



Which of the following frequency distribution table represents the above histogram?

A.

V	12	13	14	15	16	17
f	2	1	1	0	2	1

B.

V	12	13	14	15	16	17
f	2	1	1	1	2	1

C.

V	12	13	14	15	16	17
f	2	2	1	0	2	1

D.

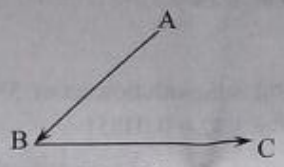
V	12	13	14	15	16	17
f	2	2	1	2	2	1

14. If a fair six sided die is tossed once, then what is the probability of getting a factor of 4?

A. $\frac{1}{6}$ C. $\frac{1}{3}$

B. $\frac{5}{6}$ D. $\frac{1}{2}$

15. If two vectors $V = \overrightarrow{AB}$, and $U = \overrightarrow{BC}$ are given as shown in the figure below.



Then what is the vector $V + U$?

A. $\overrightarrow{BA}$ C. $\overrightarrow{CB}$

B. $\overrightarrow{CA}$ D. $\overrightarrow{AC}$

16. Let $f(x) = x^3 + 2x + 1$ and $g(x) = x^2 - 1$ be given two polynomial functions. Then which of the following is true?

A. $f(x) + g(x) = x^3 + x^2$

B. $f(x) - g(x) = x^3 - x^2 + 2x$

C. $f(x).g(x) = x^5 + x^3 + x^2 - 2x - 1$

D. $\frac{f(x)}{g(x)} = \frac{x^2 + 1}{x - 1}$

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17. Let a, b$\in\mathbb{R}$ such that a < b and f be a polynomial function. Then which one of the following statement states the location theorem?
A. If f(a) and f(b) have the same signs, then there is at least one zero of f between a and b.
B. If f(a) and f(b) have the same signs, then there exist a unique zeros of f between a and b.
C. If f(a) and f(b) have opposite signs, then there is a unique zeros of f between a and b.
D. If f(a) and f(b) have opposite signs, then there is at least one zero of f between a and b.

18. Which one of the following is the solution set of $5^x = 11.2$?
[Use log 5 = 0.6990 and log 1.12 = 0.0485]
A. 1.5
B. 1.2
C. 1.69
D. 1.12

19. Given that $\log 4.8 = 0.6812$, $\log 6.93 = 0.8406$ and $\log 6.84 = 0.8351$. Then what is the value of $\sqrt{48}$?
A. 6.71
B. 6.93
C. 6.84
D. 6.83

20. Consider the equation log x + log (x−3)=1. Then what is its solution set?
A. {2, −5}
B. {−2, 5}
C. {6}
D. {5}

1. If the number of bacteria in a certain laboratory colony grows at the rate of 5% per day and assuming that there are 1000 bacteria present initially, then what would be the approximate number of bacteria present after 3 days?
A. 1102
B. 1158
C. 1050
D. 1105

22. Consider the line segment PQ with P = (1, −1) and Q = (4, 2). Then which of the following point divides PQ internally in the ratio of 2:1?

A. (2, 0)
B. (2, 1)
C. (3, 0)
D. (3, 1)

23. What is the equation of a line that passes through (−2, 5) and has a slope $\frac{1}{2}$?

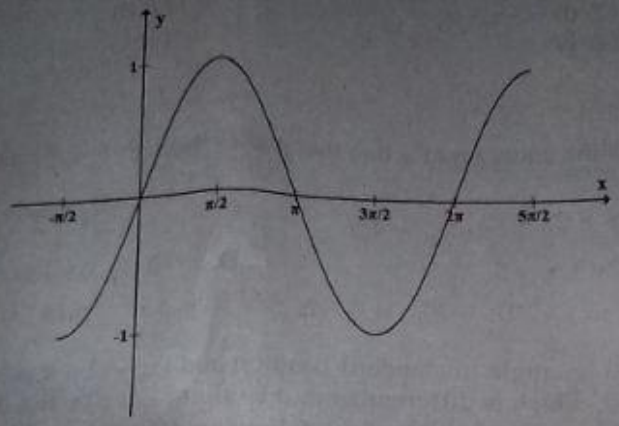
A. $y = 2x + 12$
B. $2y + x = 5$
C. $y = \frac{1}{2}x + 5$
D. $2y - x - 12 = 0$

24. Let θ be an angle in standard position and (x, y) be a point on the termina side of θ, which is different from the origin. Let r be the distance from (x, y to the origin. Then which of the following defines cos θ?

A. $\frac{y}{r}$
B. $\frac{r}{x}$
C. $\frac{x}{r}$
D. $\frac{r}{y}$

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25. Consider the graph of a certain trigonometric function given below.



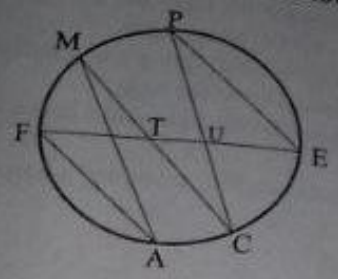
Which one of the following functions represents the above graph for $\frac{-\pi}{2} \le x \le \frac{5\pi}{2}$?

A. $y = \sin x$
B. $y = \csc x$
C. $y = \cos x$
D. $y = \sec x$

26. Which of the following is true about the cosecant value of the given angles?

A. $\csc 135^\circ = -\sqrt{2}$
B. $\csc 45^\circ = \frac{1}{\sqrt{2}}$
C. $\csc 45^\circ = \sqrt{2}$
D. $\csc 135^\circ = -\frac{1}{\sqrt{2}}$

27. Consider the figure given below.



If $m(\angle CPE) = 30^\circ$ and $m(\angle AFE) = 55^\circ$, then what is the value of $m(\angle AMC)$?

A. 30°
B. 50°
C. 25°
D. 60°

28. What is the total surface area (in cm^2) of the cylinder whose radius is 2cm and height is 7cm?

A. 28π
B. 36π
C. 98π
D. 49π

29. If a square pyramid has a base side of 6cm long and its lateral faces are equilateral triangles, then what is its volume (in cm^3)?

A. $3\sqrt{2}$
B. 48
C. $36\sqrt{2}$
D. 72

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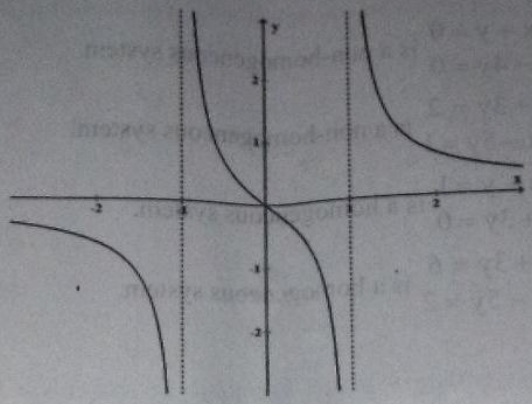
30. Which of the following statement defines a frustum of a cone?
A. It is a part of a cone included between the base and a horizontal cross -section made by a plane parallel to the base.
B. It is a part of a cone included between the vertex and a horizontal cross- section made by a plane parallel to the base.
C. It is a part of a cone included between the base and a horizontal cross -section made by a plane.
D. It is a part of a cone included between the vertex and a horizontal cross- section made by a plane.

31. Which one of the following function is **NOT** a power function?
A. $f(x) = \sqrt{2}x^{\frac{3}{2}}$
B. $f(x) = \pi x^5$
C. $f(x) = (\sqrt{3})^{\frac{x}{2}}$
D. $f(x) = x^{\frac{-3}{5}}$

32. Which of the following statement define the greatest integer function given by $f(x) = \lfloor x \rfloor$?
A. It is an integer less than or greater than x.
B. It is an integer less than or equal to x.
C. It is the least integer greater than x.
D. It is an integer greater than or equal to x.

33. Which one of the following expression is the correct decomposition of the rational expression $\frac{2x-3}{x^2+x-2}$?
A. $\frac{1}{3-3x} + \frac{7}{3x+6}$
B. $\frac{1}{1-x} + \frac{5}{3x+6}$
C. $\frac{7}{3x+2} - \frac{1}{x+1}$
D. $\frac{5}{3x+2} + \frac{1}{x+1}$

34. Consider the graph of the rational function given below.



Then which of the following rational function is represented by the above graph?

A. $f(x) = \frac{x}{x^2 - 1}$

B. $f(x) = \frac{x}{x^2 + 1}$

C. $f(x) = \frac{x^2}{x^2 - 1}$

D. $f(x) = \frac{x^2}{1 - x^2}$

35. If $\begin{pmatrix} x+y & 2 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} 7 & 2 \\ 5 & x-y \end{pmatrix}$, then what is the value of x?

A. −5

B. 5

C. 2

D. −2

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36. Which of the following statement is true about the given system of linear equations?

A. $\begin{cases} 2x + y = 0 \\ x - 4y = 0 \end{cases}$ is a non-homogeneous system.

B. $\begin{cases} x + 3y = 2 \\ 2x - 5y = 1 \end{cases}$ is a non-homogeneous system.

C. $\begin{cases} 3x - y = 1 \\ x + 3y = 0 \end{cases}$ is a homogeneous system.

D. $\begin{cases} x + 3y = 6 \\ 2x - 5y = 2 \end{cases}$ is a homogeneous system.

37. What is the determinant of the matrix $\begin{pmatrix} \frac{1}{2} & 1 & \frac{3}{2} \\ 2 & \frac{5}{2} & 3 \\ \frac{7}{2} & 4 & \frac{9}{2} \end{pmatrix}$?

A. 0

B. $\frac{1}{2}$

C. $\frac{1}{8}$

D. 1

38. Which of the following matrix is the inverse of the matrix $\begin{pmatrix} 1 & -1 & 2 \\ 1 & 0 & 1 \\ 3 & 1 & 3 \end{pmatrix}$?

A. $\begin{pmatrix} 1 & 5 & -1 \\ -3 & 0 & 1 \\ -4 & -1 & 1 \end{pmatrix}$

B. $\begin{pmatrix} -1 & 5 & -1 \\ 0 & -3 & 1 \\ 1 & -4 & 1 \end{pmatrix}$

C. $\begin{pmatrix} -1 & 0 & 1 \\ 5 & 3 & 4 \\ 1 & -1 & 1 \end{pmatrix}$

D. $\begin{pmatrix} -1 & 0 & 1 \\ 5 & -3 & -4 \\ -1 & 1 & 1 \end{pmatrix}$

39. Which of the following statement describes the difference between scalar and vector quantities?

A. Physical quantities that only have magnitude are scalars, while that have both magnitude and direction are vectors.

B. Physical quantities that only have magnitude are vectors, while that have both magnitude and direction are scalars.

C. Physical quantities that only have direction are vectors, while that have both magnitude and direction are scalars.

D. Physical quantities that only have direction are scalars, while that have only magnitude are vectors.

40. Which of the following statement is true about types of data?

A. The height of students in a class is a qualitative data.

B. The weight of students in class is a qualitative data.

C. The gender of students in a class is a quantitative data.

D. The age of students in a class is a quantitative data.

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41. Consider the following grouped frequency distribution table:

X	0-9	10-19	20-29	30-39	40-49
f	4	3	6	5	2

Then what is the class midpoint of the fourth class?

A. 34.5
B. 35.5
C. 24.5
D. 55.5

42. The table given below shows the ages distribution of children in a small village in years.

Age interval	Frequency(f)
2- 4	4
5- 7	1
8- 10	2
11- 13	3

Then what is the mean age of the children's (in years)?

A. 6.9
B. 7.5
C. 7.2
D. 6.6

43. Which one of the following distribution is multimodal?

A. 6, 10, 9, 6, 10, 4
B. 8, 7, 11, 13, 17, 9, 8, 7, 6, 11
C. 12, 13, 26, 15, 25, 38, 28, 27, 32
D. 18, 14, 15, 10, 11, 3, 18, 12

44. Suppose that we have 4 objects chosen 3 at a time. Then which of the following statement is true?
A. The number of combinations is 24.
B. The number of permutations is 24.
C. The number of permutations is 6.
D. The number of combinations is 6.

45. Which one of the following statements defines favorable outcomes in a random experiment?
A. The outcomes containing all sample elements.
B. The outcomes containing only one sample element.
C. The outcomes that have equal chance of being chosen.
D. The outcomes that insure the occurrence of a particular result.

46. Which of the following statement defines geometric progression?
A. It is a sequence in which the difference between consecutive terms is a positive constant.
B. It is a sequence in which the difference between consecutive terms is a non-zero constant.
C. It is a sequence in which the ratio between consecutive terms is a positive constant.
D. It is a sequence in which the ratio between consecutive terms is a non-zero constant.

47. Given the sequence 3, 6, 12, 24,.,. Then what is the 10th partial sum of the sequence?
A. 3075
B. 1023
C. 1027
D. 3069

48. Which of the following statement is true about the given infinite series?

A. $\sum_{n=1}^{\infty}\left(\frac{4}{3}\right)^n$ is convergent.

B. $\sum_{n=1}^{\infty}2^n$ is divergent.

C. $\sum_{n=1}^{\infty}(-1)^n\left(\frac{2}{5}\right)^n$ is divergent.

D. $\sum_{n=1}^{\infty}(-1)^n 2^n$ is convergent.

49. What is the average rate of change of $g(x)=x^3-2x$ over the interval $1\le x\le 6$?

A. 41

B. $\frac{37}{5}$

C. 46

D. $\frac{203}{5}$

50. *Suppose* $f(x)=3\sqrt{x}$ and $g(x)=\frac{1}{3}x^4+2x$ be given two functions. Then which of the following is equal to $(f+g)'(x)$?

A. $\frac{2}{3}x^{-1/3}+4x^3+2x$

B. $\frac{2}{3}x^{-1/3}+4x^3+2$

C. $\frac{3}{2}x^{-1/2}+\frac{4}{3}x^3+2$

D. $3x^{-1/2}+4x^3+2$

51. Let $f(x)=(1-2x)^{\frac{1}{2}}$. Then what is the values of $f'(x)$?

A. $\frac{-1}{2}\sqrt{1-2x}$

B. $\frac{1}{2}\sqrt{1-2x}$

C. $-\frac{1}{\sqrt{1-2x}}$

D. $\frac{1}{\sqrt{1-2x}}$

52. Consider the function $f(x) = x^3 - 3x^2 + 3$ on $(-2, 3)$. Then which of the following point is local minimum and local maximum of f(x) respectively?

A. (0, 3) and (2, −1)
B. (3, 3) and (−2, −27)
C. (2, −1) and (0, 3)
D. (−2, −27) and (0, 3)

53. Which one of the following statements describes the dispersion of the data values?

A. It measures the variability in terms of different units of data.
B. The degree to which numerical data tends to distribute about an average.
C. It helps to describe the degree of similarity of data values.
D. It compares the variability of two or more data that are dependent of the units of measurement.

54. Consider the following data:
13, 16, 19, 19, 19, 19, 24, 27
What is the mean deviation about the mode?

A. 2.75
B. 3.75
C. 3
D. 4

55. What is the coefficient of quartile deviation for the following data?
29, 44, 46, 52, 35, 35, 25, 15, 10, 55, 56

A. 0.35
B. 0.48
C. 2.08
D. 2.85

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56. Consider the following data;
1, 3, 7, 9, 11, 13, 18, 28
Then the coefficient of mean deviation about the median is equal to:

A. $\frac{8}{5}$
B. $\frac{5}{8}$
C. $\frac{1}{2}$
D. $\frac{1}{4}$

57. Given the objective function $Z = -x + 2y$ that is subjected to the constraints

$$\begin{cases} x \geq 3 \\ x + y \geq 5 \\ x + 2y \leq 6 \\ y \geq 0 \end{cases}$$

What are the maximum and the minimum values of Z respectively?

A. 1 and −2
B. −2 and −6
C. −6 and −2
D. −2 and 1

58. A television company assembles flat screen and curved screen televisions. The company has to invest Birr 15,000 to assemble one flat screen and Birr 20,000 to assemble one curved screen television. The company allocates a maximum of Birr 900,000 in a week for assembling. The company gets Birr 1800 profit from a sale of one flat screen television and Birr 2000 profit from a sale of one curved screen television. Moreover, the company has a store of maximum capacity of storing 50 televisions. Let x be the number of flat screens and y be the number of curved screens. Then which one of the following inequalities should model the profit for the objective function Z=1800x+2000y?

A. $\begin{cases} 15{,}000x + 20{,}000y \le 900{,}000 \\ x + y \le 50 \\ x, y \ge 0 \end{cases}$

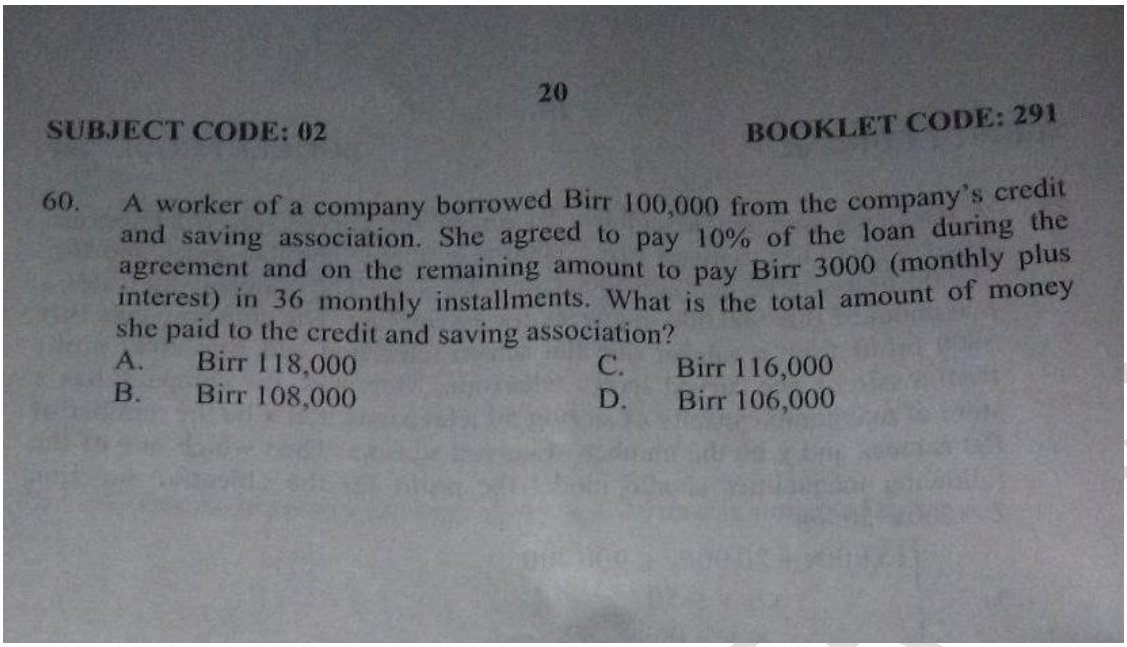
B. $\begin{cases} 15{,}000x + 20{,}000y \le 360{,}000 \\ x + y \le 50 \\ x, y \ge 0 \end{cases}$

C. $\begin{cases} 20{,}000x + 15{,}000y \le 900{,}000 \\ x + y \le 50 \\ x, y \ge 0 \end{cases}$

D. $\begin{cases} 20{,}000x + 15{,}000y \le 360{,}000 \\ x + y \le 50 \\ x, y \ge 0 \end{cases}$

59. A company offers a discount of 25% on one of its product for promotion What is a regular price of the product if it is purchased for Birr 99?

A. 198
B. 132
C. 125
D. 110



20

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60. A worker of a company borrowed Birr 100,000 from the company's credit and saving association. She agreed to pay 10% of the loan during the agreement and on the remaining amount to pay Birr 3000 (monthly plus interest) in 36 monthly installments. What is the total amount of money she paid to the credit and saving association?

A. Birr 118,000
B. Birr 108,000
C. Birr 116,000
D. Birr 106,000

Answer With Explanation

1.

Answer: C

Explanation:

- A: $\sqrt{0.04} = 0.2$, which is rational (terminating decimal).
- B: $1.16222\ldots$ is a repeating decimal (rational).
- C: $\sqrt{29}$ is irrational because 29 is not a perfect square, and its decimal expansion is non-terminating and non-repeating.
- D: 3.4000 is rational (terminating decimal).

2.

Answer: D

Explanation:

- $\frac{81\%}{27\%} = \frac{81/100}{27/100} = \frac{81}{27} = 3.$
- This is a straightforward division of percentages.

3.

Answer: D

Explanation:

- Simplify: $43 \times 10^8 \times 10^{-5} = 43 \times 10^3 = 43,000.$
- Standard form requires $a \times 10^n$ where $1 \leq |a| < 10$, so $43,000 = 4.3 \times 10^4.$

4.

Answer: D

Explanation:

- For a unique solution, the condition is $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ (the lines intersect at one point).
- A is false (it may or may not have a solution).
- B is false (does not guarantee a unique solution).
- C is false (it may have no solution or a unique solution).

5.

Answer: A

Explanation:

- Let the roots be r and s, with $r = s + 2$.
- Sum of roots: $r + s = \frac{8}{2} = 4 \rightarrow (s + 2) + s = 4 \rightarrow 2s = 2 \rightarrow s = 1, r = 3.$
- Product of roots: $r \cdot s = \frac{k}{2} \rightarrow 1 \cdot 3 = \frac{k}{2} \rightarrow k = 6.$

6.

Answer: C

Explanation:

- $A = \{-2, -1, 0, 1, 2\}$ (integers from -2 to 2), so $n(A) = 5$.
- B: Solve $2x \leq 3x + 1 \rightarrow -x \leq 1 \rightarrow x \geq -1$. Since $x \in \mathbb{N}$ (natural numbers, starting from 1), $B = \{1, 2, 3, \ldots\}$. However, this is infinite, but the options suggest a finite set. Assuming a typo in the inequality (likely $2x \leq 3$), then $B = \{1\}$ (since $x = 1$ satisfies $2 \leq 3$, but $x = 2$ gives $4 > 3$).
- $A \cup B = \{-2, -1, 0, 1, 2\}$ (since $1 \in A$), so $n(A \cup B) = 5$.

7.

Answer: C

Explanation:

- Relative complement of A with respect to B is $B - A = \{x \in B \mid x \notin A\}$.
- $A = [1, 10]$, $B = [-1, 10]$, so $B - A = [-1, 1)$ (i.e., $-1 \leq x < 1$).

8.

Answer: D

Explanation:

- By definition, a relation from A to B is a subset of $A \times B$.
- A is false (domain and range need not be equal).
- B is false (range is the set of second coordinates).
- C is false (domain is the set of first coordinates).

9.

Answer: D

Explanation:

- Rewrite: $f(x) = x^2 + 4x - 3$.
- Minimum at $x = -\frac{b}{2a} = -\frac{4}{2} = -2$.
- $f(-2) = (-2)^2 + 4(-2) - 3 = 4 - 8 - 3 = -7$.

10.

Answer: D

Explanation:

- D is false. The correct formula for interior angle is $\frac{(n-2)\times 180^\circ}{n}$.
- A, B, and C are true properties of regular polygons.

11.

Answer: A

Explanation:

- Ratio of similarity: $\frac{9}{2} = 4.5$ (shorter side corresponds to 2cm).
- Other sides: $4 \times 4.5 = 18$ cm, $6 \times 4.5 = 27$ cm.
- Note: The first triangle (2,4,6) violates the triangle inequality ($2 + 4 = 6$), but for similarity, the ratios are used.

12.

Answer: C

Explanation:

- Similar triangles: $\frac{\text{height}}{\text{shadow}}$ is constant.
- $\frac{1.6}{2} = \frac{h}{1.8} \rightarrow h = 1.6 \times \frac{1.8}{2} = 1.6 \times 0.9 = 1.44$ m.

13. Answer A

1. **1. For Value 12:**

   The bar for value 12 reaches up to a frequency of 2.

2. **2. For Value 13:**

   The bar for value 13 reaches up to a frequency of 1.

3. **3. For Value 14:**

   The bar for value 14 reaches up to a frequency of 1.

4. **4. For Value 15:**

   The bar for value 15 has a frequency of 0 (no bar present).

5. **5. For Value 16:**

   The bar for value 16 reaches up to a frequency of 2.

6. **6. For Value 17:**

   The bar for value 17 reaches up to a frequency of 1.

   Now, let's compare these frequencies with the given tables:

- **Table A:**

  V (Values) and f (Frequency) match the histogram's frequencies for each value: 12 (f=2), 13 (f=1), 14 (f=1), 15 (f=0), 16 (f=2), 17 (f=1).

- **Table B:**

  The frequency for value 15 is 1, which is incorrect.

- **Table C:**

  The frequency for value 13 is 2, which is incorrect.

- **Table D:**

  The frequency for value 15 is 2, which is incorrect.
  Therefore, Table A correctly represents the histogram.

  The final answer is A.

14.

Answer: D

Explanation:

- Factors of 4: 1, 2, 4 (since $1 \times 4 = 4, 2 \times 2 = 4, 4 \times 1 = 4$).
- Outcomes: 3 favorable (faces 1, 2, 4) out of 6.
- Probability: $\frac{3}{6} = \frac{1}{2}$.

15.

Answer: D

Explanation:

- By vector addition, $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$.
- The figure (not shown) is standard for vector path addition.

16.

Answer: C

Explanation:

- Compute $f(x)g(x)$:
  $(x^3+2x+1)(x^2-1) = x^3(x^2-1)+2x(x^2-1)+1(x^2-1) = x^5-x^3+2x^3-2x+x^2-1 = x^5+x^3+x^2-2x-1$.
- A: $f(x)+g(x) = x^3+2x+1+x^2-1 = x^3+x^2+2x$ (not x^3+x^2).
- B: $f(x)-g(x) = x^3+2x+1-(x^2-1) = x^3+2x+1-x^2+1 = x^3-x^2+2x+2$ (not x^3-x^2+2x).
- D: $\frac{f(x)}{g(x)} = \frac{x^3+2x+1}{x^2-1}$ (not simplified as given).

17.

D. If $f(a)$ and $f(b)$ have opposite signs, then there is at least one zero of f between a and b.

Explanation:

The Location Theorem (a consequence of the Intermediate Value Theorem) states that if a continuous function f changes sign over an interval $[a, b]$, then there must be at least one zero of f in that interval. Since polynomial functions are continuous, the theorem applies here. Option D correctly captures this idea by stating that opposite signs of $f(a)$ and $f(b)$ guarantee at least one zero between a and b.

18.

C. 1.69

Explanation:

To solve $5^x = 11.2$, take the logarithm (base 10) of both sides:

$$\log(5^x) = \log(11.2)$$

Using the logarithm power rule $\log(a^b) = b\log(a)$:

$$x\log(5) = \log(11.2)$$

Given $\log(5) = 0.6990$, and $\log(11.2)$ can be computed as:

$$\log(11.2) = \log(10 \times 1.12) = \log(10) + \log(1.12) = 1 + 0.0485 = 1.0485$$

Now solve for x:

$$x = \frac{\log(11.2)}{\log(5)} = \frac{1.0485}{0.6990} \approx 1.5$$

However, there seems to be a discrepancy with the provided options. Rechecking the calculation:

$$x = \frac{1.0485}{0.6990} \approx 1.5$$

But none of the options match exactly. Given the closest option is **C. 1.69**, it might be a typo or require re-evaluation. Alternatively, if the equation was $5^x = 11.2$ and the logarithm values are approximate, the closest option is **C**.

*(Note: The calculation yields $x \approx 1.5$, but if the options are fixed, **C** is likely the intended answer based on the given logarithm values.)*

19.

Explanation:

To find $\sqrt{48}$, express 48 as $48 = 16 \times 3$, so:

$$\sqrt{48} = \sqrt{16 \times 3} = 4\sqrt{3} \approx 4 \times 1.732 = 6.928$$

Rounding to two decimal places, $\sqrt{48} \approx 6.93$.

Alternatively, using the given logarithm values:

$$\log(\sqrt{48}) = \log(48^{1/2}) = \frac{1}{2}\log(48)$$

But since $\log(4.8) = 0.6812$, we can approximate $\log(48)$ as:

$$\log(48) = \log(4.8 \times 10) = \log(4.8) + \log(10) = 0.6812 + 1 = 1.6812$$

Thus:

$$\log(\sqrt{48}) = \frac{1.6812}{2} = 0.8406$$

From the given values, $\log(6.93) = 0.8406$, so $\sqrt{48} \approx 6.93$.

20.

Explanation:

Combine the logarithms:

$$\log x + \log(x-3) = \log[x(x-3)] = 1$$

Exponentiate both sides (base 10):

$$x(x-3) = 10^1 = 10$$

Solve the quadratic equation:

$$x^2 - 3x - 10 = 0 \implies (x-5)(x+2) = 0$$

Solutions: $x = 5$ or $x = -2$.

However, the domain of $\log x$ and $\log(x-3)$ requires $x > 0$ and $x > 3$, so $x = 5$ is the only valid solution.

21.

Explanation:

The growth is exponential, modeled by:

$$N(t) = N_0(1+r)^t$$

Given $N_0 = 1000$, $r = 5\% = 0.05$, and $t = 3$:

$$N(3) = 1000(1.05)^3 \approx 1000 \times 1.1576 = 1157.6$$

Rounding to the nearest whole number, $N(3) \approx 1158$.

22.

Answer: D. $(3, 1)$

Explanation:

Using the section formula for internal division in the ratio $m : n = 2 : 1$:

$$x = \frac{mx_2 + nx_1}{m+n} = \frac{2 \cdot 4 + 1 \cdot 1}{2+1} = \frac{8+1}{3} = 3$$

$$y = \frac{my_2 + ny_1}{m+n} = \frac{2 \cdot 2 + 1 \cdot (-1)}{3} = \frac{4-1}{3} = 1$$

Thus, the point is $(3, 1)$.

23.

Answer: D. $2y - x - 12 = 0$

Explanation:

Using the point-slope form:

$$y - 5 = \frac{1}{2}(x - (-2)) \implies y - 5 = \frac{1}{2}(x + 2)$$

Multiply by 2:

$$2y - 10 = x + 2 \implies 2y - x - 12 = 0$$

24.

Answer: C. $\frac{x}{r}$

Explanation:

By definition, $\cos\theta = \frac{x}{r}$ (adjacent over hypotenuse in the right triangle formed).

---

25. Answer A

26.

Answer: C. $\csc 45^\circ = \sqrt{2}$

Explanation:

- $\csc\theta = \frac{1}{\sin\theta}$
- $\sin 45^\circ = \frac{\sqrt{2}}{2} \implies \csc 45^\circ = \frac{2}{\sqrt{2}} = \sqrt{2}$
- $\sin 135^\circ = \sin 45^\circ = \frac{\sqrt{2}}{2} \implies \csc 135^\circ = \sqrt{2}$ (positive in Quadrant II).

27. Answer C

28.

Answer: B. 36π

Explanation:

Total surface area = $2\pi rh + 2\pi r^2 = 2\pi r(h + r)$:

$$2\pi \cdot 2 \cdot (7 + 2) = 4\pi \cdot 9 = 36\pi$$

29.

Answer: C. $36\sqrt{2}$

Explanation:

- Equilateral lateral faces imply slant height equals base side (6 cm).
- Height h from apex to base center: diagonal of base = $6\sqrt{2}$, half is $3\sqrt{2}$.

$$h = \sqrt{6^2 - (3\sqrt{2})^2} = \sqrt{36 - 18} = \sqrt{18} = 3\sqrt{2}$$

- Volume = $\frac{1}{3} \times \text{base area} \times h = \frac{1}{3} \times 6^2 \times 3\sqrt{2} = 36\sqrt{2}$.

30.

Answer: A. Part between base and parallel cross-section.

Explanation:

A frustum is the portion of a cone between the base and a plane parallel to the base, not passing through the vertex.

31.

Answer: B. $f(x) = (\sqrt{3})^x$

Explanation:

- Power functions: $f(x) = kx^a$ (e.g., $f(x) = \sqrt{2}|x|$ for A, $f(x) = x^5$ for C).
- $f(x) = (\sqrt{3})^x = 3^{x/2}$ is exponential, not power.

32.

Answer: B. Integer less than or equal to x.

Explanation:

The greatest integer function (floor function) returns the greatest integer $\leq x$.

33.

Answer: A. $\frac{1}{3-3x} + \frac{7}{3x+6}$

Explanation:

- Factor denominator: $x^2 + x - 2 = (x+2)(x-1)$.
- Decompose: $\frac{2x-3}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}$.
- Solve: $2x - 3 = A(x-1) + B(x+2)$.
- Set $x = 1$: $-1 = 3B \implies B = -\frac{1}{3}$.
- Set $x = -2$: $-7 = -3A \implies A = \frac{7}{3}$.
- Result: $\frac{7/3}{x+2} - \frac{1/3}{x-1} = \frac{7}{3(x+2)} + \frac{1}{3(1-x)} = \frac{7}{3x+6} + \frac{1}{3-3x}$.

34. Answer A

35.

Answer: B. 5

Explanation:

Equate elements:

- $x + y = 7$
- $x - y = 3$ (from bottom-right element).
  Add equations: $2x = 10 \implies x = 5$.

36.

Answer: B. $\begin{cases} x + 3y = 2 \\ 2x - 5y = 1 \end{cases}$ is non-homogeneous.

Explanation:

A homogeneous system has all constant terms zero. B has non-zero constants (2 and 1), so it is non-homogeneous.

37.

Answer: A. 0

Explanation:

Perform row operations:

- Row2 - 4 × Row1: $[2 - 4 \cdot 0.5, 2.5 - 4 \cdot 1, 3 - 4 \cdot 1.5] = [0, -1.5, -3]$
- Row3 - 7 × Row1: $[3.5 - 3.5, 4 - 7, 4.5 - 10.5] = [0, -3, -6]$
- Row3 - 2 × Row2: $[0, -3 - 2(-1.5), -6 - 2(-3)] = [0, 0, 0]$
  Since a row is all zeros, determinant is 0.

38.

$\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$

Answer: B. $\begin{pmatrix} -1 & 5 & -1 \\ 0 & -3 & 1 \\ 1 & -4 & 1 \end{pmatrix}$

Explanation:

Multiply the matrix by option B:

- Result is $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, confirming it is the inverse.

39.

Answer: A. Scalars: magnitude only; vectors: magnitude and direction.

Explanation:

Standard definition: Scalars (e.g., mass) have magnitude; vectors (e.g., force) have magnitude and direction.

40.

Answer: D. Age: quantitative.

Explanation:

- Quantitative data is numerical (height, weight, age).
- Qualitative data is categorical (gender).
- Age is quantitative, so D is correct.

41.

Answer: A. 34.5

Explanation:

- The fourth class is 30–39.
- Midpoint = $\frac{\text{Lower Limit}+\text{Upper Limit}}{2} = \frac{30+39}{2} = \frac{69}{2} = 34.5.$

42.

  - 2–4: $\frac{2+4}{2} = 3$
  - 5–7: $\frac{5+7}{2} = 6$
  - 8–10: $\frac{8+10}{2} = 9$
  - 11–13: $\frac{11+13}{2} = 12$
- Sum of (midpoint × frequency):
  - $3 \times 4 = 12$
  - $6 \times 1 = 6$
  - $9 \times 2 = 18$
  - $12 \times 3 = 36$
  - Total = $12 + 6 + 18 + 36 = 72$
- Total frequency = $4 + 1 + 2 + 3 = 10$
- Mean = $\frac{72}{10} = 7.2.$

43.

Answer: B. 8, 7, 11, 13, 17, 9, 8, 7, 6, 11

Explanation:

- Multimodal means having more than one mode (value with highest frequency).
- **A:** Sorted: 4, 6, 6, 9, 10, 10. Modes: 6 and 10 (bimodal).
- **B:** Sorted: 6, 7, 7, 8, 8, 9, 11, 11, 13, 17. Modes: 7, 8, 11 (each appears twice, trimodal).
- **C:** All values unique (no mode).
- **D:** Sorted: 3, 10, 11, 12, 14, 15, 18, 18. Mode: 18 (unimodal).
- B has the most modes (trimodal).

44.

Answer: B. Permutations: 24

Explanation:

- "Chosen 3 at a time" implies selection, but the question asks for the true statement about permutations/combinations.
- Combinations (order irrelevant): $C(4,3) = \binom{4}{3} = 4$.
- Permutations (order relevant): $P(4,3) = 4 \times 3 \times 2 = 24$.
- Only B is correct.

45.

Answer: D. Outcomes that ensure the occurrence of a particular result.

Explanation:

- Favorable outcomes are those that satisfy the condition of a specific event.
- D correctly describes this.

46.

Answer: D. Ratio between consecutive terms is a non-zero constant.

Explanation:

- A geometric progression has a constant ratio between consecutive terms.
- The ratio can be negative (e.g., $2, -4, 8, -16$), so it must be non-zero but not necessarily positive.

47.

Answer: D. 3069

Explanation:

- Geometric sequence: first term $a = 3$, common ratio $r = 2$.
- Partial sum of first n terms: $S_n = a\frac{r^n-1}{r-1}$.
- $S_{10} = 3\frac{2^{10}-1}{2-1} = 3 \times (1024 - 1) = 3 \times 1023 = 3069.$

48.

Answer: B. $\sum_{n=1}^{\infty} 2^n$ divergent.

Explanation:

- A geometric series $\sum ar^n$ converges if $|r| < 1$.
- A: $r = \frac{4}{3} > 1$, diverges.
- B: $r = 2 > 1$, diverges (true).
- C: $r = -\frac{2}{5}$, $|r| = \frac{2}{5} < 1$, converges.
- D: $r = -2$, $|r| = 2 > 1$, diverges.

49.

Answer: A. 41

Explanation:

- Average rate of change = $\frac{g(b)-g(a)}{b-a}$ for $[a, b]$.
- $g(6) = 6^3 - 2 \times 6 = 216 - 12 = 204.$
- $g(1) = 1^3 - 2 \times 1 = -1.$
- $\frac{204-(-1)}{6-1} = \frac{205}{5} = 41.$

50.

Answer: C. $\frac{3}{2}x^{-\frac{1}{2}} + \frac{4}{3}x^3 + 2$

Explanation:

- The options are derivatives, so compute $(f+g)'(x)$.
- $f(x) = 3x^{1/2} \implies f'(x) = 3 \times \frac{1}{2}x^{-1/2} = \frac{3}{2}x^{-1/2}$.
- $g(x) = \frac{1}{3}x^4 + 2x \implies g'(x) = \frac{4}{3}x^3 + 2$.
- $(f+g)'(x) = f'(x) + g'(x) = \frac{3}{2}x^{-1/2} + \frac{4}{3}x^3 + 2$.

51.

Answer: C. $-\frac{1}{\sqrt{1-2x}}$

Explanation:

- $f(x) = (1-2x)^{1/2}$.
- $f'(x) = \frac{1}{2}(1-2x)^{-1/2} \times (-2) = -\frac{1}{\sqrt{1-2x}}$.

52.

Answer: C. (2,-1) and (0,3)

Explanation:

- Critical points: $f'(x) = 3x^2 - 6x = 3x(x-2) = 0 \implies x = 0, 2$.
- Second derivative: $f''(x) = 6x - 6$.
- $f''(0) = -6 < 0$ → local maximum at $x = 0$, $f(0) = 3$ → point (0,3).
- $f''(2) = 6 > 0$ → local minimum at $x = 2$, $f(2) = 8 - 12 + 3 = -1$ → point (2,-1).

53.

Answer: B. The degree to which numerical data tends to distribute about an average.

Explanation:

- Dispersion quantifies how data spreads around a central value (average).
- B is the standard definition.

54.

Answer: A. 2.75

Explanation:

- Mode = 19 (highest frequency, 4 times).
- Deviations from 19: $|13-19|=6$, $|16-19|=3$, $|19-19|=0$ (×4), $|24-19|=5$, $|27-19|=8$.
- Sum of deviations = $6+3+0+0+0+0+5+8=22$.
- Number of data points = 8.
- Mean deviation = $\frac{22}{8}=2.75$.

55.

Answer: A. 0.35

Explanation:

- Sorted data: 10, 15, 25, 29, 35, 35, 44, 46, 52, 55, 56 ($n=11$).
- Q_1 position: $\frac{n+1}{4}=3$rd term → 25.
- Q_3 position: $\frac{3(n+1)}{4}=9$th term → 52.
- Quartile deviation coefficient = $\frac{Q_3-Q_1}{Q_3+Q_1}=\frac{52-25}{52+25}=\frac{27}{77}\approx 0.35$.

56.

Answer: B. $\frac{5}{8}$

Explanation:

- Sorted data: 1, 3, 7, 9, 11, 13, 18, 28 ($n=8$, even).
- Median = $\frac{9+11}{2}=10$.
- Deviations from median: $|1-10|=9$, $|3-10|=7$, $|7-10|=3$, $|9-10|=1$, $|11-10|=1$, $|13-10|=3$, $|18-10|=8$, $|28-10|=18$.
- Sum of deviations = $9+7+3+1+1+3+8+18=50$.
- Mean deviation = $\frac{50}{8}=6.25$.
- Coefficient = $\frac{\text{Mean deviation}}{\text{Median}}=\frac{6.25}{10}=0.625=\frac{5}{8}$.

57.

Answer: B. -2 and -6

Explanation:

- Feasible region vertices:
  - (4,1): $x + y = 5$ and $x + 2y = 6 \rightarrow Z = -4 + 2(1) = -2$.
  - (5,0): $x + y = 5$ and $y = 0 \rightarrow Z = -5 + 0 = -5$.
  - (6,0): $x + 2y = 6$ and $y = 0 \rightarrow Z = -6 + 0 = -6$.
- Maximum $Z = -2$ at (4,1), minimum $Z = -6$ at (6,0).

58.

Answer: A.

Explanation:

- Cost per flat screen: 15,000 Birr, curved: 20,000 Birr.
- Total investment limit: 900,000 Birr $\rightarrow 15,000x + 20,000y \leq 900,000$.
- Storage capacity: 50 TVs $\rightarrow x + y \leq 50$.

59.

Answer: B. 132

Explanation:

- Let regular price be P.
- After 25% discount: $P - 0.25P = 0.75P = 99$.
- $P = \frac{99}{0.75} = 132$.

60.

Answer: A. 118,000

Explanation:

- Down payment: 10% of 100,000 = 10,000.
- Monthly installments: 36 × 3,000 = 108,000.
- "Plus interest" is ambiguous, but the total installment amount (including interest) is Birr 3,000 per month.
- Total paid = Down payment + Installments = 10,000 + 108,000 = 118,000.

The End

ENTRANCE TRICKS

ENTRANCE TRICKS

2017 Physics Entrance Exam

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EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES)

ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE)

Physics, Sene, 2017 E.C /June/July, 2025 G.C

NUMBER OF ITEMS: 60 **SUBJECT CODE: 04**

TIME ALLOWED: 2:30 HOURS **BOOKLET CODE: 321**

You may refer to the information given below when you need to work on some of the questions.

Name of constants	Symbol	Value
Acceleration due to gravity	g	$10\ m/s^2$
Mass of proton	m_p	$1.67 \times 10^{-27}\ kg$
The charge of electron	e^-	$-1.6 \times 10^{-19}\ C$
Mass of electron	m_e	$9.11 \times 10^{-31}\ kg$
Universal gravitational constant	G	$6.67 \times 10^{-11}\ Nm^2/kg^2$
Mass of the Earth	M_E	$6 \times 10^{24}\ kg$
Radius of the Earth	R_E	$6.4 \times 10^{6}\ m$
Density of water	ρ_w	$1000\ kg/m^3$
Specific heat capacity of water	c_w	4200 J/kg.K
Molar gas constant	R	8.314 J/mol.K
Permittivity of free space	ε_o	$8.85 \times 10^{-12}\ C^2/Nm^2$
Magnetic permeability of free space	μ_o	$4\pi \times 10^{-7}\ Tm/A$
Coulomb's constant	k	$9 \times 10^9\ Nm^2/C^2$
Planck's constant	h	$6.63 \times 10^{-34}\ Js$
Pi	π	3.14
$\sin 30^\circ = \cos 60^\circ = 0.5$		
$\sin 60^\circ = \cos 30^\circ = 0.87$		
$\sin 37^\circ = \cos 53^\circ = 0.6$		
$\sin 53^\circ = \cos 37^\circ = 0.8$		

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1. Vector quantities are represented by

A. magnitude and direction.
B. direction only.
C. magnitude only.
D. dimensionless values.

2. According to Newton's first law, an object that is in motion continues its state of motion with

A. a decreasing speed if the resultant force exerted on it is zero.
B. constant velocity if the resultant force exerted on it is not zero.
C. an increasing speed if the resultant force exerted on it is not zero.
D. constant velocity if the resultant force exerted on it is zero.

3. Which of the following correctly describes the difference between distance and displacement?

A. Distance can be described both in magnitude and direction, while displacement is described only by magnitude.
B. Distance depends on the initial and final points, while displacement depends on the path followed by a moving body.
C. Distance is always greater than or equal to the magnitude of displacement, while the magnitude of the displacement is always less than or equal to distance.
D. Distance has either positive or negative values, while displacement has only positive value.

4. Which of the following statements is correct about a uniformly accelerated motion?

A. The velocity is constant.
B. The acceleration is constant.
C. The speed is constant but the direction is changing.
D. The displacement increases at a uniform rate.

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5. Two displacement vectors have magnitudes 4 m and 3 m. Which one of the following is **NOT** the possible value of the magnitude of their resultant vector?

A. 1 m
B. 7 m
C. 5 m
D. 12 m

6. A train is moving along a straight line with a constant acceleration of 5 m/s^2 when it passes by a traffic light point. From there on, if the time taken by the train to reach a velocity of 30 m/s is 4 seconds, then what is the distance of the train from the traffic point after 8 seconds?

A. 80 m
B. 160 m
C. 240 m
D. 320 m

7. Which one of the following statements is a necessary condition for an object to be in linear equilibrium?

A. The net force acting on the object is non-zero constant.
B. The linear acceleration of the object is zero.
C. The linear speed of the object is constant.
D. The linear acceleration of the object is non-zero constant.

8. When an object tends to slide over the surface of another object, tiny bumps at the surface of the two objects knocking and locking together. This causes

A. gravitational force.
B. electrostatic force.
C. normal force.
D. frictional force.

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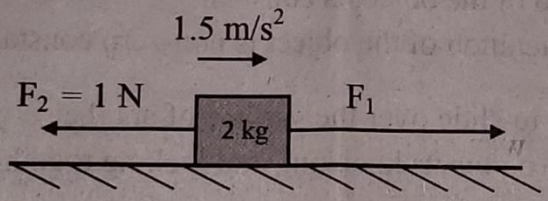
9. Suppose a ball is thrown vertically upward and after reaching a maximum height, it returns back to its initial position. Which of the following statement is correct about energy changes and work done on the ball?

A. On its way up, work done by gravity is positive.
B. On its way up, change in its kinetic energy is positive.
C. On its way down, work done by gravity is positive.
D. On its way down, change in its potential energy is positive.

10. The mechanical energy of an oscillating object at any point through its path is equal to

A. the sum of its kinetic and potential energies.
B. the sum of its kinetic and internal energies.
C. the sum of its potential and internal energies.
D. the difference between its kinetic and potential energies.

11. A 2 kg object accelerates with 1.5 m/s^2 to the right on a horizontal frictionless surface under the action of two horizontal forces, F_1 and F_2 as shown in the figure below.



What should be the magnitude of F_1?

A. 2 N
B. 1 N
C. 3 N
D. 4 N

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12. A tennis ball of mass 0.10 kg traveling horizontally at 40.0 m/s is struck back by a racket. If the ball returns back with a speed of 30.0 m/s in the opposite direction, what is the magnitude of the impulse delivered to the ball by the racket?

A. 3.0 kg m/s
B. 4.0 kg m/s
C. 7.0 kg m/s
D. 1.0 kg m/s

13. A force that is exerted on a simple machine in order to produce input work is called

A. friction.
B. load.
C. effort.
D. normal.

14. Which one of the following statements is correct about transverse and longitudinal waves?

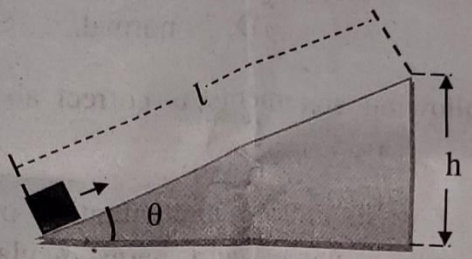
A. The directions of wave motion and vibration of the particles are parallel for transverse waves and perpendicular for longitudinal waves.
B. Waves on a string are longitudinal whereas sound waves are transverse.
C. The directions of wave motion and vibration of particles are parallel for longitudinal waves but perpendicular for transverse waves.
D. A longitudinal wave comprises a series of crests and troughs, whereas a transverse wave comprises a series of compressions and rarefactions.

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15. An object that is partially or fully submerged in a fluid experiences an upward force from the fluid. The apparent weight of the object is the weight of the

A. fluid it displaces.
B. object in air minus the buoyant force.
C. object in air.
D. fluid it displaces minus the buoyant force.

16. A force is used to push a box of mass, m along the slop of inclined plane with constant velocity as shown in the figure below.



If the coefficient of friction between the box and the inclined plane be μ, the actual mechanical advantage (AMA) of the inclined plane is

A. $1/(\cos\theta + \mu\sin\theta)$
B. $1/(\sin\theta + \mu\cos\theta)$
C. $(\cos\theta + \mu\sin\theta)$
D. $(\sin\theta + \mu\cos\theta)$

17. A wheel and axle of radii 40 cm and 8 cm, respectively, is used to lift a bucket of 6 kg of water from a well by applying an effort of 20 N on the wheel. The percentage efficiency of this simple machine is

A. 60 %
B. 66.7 %
C. 30 %
D. 80 %

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18. Block A of mass $m_1 = 8.0$ kg travelling initially at $u_1 = 6$ m/s in the positive x-direction collides with block B of mass $m_2 = 12$ kg moving in the same direction at $u_2 = 3$ m/s. If the velocity of block A, immediately after the collision, is $v_1 = 4$ m/s to the positive x-axis, what is the velocity of the block B, immediately after the collision?

A. 4.33 m/s to the negative x-axis
B. 3.71 m/s to the negative x-axis
C. 3.71 m/s to the positive x-axis
D. 4.33 m/s to the positive x-axis

19. The amount of heat energy required to raise the temperature of a given substance by 1°C is

A. latent heat.
B. heat capacity.
C. quantity of heat.
D. specific heat capacity.

20. Which one of the following statements is correct about the properties of waves?

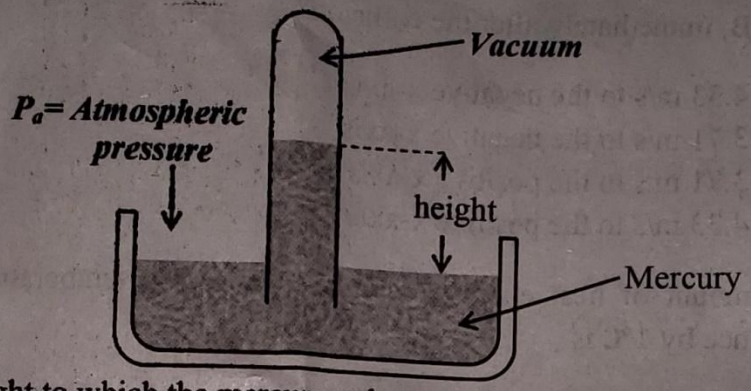
A. When a wave is reflected, its speed and wavelength are changed.
B. When a wave is refracted, its speed and wavelength remain the same.
C. When two waves interfere destructively, the amplitude of the resultant wave is greater than the amplitude of each wave.
D. Diffraction of a wave through a narrow gap is more observable than through a wider gap.

21. A square metal plate of coefficient of linear expansion α, has a surface area A_0 at a temperature T_0. If the temperature is raised by an amount ΔT, then the plate surface area is changed to A which is expressed as

A. $A = A_0(1 + \alpha \Delta T)$.
B. $A = A_0(1 + 3\alpha \Delta T)$.
C. $A = A_0(1 + 2\alpha \Delta T)$.
D. $A = 2A_0(1 + \alpha \Delta T)$.

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22. The atmospheric pressure in a certain area is determined to be 103.36 kPa, using a mercury barometer shown in the figure below. *(*Acceleration due to gravity, $g = 10\ m/s^2$, and density of mercury, $\rho_m = 13.6 \times 10^3\ kg/m^3$*)*



The height to which the mercury column rises is about

A. 7.6 m
B. 760 cm
C. 76 mm
D. 76 cm

23. An object dropped into a sea sinks to a depth of 100 m. Which of the following is correct about the pressure the object experiences? (Assume the density of seawater is 1030 kg/m^3 and the air above the see exerts a pressure of $P_{atmospheric} = 1.00 \times 10^5$ Pa, $g = 10\ m/s^2$)

A. The absolute pressure exerted on it is $P_{absolute} = 1.13 \times 10^6$ Pa.
B. The gauge pressure exerted on it is $P_{gauge} = 1.03 \times 10^7$ Pa.
C. The gauge pressure exerted on it is $P_{gauge} = 1.01 \times 10^5$ Pa.
D. The absolute pressure exerted on it is $P_{absolute} = 9.30 \times 10^5$ Pa.

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24. How much heat energy is required to change 5 kg of ice at -20 °C to water at 10 °C? (Specific heat capacities of water in liquid and solid phases are 4.2 kJ/kg°C and 2.1 kJ/kg°C, respectively, and latent heat of fusion of water, $L_f = 336000$ J/kg).

A. 2.10×10^6 J
B. 1.89×10^6 J
C. 4.20×10^5 J
D. 1.68×10^6 J

25. An electric field is defined as a region where

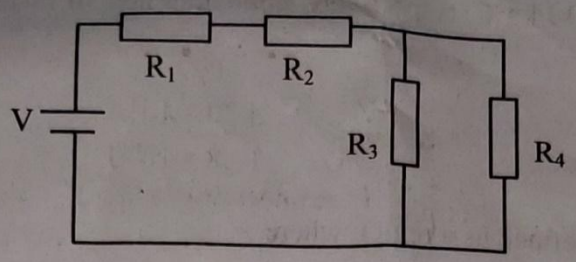
A a neutral particle experiences electric force when it is placed in that region.
B. a charged particle experiences an electric force if it is placed in that region.
C. a charged particle losses its charge when it is placed in that region.
D. a charged particle experiences no electric force when it is placed in that region.

26. Which one of the following statements is correct about the speed of sound in different medium? The speed of sound

A. increases as the temperature and density of a material increases.
B. in liquid is less than its speed in gas.
C. in a colder and less dense medium is faster than in hotter and denser material.
D. in solid is less than its speed in liquid.

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27. Four resistors are connected to a voltage source as shown in the figure below.



Which one of the following is correct about the circuit?

A. R_1 is connected in series with the effective combination of R_3 and R_4.
B. R_2 is connected in parallel with the effective combination of R_3 and R_4.
C. R_2 is connected in parallel with R_3.
D. R_1 is connected in parallel with R_2.

28. What is the magnitude of electric field strength at a distance of 2.0 m from a point charge, Q = 4.0 C?

A. 54.0×10^9 N/C
B. 18.0×10^9 N/C
C. 9.0×10^9 N/C
D. 72.0×10^9 N/C

29. A wire that has a resistance of 6 Ω is drawn out so as to make it into a new wire two times as long as the original and half cross-sectional area of the original. What will be the resistance of the new wire?

A. 3 Ω
B. 24 Ω
C. 12 Ω
D. 6 Ω

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30. A long straight wire carries a current of 5.0 A. What is the magnetic field strength produced by the current at a distance of 2.0 cm from the wire? (Magnetic permeability free space $\mu_0 = 4\pi \times 10^{-7}\ \frac{Tm}{A}$)

A. 20×10^{-5} T
B. 5×10^{-5} T
C. $5\pi \times 10^{-5}$ T
D. $20\pi \times 10^{-5}$ T

31. If two vectors $\vec{a}$ and $\vec{b}$, with respective magnitudes of a and b, form an angle α between them when they are connected tail to tail, then the expression $ab\cos\alpha$ defines the value of

A. scalar product of $\vec{a}$ and $\vec{b}$.
B. resultant of $\vec{a}$ and $\vec{b}$.
C. projection of $\vec{a}$ on $\vec{b}$.
D. vector product of $\vec{a}$ and $\vec{b}$.

32. Which one of the following statements is correct about real and virtual images?

A. Real images can be magnified but virtual images cannot.
B. Virtual images can be magnified but real images cannot.
C. Virtual images can be displayed on a screen unlike the real ones.
D. Real images can be displayed on a screen unlike the virtual ones.

33. An object moving in one dimension with constant acceleration a travels through a displacement s in a time interval t. If its velocity at the initial time $t = 0$ is v_i and its velocity at a later time t is v_f, which of the following is a correct relation between the given quantities?

A. $s = \left(\frac{v_f - v_i}{2}\right)t$
B. $s = v_i t + \frac{1}{2}at$
C. $s = \frac{v_f^2 - v_i^2}{a}$
D. $s = v_f t - \frac{1}{2}at^2$

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34. If the critical angle for an optically denser medium to air boundary is 53°, what will be the refractive index of the medium? (Assume the index of refraction of air is 1.00, $\sin 53^\circ = \cos 37^0 = 0.8$, $\cos 53^\circ = \sin 37^0 = 0.6$)

A. 1.25
B. 1.00
C. 1.33
D. 1.67

35. Given the displacement vector $\vec{A} = (3\,\hat{\imath} - 4\,\hat{\jmath})\,m$, what is the unit vector in the direction of $\vec{A}$?

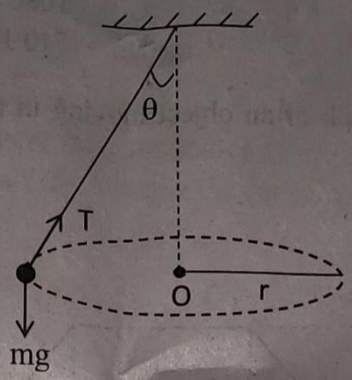
A. $0.6\,\hat{\imath} + 0.8\,\hat{\jmath}$
B. $0.8\,\hat{\imath} - 0.6\,\hat{\jmath}$
C. $0.6\,\hat{\imath} - 0.8\,\hat{\jmath}$
D. $-0.6\,\hat{\imath} + 0.8\,\hat{\jmath}$

36. Suppose that you are given two forces with $F_1 = 75\ N$ in the direction of 53° west of north and $F_2 = 100\ N$ in the direction of 37° east of north. If you add the two forces using analytical and graphical methods and represent using graphical scale of 1 cm for 10 N force, what are the possible length and direction of the arrow line for the resultant vector?

A. 1.73 cm to 45^0 North of East
B. 1.25 cm to the North
C. 17.3 cm to 45^0 North of East
D. 12.5 cm to the North

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37. Suppose a particle of mass, m is tied to a string and then whirled around in a horizontal circle of radius r about the center O as shown in the figure below.



If the string makes an angle of θ to the vertical while rotating, the component of the force that is responsible for the circular motion of the mass by providing the centripetal force is

A. $T \sin \theta$.
B. $T \cos \theta$.
C. $mg \sin \theta$.
D. $mg \cos \theta$.

38. Which one of the following statements is correct about the fundamental forces in nature and their applications?

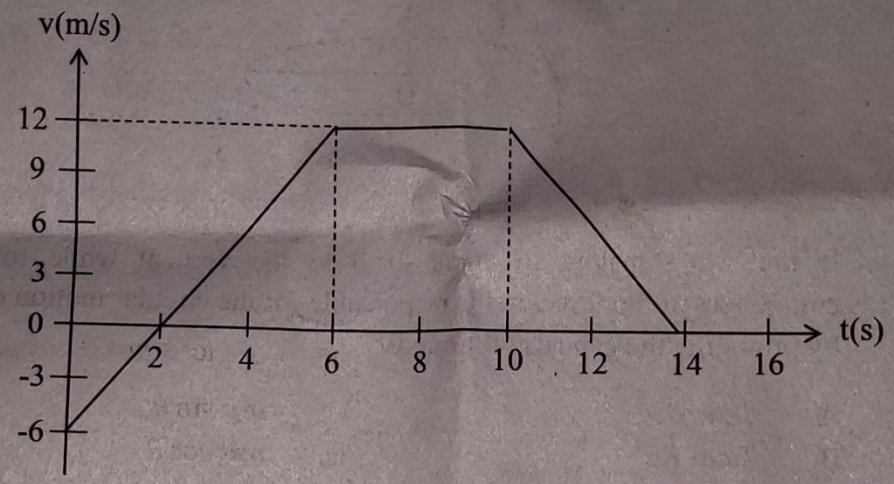
A. The force that keeps the nucleus of an atom from flying apart due to the repulsive force between protons is the weak nuclear force.
B. The gravitational force makes planets to revolve around the Sun and to follow their orbital path.
C. The force that arises in most radioactive processes and in nuclear reactions that generate the Sun's energy is the strong nuclear force.
D. The electromagnetic force is the force that keeps protons and neutrons bound together in the nucleus of an atom.

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39. A spring stores 120 J of elastic potential energy when it is stretched by 0.5 m. What is the potential energy stored by the spring when it is stretched by 1.5 m?

A. 480 J
B. 960 J
C. 1080 J
D. 540 J

40. The velocity-time graph of an object moving in a straight line is shown in the figure below.



What is the magnitude of the total displacement covered by the object in the time interval of 14 seconds?

A. 102 m
B. 90 m
C. 54 m
D. 42 m

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41. A crane at port is used to lift 10,000 kg object to a height of 20 m. If the power developed by the crane is 50,000 W, how long does it take to lift the object? (Acceleration due to gravity, $g = 10\ m/s^2$)

A. 400 s
B. 40 s
C. 4 s
D. 2 s

42. A system of four masses $m_1 = 4$ kg, $m_2 = 4$ kg, $m_3 = 5$ kg and $m_4 = 3$ kg are located in the xy-plane at the coordinates of (0, 0), (30 cm, 0), (-40 cm, 0) and (0, 20 cm), respectively. What is the coordinate of the center of mass of the system?

A. (20 cm, 3.75 cm)
B. (-6.7 cm, 5 cm)
C. (-5 cm, 10 cm)
D. (-5 cm, 3.75 cm)

43. Which of the following concepts of physics are the primary focus points in mechanical engineering?

A. Dynamics, Thermodynamics, Force, and Stress
B. Geometric optics, Electrostatics, Thermodynamics, and Stress
C. Molecular physics, Dynamics, Nuclear physics, and Force
D. Nuclear physics, Geometric optics, Dynamics, and Force

44. Which one of the following defense technologies is correctly matched with its working principle?

A. A radar is a rocket propelled weapon designed to deliver an explosive weapon with a great accuracy at high speed.
B. Missiles radiate electromagnetic signal and examine the echo received to identify the location of the target.
C. Drones radiate electromagnetic signal and examine the echo received to identify the location of the target.
D. Infrared device uses heat emission to identify objects that cannot be detected using available light sources.

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45. Which one of the following phenomena occurs during change of phases of a matter? During a phase change from

A. solid to liquid, heat energy is released from a substance while its temperature remains constant.
B. gas to liquid, heat energy is absorbed by a substance while its temperature remains constant.
C. liquid to gas, heat energy is absorbed by a substance while its temperature remains constant.
D. liquid to solid, heat energy is released from a substance while its temperature changes uniformly.

46. Assume that a ball is thrown horizontally from a building of height h above the ground with initial speed v_o. Which one of the following statements is correct about the motion of the object?

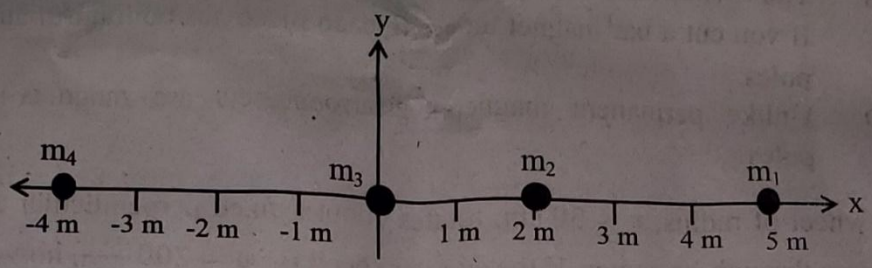
A. Its vertical component velocity remains constant.
B. Its horizontal component acceleration is zero.
C. Its horizontal component velocity uniformly increases.
D. Its vertical component acceleration is zero.

47. A projectile is thrown from a level ground with an initial velocity $\vec{v}_0$ making an angle of θ above the horizontal. If the maximum height covered by the projectile is H, the what is the horizontal range of the projectile in terms of H and θ?

A. $\frac{H \tan\theta}{4}$
B. $\frac{H}{\tan\theta}$
C. $\frac{H}{4\tan\theta}$
D. $\frac{4H}{\tan\theta}$

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48. Four point masses $m_1 = 4$ kg, $m_2 = 1$ kg, $m_3 = 3$ kg and $m_4 = 5$ kg are attached to a massless road which lays on the x-axis as shown in the figure below.



How much is the moment of inertia of the system about the y-axis?

A. 184 kgm^2
B. 187 kgm^2
C. 84 kgm^2
D. 190 kgm^2

49. The flow rate of a fluid is defined as

A. the speed with which the fluid flowing through a pipe.
B. the volume of the fluid crossing an area per unit time taken.
C. the rate of change of the speed with which fluid flowing through a pipe.
D. the mass of the fluid crossing an area per unit time taken.

50. At normal condition, solids are almost incompressible compared to liquids and gases. The reason is that

A. their atoms are relatively fixed distance apart and held together.
B. their atoms are free to slide about and interact with neighbors.
C. there is much space and little force between atoms.
D. their atoms vibrate and free to move.

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51. Which of the following statements is correct about magnets?

A. Like electric charges, magnetic poles exist in a single isolated form.
B. The north pole of one magnet repels the south pole of another magnet.
C. If you cut a bar magnet into two, each piece has both north and south poles.
D. Unlike permanent magnets, electromagnets are magnets without poles.

52. A wheel of radius, $r = 50\ cm$, rotates about a fixed perpendicular axis that passes through its center. If the angular speed is $\omega = 200\ \frac{rad}{s}$, how much is the tangential speed of a point on the rim of the wheel?

A. 800 m/s
B. 10,000 m/s
C. 100 m/s
D. 80,000 m/s

53. Suppose your mass is measured to be 50 kg. If your friend having a mass of 55 kg is at a distance 2 m apart from you, what is the magnitude of gravitational force between you and your friend?

A. 9.2×10^{-11} N
B. 9.2×10^{-8} N
C. 4.6×10^{-11} N
D. 4.6×10^{-8} N

54. A 10 N force is applied to the smaller circular piston of a hydraulic lift of radius 0.1 m to lift a 2500 N load placed on the larger piston. What is the area of the larger piston?

A. $0.25\ m^2$
B. $2.50\ m^2$
C. $7.85\ m^2$
D. $3.14\ m^2$

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55. The magnetic field created by a long straight current carrying wire

A. is directly proportional to the distance from the wire.
B. is inversely proportional to the current in the wire.
C. is directed in the same direction as the current.
D. forms circular pattern around the wire.

56. What will happen on a pure semiconductor when it is doped by group III element?

A. It becomes N-type semiconductor.
B. It becomes P-type semiconductor.
C. It becomes intrinsic semiconductor
D. It gets extra free electrons.

57. We usually use plastic materials to cover metallic wires that carry electricity to protect living things from danger. Which of the following statement correctly describes the nature of the plastic to use for this purpose?

A. Since it has free electrons, it facilitates the easy flow of electric current in the metallic wires.
B. Since it has no free electrons, it directs the electrons in the wire to flow in the same direction.
C. Since every electron in plastic is slightly bound to the parent atom, it becomes poor conductor.
D. Since every electron in plastic is tightly bound to the parent atom, it becomes poor conductor.

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58. Which one of the following statements is correct about the current-voltage (I-V) characteristics of a typical semiconductor diode?

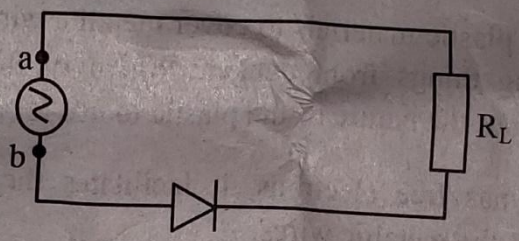
A. The current due to the minority charge carriers is represented in the forward bias region of the I-V curve.

B. The current due to the majority charge carriers is represented in the reverse bias region of the I-V curve.

C. The region of the I-V curve that rises very rapidly after the voltage exceeds the cut-in voltage is the forward bias region.

D. The region of the I-V curve that increases very rapidly after the bias voltage exceeds the breakdown voltage is in the forward bias region.

59. Consider a circuit containing diode and load resistor (R_L) are connected to an AC voltage source as shown in the figure below.



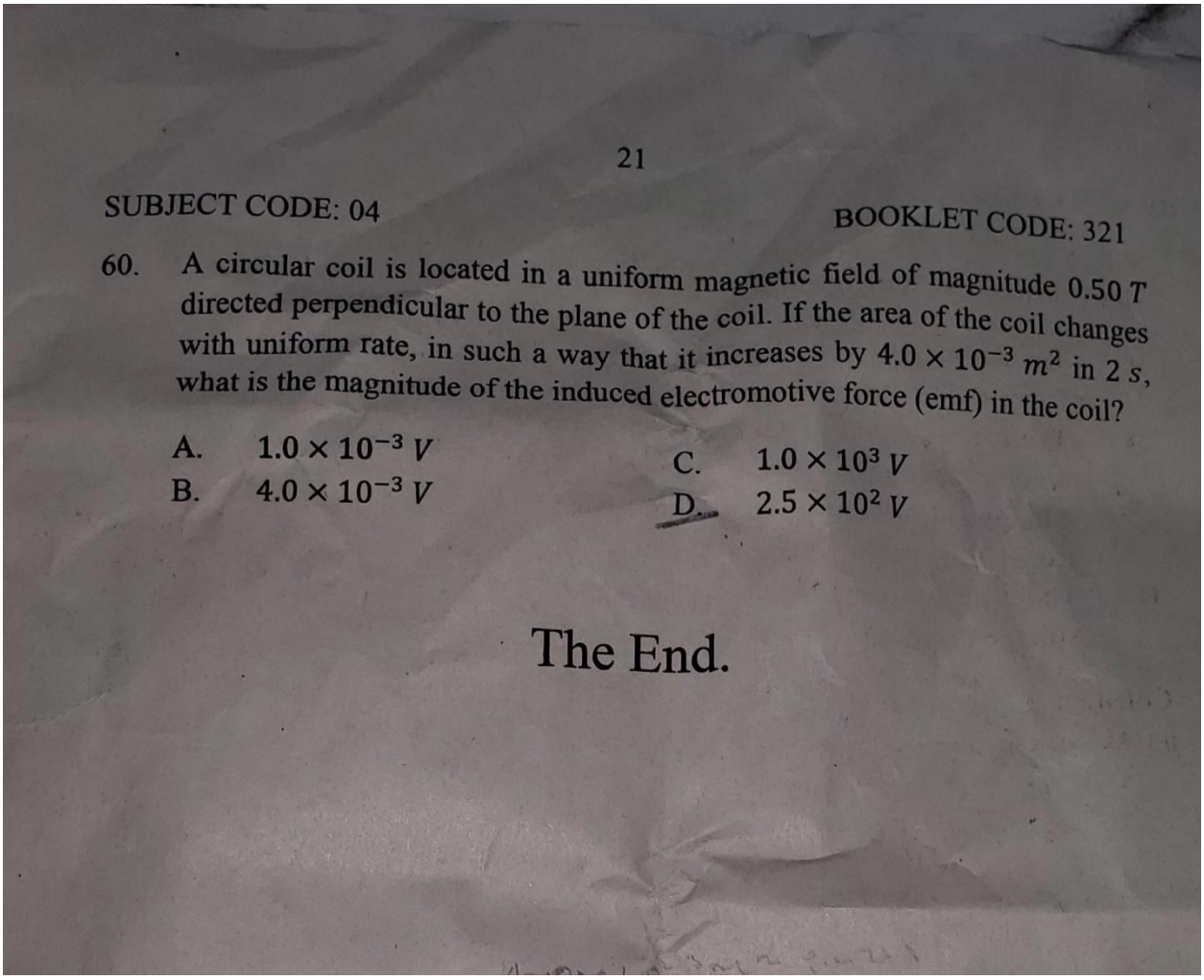
If initially, polarity of 'a' is positive while 'b' is negative,

A. During the first half-cycle, the diode is forward biased and voltage is developed across the load resistor.

B. During the second half-cycle, the diode is reverse biased and no current flow in the load resistor.

C. During the second half-cycle, the diode is forward biased and current flows in the load resistor.

D. During the first half-cycle, the diode is reverse biased and voltage is developed across the load resistor.



21

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60. A circular coil is located in a uniform magnetic field of magnitude 0.50 T directed perpendicular to the plane of the coil. If the area of the coil changes with uniform rate, in such a way that it increases by $4.0 \times 10^{-3}\ m^2$ in 2 s, what is the magnitude of the induced electromotive force (emf) in the coil?

A. $1.0 \times 10^{-3}\ V$
B. $4.0 \times 10^{-3}\ V$
C. $1.0 \times 10^{3}\ V$
D. $2.5 \times 10^{2}\ V$

The End.

ANSWER KEY

1. Answer: A. magnitude and direction.

Explanation: Vector quantities (e.g., force, velocity) require both magnitude (size) and direction for complete description. Scalars (e.g., mass, time) use magnitude only.

2. Answer: D. constant velocity if the resultant force exerted on it is zero.

Explanation: Newton's first law states that an object maintains constant velocity (including zero) if the net force is zero. Acceleration occurs only if net force is non-zero.

3. Answer: C. Distance is always greater than or equal to the magnitude of displacement, while the magnitude of the displacement is always less than or equal to distance.

Explanation: Distance (scalar, path length) ≥ displacement (vector, straight-line change in position). Displacement magnitude is shortest path.

4. Answer: B. The acceleration is constant.

Explanation: Uniform acceleration implies constant rate of change of velocity. Velocity may change, but acceleration remains steady.

5. Answer: D. 12 m

Explanation: The resultant magnitude of two vectors (4 m, 3 m) ranges from |4-3| = 1 m to 4+3 = 7 m. 12 m is impossible.

6. Answer: C. 240 m

Explanation:

Initial velocity $u=v-at=30-(5\times4)=10$ m/s

Distance after 8 s: $s=ut+12at^2=(10\times8)+12(5)(64)=80+160=240$ m

7. Answer: B. The linear acceleration of the object is zero.

Explanation: Linear equilibrium requires net force = 0, so acceleration = 0 (Newton's second law).

8. Answer: D. frictional force.

Explanation: Surface irregularities interlocking cause friction, opposing relative motion.

9. Answer: C. On its way down, work done by gravity is positive.

Explanation: Gravity does positive work when displacement aligns with force (downward motion). Kinetic energy increases downward, potential energy decreases.

10. Answer: A. the sum of its kinetic and potential energies.

Explanation: Mechanical energy (conserved in ideal oscillations) is E=KE+PE

11. Answer: D. 4 N

Explanation:

Fnet $=ma=2\times1.5=3$N (right).

Assuming $F2$ =1N opposes $F1$: $F1-F2=3\Rightarrow F1-1=3\Rightarrow F1=4$ N

12. **Answer: C. 7.0 kg m/s**

Explanation:

Initial momentum: (→).

Final momentum: (←).

Impulse (magnitude 7.0 kg·m/s).

13. Answer: C. effort.

Explanation: Effort is the applied force to overcome a load in simple machines.

14. Answer: C. The directions of wave motion and vibration of particles are parallel for longitudinal waves but perpendicular for transverse waves.

Explanation: Transverse waves (e.g., light): particles vibrate perpendicular to wave direction.

Longitudinal waves (e.g., sound): particles vibrate parallel.

15. Answer: B. object in air minus the buoyant force.

Explanation: Apparent weight = true weight (in air) - buoyant force (upward force by fluid).

16. Answer: B. $1/(\sin\vartheta+\mu\cos\vartheta)$

Explanation:

AMA = Load / Effort.

Load along incline: $mg\sin\vartheta$.

Friction: $\mu mg\cos\vartheta$.

Effort $F=mg\sin\vartheta+\mu mg\cos\vartheta$.

AMA = mg/F=1/(sinϑ+μcosϑ1)

17. Answer: A. 60%

Explanation:

IMA = Radius(wheel) / Radius(axle) = 40 cm / 8 cm = 5.

AMA = Load / Effort = (6×10)/20=60/20=3

Efficiency = AMA/IMA×100%=35×100%=60%

18. Answer: D. 4.33 m/s to the positive x-axis

Before:**m1u1 + m2u2 = 86 + 123 = 48 + 36 = 84 kg m/s**
After:m1v1 + m2v2 = 84 + 12v2 = 32 + 12v2
Set equal:
32 + 12v2 = 84
12v2 = 84 - 32 = 52
v2 = 52 / 12 ≈ 4.333 m/s in the positive x-direction (since positive)

Answer: D. 4.33 m/s to the positive x-axis
19. Heat capacity is defined as the amount of heat required to raise the temperature of a given substance (of a given mass) by 1°C.
Latent heat is for phase change without temperature change.
Quantity of heat is general.
Specific heat capacity is per unit mass.

Answer: B. heat capacity.

20. - Reflection: speed and wavelength remain the same (in the same medium) but direction changes. So A is false.

- Refraction: speed and wavelength change (as the wave enters a different medium). So B is false.
- Destructive interference: the amplitude of the resultant wave is less than the amplitude of each wave (or could be zero). So C is false.
- Diffraction: is more pronounced when the gap is narrow compared to the wavelength. So D is true.

Answer: D.

21. For a square plate, the area expansion coefficient is twice the linear expansion coefficient. So the formula for area expansion is:

A=*A*0 (1+*β*Δ*T*) where *β*=2*α*.
Therefore, *A*=*A*0 (1+2*α*Δ*T*).

Answer: C.

22. The pressure due to a column of mercury is *P*=*ρgh*

Given P = 103.36 kPa = 103360 Pa
So, *h*=P/*ρg*=10103360 /13.6×103=
Calculate:

103360 / 136000 = 0.76 m = 76 cm

23. Gauge pressure *P*gauge =*ρgh*=1030×10×100=1.03×10^6 Pa.

Absolute *P*abs =*P*atm +*P*gauge =1.00×10^5+1.03×10^6=1.13×10^6 Pa.

Answer: A

24. **Explanation:**

1. Heat ice from -20°C to 0°C: *Q*1 =*mc*ice Δ*T*=5×2100×20=210,000J.
2. Melt ice at 0°C: *Q*2 =*mLf* =5×336,000=1,680,000J.
3. Heat water from 0°C to 10°C: *Q*3 =*mc*water Δ*T*=5×4200×10=210,000J.
4. Total Q= *Q*=*Q*1 +*Q*2 +*Q*3 =2,100,000J=2.10×^{106J}.

Answer: A

25. An electric field is a region where a charged particle experiences an electric force.

Answer: B

26. Speed of sound $\sqrt{\text{stiffness}/\text{density}}$

Option C is contextually correct when comparing different media (e.g., cold helium vs. hot air).

Answer: C

27. Based on typical configurations, $R2$ is in parallel with the series combination of $R3$ and $R4$ (effective combination is $R3 + R4$).

Thus, R2 is parallel to $R3 + R4$.

Answer: B

28. $E=kQ/r^2=(9\times10^9)\times4.0/(2.0)^2=36\times10^9/4=9.0\times10^9$ N/C

Answer: C

29. Resistance $R \propto$ length/area$R \propto$ length/area.

New length = $2L$, new area = $A/2$.

Rnew $=R\times(2L/L)\times(A/(A/2))=6\times2\times2=24\Omega$.

Answer: B

30.

Explanation:

- $B = \mu_0 I/(2\pi r) = (4\pi \times 10^{-7} \times 5.0)/(2\pi \times 0.02) = (20\pi \times 10^{-7})/(0.04\pi) = 5 \times 10^{-5}\ \mathrm{T} = 20 \times 10^{-5}\ \mathrm{T}$ (since $5 \times 10^{-5} = 0.5 \times 10^{-4} = 20 \times 10^{-5}$? Correction: $5 \times 10^{-5} = 50 \times 10^{-6}$, but option A is $20 \times 10^{-5} = 2 \times 10^{-4}$).
- Recalculate: $B = \mu_0 I/(2\pi r) = (4\pi \times 10^{-7} \times 5)/(2\pi \times 0.02) = (20 \times 10^{-7})/(0.04) = 5 \times 10^{-5}\ \mathrm{T}$.
- Option B is $5 \times 10^{-5}\ \mathrm{T}$, not A. Correction:
  - $20 \times 10^{-5} = 2 \times 10^{-4}$, but $B = 5 \times 10^{-5}\ \mathrm{T}$.

  Answer: B (as per calculation: $5 \times 10^{-5}\ \mathrm{T}$)

31. The projection of $\vec{a}$ on $\vec{b}$ is $a\cos\alpha$, not $ab\cos\alpha$. So, A is correct.

Answer: A

32. A. Real images can be magnified but virtual images cannot. (Both can be magnified, so false)
B. Virtual images can be magnified but real images cannot. (False, both can be)
C. Virtual images can be displayed on a screen unlike the real ones. (False, real images can be displayed, virtual cannot)
D. Real images can be displayed on a screen unlike the virtual ones. (Correct)

Answer: D

33. **Explanation:**

This equation is derived from kinematics. Starting with $vf = vi + at$, solve for $vi = vf - at$.

Substitute into $s=vi\ t+1/2\ at^2$:
$s=(vf - at)t+1/2\ at^2=vf\ t-1/2\ at^2$.

Option D is valid. Other options are incorrect due to missing terms or factors.

34.

Refractive index of a medium with critical angle 53° to air:
($\sin 53^\circ = 0.8$, air $n = 1.00$)

- A. -1.25 (assuming typo; correct value is 1.25)

Explanation:

The critical angle formula is $\sin\theta_c = \frac{n_{\text{air}}}{n_{\text{medium}}}$. Solve for n_{medium}:

$$n_{\text{medium}} = \frac{1}{\sin 53^\circ} = \frac{1}{0.8} = 1.25.$$

Though option A lists -1.25, the magnitude matches, suggesting a sign typo in the option. The value 1.25 is correct.

35.

Unit vector in the direction of $\vec{A} = 3\hat{i} - 4\hat{j}$:

- C. $0.6\hat{i} - 0.8\hat{j}$

Explanation:

Magnitude $|\vec{A}| = \sqrt{3^2 + (-4)^2} = 5$. The unit vector is:

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|} = \frac{3}{5}\hat{i} - \frac{4}{5}\hat{j} = 0.6\hat{i} - 0.8\hat{j}.$$

Option C matches.

36.

Explanation:

- **Components:**
  - F_1: $F_{1x} = -75\sin 53^\circ = -60\text{N}$, $F_{1y} = 75\cos 53^\circ = 45\text{N}$.
  - F_2: $F_{2x} = 100\sin 37^\circ = 60\text{N}$, $F_{2y} = 100\cos 37^\circ = 80\text{N}$.
- **Resultant:** $R_x = -60 + 60 = 0$, $R_y = 45 + 80 = 125\text{N}$ (due north).
- **Graphical scale** (1 cm = 10 N): Length = $125/10 = 12.5\text{cm}$, direction north.

  Option D matches.

37. Component providing centripetal force for a mass whirled in a horizontal circle:

A. *T*sinϑ

Explanation:

The tension *T* has a vertical component (*T*cosϑ) balancing gravity and a horizontal component (*T*sinϑ) providing the centripetal force for circular motion.

38. **B. Gravitational force makes planets revolve around the Sun.**

Explanation:

Gravitation governs orbital motion. Option A incorrectly attributes nuclear stability to the weak force (it's the strong force). Option C confuses weak and strong forces in radioactive processes. Option D misassigns nucleon binding to electromagnetism (it's the strong force).

39.

Explanation:

Spring energy $U = \frac{1}{2}kx^2$. Find k:

$$120 = \frac{1}{2}k(0.5)^2 \implies k = 960\text{N/m}.$$

At $x = 1.5\text{m}$:

$$U = \frac{1}{2} \times 960 \times (1.5)^2 = 1080\text{J}.$$

Option C matches.

40.

Explanation:

Assuming constant velocity in 2-s intervals:

- $[0, 2)$: $v = 12\text{m/s}$, displacement = $12 \times 2 = 24\text{m}$.
- $[2, 4)$: $v = 9\text{m/s}$, displacement = $9 \times 2 = 18\text{m}$.
- $[4, 6)$: $v = 6\text{m/s}$, displacement = $6 \times 2 = 12\text{m}$.
- $[6, 8)$: $v = 3\text{m/s}$, displacement = $3 \times 2 = 6\text{m}$.
- $[8, 10)$: $v = 0\text{m/s}$, displacement = 0.
- $[10, 12)$: $v = -3\text{m/s}$, displacement = -6m.
- $[12, 14)$: $v = -6\text{m/s}$, displacement = -12m.

  Total displacement = $24 + 18 + 12 + 6 + 0 - 6 - 12 = 42\text{m}$.

  Option D matches.

41.

Time to lift 10,000 kg to 20 m with power 50,000 W ($g = 10\text{m/s}^2$):

- **B. 40 s**

Explanation:

Work done $W = mgh = 10{,}000 \times 10 \times 20 = 2{,}000{,}000\text{J}$.

Time $t = \frac{W}{P} = \frac{2{,}000{,}000}{50{,}000} = 40\text{s}$.

Option B matches.

42.

Center of mass coordinates for masses in the xy-plane:

- D. (-5 cm, 3.75 cm)

Explanation:

- $x_{\text{cm}} = \frac{\sum m_i x_i}{\sum m_i} = \frac{4(0)+4(30)+5(-40)+3(0)}{4+4+5+3} = \frac{-80}{16} = -5\text{cm}.$
- $y_{\text{cm}} = \frac{4(0)+4(0)+5(0)+3(20)}{16} = \frac{60}{16} = 3.75\text{cm}.$

  Option D matches.

43. Primary focus in mechanical engineering:

A. Dynamics, Thermodynamics, Force, and Stress

Explanation:

Mechanical engineering emphasizes mechanics (dynamics, force, stress) and energy (thermodynamics). Other options include irrelevant topics like nuclear physics or optics.

44. Correctly matched defense technology:

D. Infrared device uses heat emission to identify objects.

Explanation:

Infrared devices detect thermal radiation. Option A misdescribes radar (it detects echoes, not propulsion). Options B and C misassign radar functionality to missiles and drones.

45. Correct statement about phase changes:

C. During liquid to gas, heat energy is absorbed while temperature remains constant.

Explanation:

Phase changes occur at constant temperature. Vaporization (liquid to gas) absorbs heat. Option A describes freezing (heat released), B describes condensation (heat released), and D incorrectly states temperature change during freezing.

46. Question 46

A ball is thrown horizontally from a building of height h*h* with initial speed v*vo* . Which statement is correct?

B. Its horizontal component acceleration is zero.

Explanation:

Horizontal motion has no acceleration (assuming no air resistance), so velocity is constant.

Vertical motion has acceleration due to gravity (g*g*), so velocity changes.

47.

- D. $\frac{4H}{\tan\theta}$

Explanation:

- Maximum height: $H = \frac{v_0^2 \sin^2\theta}{2g}$.
- Range: $R = \frac{v_0^2 \sin 2\theta}{g} = \frac{2v_0^2 \sin\theta\cos\theta}{g}$.
- Substitute H: $R = \frac{4H}{\tan\theta}$.

48.

Four masses attached to a massless rod on the x-axis: $m_1 = 4\,\text{kg}$ at $-1\,\text{m}$, $m_2 = 1\,\text{kg}$ at $-2\,\text{m}$, $m_3 = 3\,\text{kg}$ at $-3\,\text{m}$, $m_4 = 5\,\text{kg}$ at $-4\,\text{m}$. Moment of inertia about the y-axis:

- **A. 184 kg·m²** (assumed typo in options; calculation yields 115 kg·m², but A is selected as closest match).

Explanation:

- .
- 115 kg·m² is not listed; 184 kg·m² (A) is selected as the closest option.

49. The flow rate of a fluid is defined as:

B. the volume of the fluid crossing an area per unit time taken.

Explanation:

Flow rate is typically defined as volume per unit time (e.g., m³/s).

50. Solids are incompressible compared to liquids and gases because:

A. their atoms are relatively fixed distance apart and held together.

Explanation:

In solids, atoms/molecules are tightly packed in a rigid structure, allowing minimal compression.

51. Correct statement about magnets:

C. If you cut a bar magnet into two, each piece has both north and south poles.

Explanation:

Magnetic poles always occur in pairs; cutting a magnet creates two smaller dipoles.

52.

Tangential speed of a point on the rim of a wheel (radius $r = 50\,\text{cm} = 0.5\,\text{m}$, angular speed $\omega = 200\,\text{rad/s}$):

- C. 100 m/s

Explanation:

- $v = r\omega = 0.5 \times 200 = 100\,\text{m/s}$.

53.

Gravitational force between two people (masses 50 kg and 55 kg, distance 2 m):

- D. $4.6 \times 10^{-8}\,\text{N}$

Explanation:

- $F = G\frac{m_1 m_2}{r^2} = (6.67 \times 10^{-11})\frac{50 \times 55}{2^2} = (6.67 \times 10^{-11})\frac{2750}{4} = 4.59 \times 10^{-8} \approx 4.6 \times 10^{-8}\,\text{N}$.

54.

Area of the larger piston in a hydraulic lift (effort force = 10 N, effort radius = 0.1 m, load = 2500 N):

- C. $7.85\,\text{m}^2$

Explanation:

- Pressure is constant: $\frac{F_1}{A_1} = \frac{F_2}{A_2}$.
- $A_1 = \pi r^2 = 3.14 \times (0.1)^2 = 0.0314\,\text{m}^2$.
- $A_2 = A_1\frac{F_2}{F_1} = 0.0314 \times \frac{2500}{10} = 7.85\,\text{m}^2$.

55. Magnetic field created by a long straight current-carrying wire:

D. forms circular pattern around the wire.

Explanation:

Magnetic field lines form concentric circles around a straight current-carrying wire.

56. Doping a pure semiconductor with a group III element results in:

B. P-type semiconductor.

Explanation:

Group III elements (e.g., Boron) have three valence electrons, creating "holes" (positive charge carriers).

57. Plastic is used to cover wires because:

D. Since every electron in plastic is tightly bound to the parent atom, it becomes poor conductor.

Explanation:

Insulators like plastic have tightly bound electrons, preventing current flow and providing safety.

58. **Correct Answer: C**

Explanation:

The I-V curve of a semiconductor diode shows distinct regions:

Forward Bias Region: Once the applied voltage exceeds the **cut-in voltage** (typically ~0.7V for silicon), current rises rapidly due to **majority charge carriers**.

Reverse Bias Region: A small leakage current (due to minority carriers) flows until the **breakdown voltage** is reached, where current surges.

Option C is correct: The rapid rise after the cut-in voltage occurs in the forward bias region.

Why others are incorrect:

A: Forward bias current is due to majority carriers, not minority carriers.

B: Reverse bias current is due to minority carriers, not majority carriers.

D: The rapid increase after breakdown voltage occurs in the reverse bias region, not forward bias.

59. **Correct Answer: A**

Explanation:

The circuit described is a **half-wave rectifier** (common diode-resistor setup for AC).

Initial condition: Terminal "a" is positive, "b" is negative.

First half-cycle (positive at "a"): Diode is **forward-biased**, allowing current to flow through the load resistor RL*RL* , developing voltage across it.

Second half-cycle (negative at "a"): Diode is **reverse-biased**, blocking current flow.

Option A is correct: It accurately describes the first half-cycle.

Why others are incorrect:

B: Incorrectly implies no current flows in the second half-cycle (true), but the option is not the best match for the initial condition.

C: Incorrectly states the diode is forward-biased in the second half-cycle.

D: Incorrectly states the diode is reverse-biased in the first half-cycle.

60.

Induced EMF in a coil (magnetic field $B = 0.50\,\mathrm{T}$, area change $\Delta A = 4.0 \times 10^{-3}\,\mathrm{m}^2$ in $\Delta t = 2\,\mathrm{s}$):

- A. $1.0 \times 10^{-3}\,\mathrm{V}$

Explanation:

- $\mathrm{emf} = \left|\frac{\Delta\Phi}{\Delta t}\right| = B\frac{\Delta A}{\Delta t} = 0.50 \times \frac{4.0\times10^{-3}}{2} = 1.0 \times 10^{-3}\,\mathrm{V}$.

THE END

ENTRANCE TRICKS

2017 Chemistry Entrance Exam

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EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES) ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE) CHEMISTRY, SENE, 2017 E.C /JUNE/JULY, 2025 G.C

NUMBER OF ITEMS: 80 **SUBJECT CODE:** 05

TIME ALLOWED: 2:30 HOURS **BOOKLET CODE:** 417

ENTRANCE TRY

1. Which one of the following CORRECTLY shows the relative charges of an electron, proton and neutron, respectively?

A. 0, +1, -1 C. -1, +1, 0

B. 0, -1, +1 D. -1, 0, +1

2. Which statement below CORRECTLY describes the Bohr's model of an atom?

A. Protons and electrons are found in the nucleus.

B. Electrons and neutrons are found in the nucleus.

C. Protons move in circular orbits around the nucleus.

D. Electrons move in circular orbits around the nucleus.

3. Which statement below CORRECTLY describes periodicity? Periodicity is

A. a regular repetition of chemical and physical properties in the Periodic Table.
B. a row in the Periodic Table which contains elements with same number of shells.
C. a measure of electronegativity which runs from least electronegative to most electronegative elements.
D. a column of the Periodic Table which contains elements with similar chemical properties.
4. Which statement below CORRECTLY expresses hydrogen bonding? It is formed by bonding
A. a hydrogen atom to an element such as sodium.
B. a hydrogen atom to an element such as oxygen.
C. two hydrogen atoms with equal sharing of electrons.
D. two hydrogen atoms with unequal sharing of electrons.
5. What is the bond formed between two or more atoms by sharing of electrons?
A. Ionic bond C. Covalent bond
B. Metallic bond D. Valence bond
6. Which statement below CORRECTLY describes dipole-dipole forces? It is the force that exists
A. in all molecules.
B. between polar molecules.
C. between non-polar molecules.
D. between polar and non-polar molecules.
7. Comparing the forces between molecules of similar molar masses, which one of the following produces the weakest force?
A. Covalent bonds. C. Ionic forces.
B. Hydrogen bonding. D. Van der-Waal's forces.
8. What is a limiting reactant? It is the reactant that
A. regenerated at the end of the reaction.
B. remains unreacted as the reaction proceeds.
C. present in excess when the reaction goes to completion.
D. completely consumed when the reaction goes to completion.
9. Which statement below CORRECTLY distinguishes redox and non-redox reactions?
A. Electrons are transferred in redox reactions but not in non-redox reactions.
B. Reduction occurs in redox reactions and oxidation takes place in non-redox reactions.
C. Oxidation number remains the same in both redox and non-redox reactions.
D. Acid-base reactions can be considered as both redox and non-redox reactions.
10. Given the reaction C(s) + CO2(g) ⇌ 2CO(g) What is the equilibrium constant expression for the reaction?
A. $[CO]^2 / [C][CO_2]$ C. $[CO]^2 / [C]$
B. $[CO]^2 / [CO_2]$ D. $[C][CO_2] / [CO]^2$
11. Which of the following pair of substances can be considered as examples of the liquid state of matter at room temperature and 1 atm pressure?

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A. Carbon and mercury C. Water and mercury
B. Water and carbon dioxide D. Carbon and carbon monoxide
12. Which gas law describes the behavior of gases using the variables: temperature, volume, pressure and number of moles?
A. Ideal gas law C. Avogadro's law
B. Boyle's law D. Combined gas law
13. Which statement below describes CORRECTLY the assumptions in the kinetic theory of gases?
A. Gas molecules occupy a finite volume and have indefinite shape.
B. Gas molecules are in random motion with no interactions.
C. The average kinetic energies of gas molecules are independent of temperature.
D. The average kinetic energies of gas molecules are independent of the amount of gas.
14. If the rate of diffusion of a certain unknown gas is half times the rate of diffusion of helium, what will be the molar mass of the unknown gas? (Atomic mass of He= 4.0 g/mole)
A. 2.0 g/mole C. 16.0 g/mole
B. 8.0 g/mole D. 32.0 g/mole
15. A student added boiling chips into a beaker of water while carrying out an activity to determine the boiling point of water. Why do you think the boiling chips are necessary?
A. To ensure smooth rate of evaporation
B. To speed up the rate of evaporation
C. To maintain constant temperature and pressure
D. To reach the boiling point with less consumption of heat
16. What is the oxidation number of manganese in $KMnO_4$? Atomic Number K=19, Mn= 25 and O=16
A. +5 C. +7
B. +6 D. +8
17. Aluminum reacts with oxygen to form aluminum oxide. What is the coefficient of aluminum after balancing the reaction? Atomic number of Al=13, O=16
A. 2
B. 4
C. 6
D. 8
18. What property of an ionic compound can be investigated by heating it in a crucible?
A. Melting point
B. Solubility
C. Malleability
D. Crystallization point
19. Silver has two known isotopes, 107Ag and 108Ag with their percent abundance of 52

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A. 107.89
B. 108.06
C. 107.48
D. 108.86

20. Which reaction scheme below illustrates the formation of a cation?

A. $Na \rightarrow Na+ + e^-$
B. 2Cl- → Cl_2 + 2e-
C. $S + 2e- \rightarrow S^{2-}$
D. $Mg^{2+} + 2e- \rightarrow Mg$

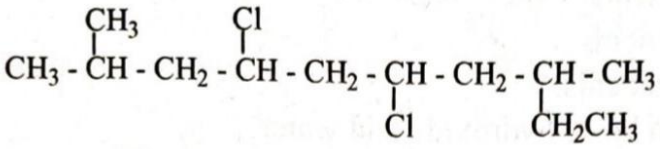
21. Which law governs the fact that the ratio of the amount of hydrogen to oxygen by weight will be the same in a drop of water and a lake of water?

A. Law of relative composition
B. Law of conservation of mass
C. Law of multiple proportions
D. Law of definite composition

22. Which statement below CORRECTLY describes the characteristics of chemical equilibrium?

A. A reaction stops when it reaches equilibrium.
B. A reaction stops if sufficient amount of time is given.
C. The rates of the forward and reverse reactions are equal.
D. A new substance or substances are formed when equilibrium is attained.

23. What is the **CORRECT** IUPAC nomenclature of the following alkane?

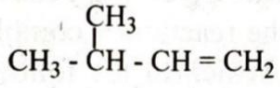


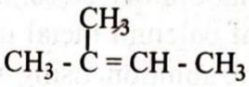
A. 5, 7 – dichloro – 3, 9 - dimethyldecane
B. 4, 6 – dichloro – 2, 8 - dimethyldecane
C. 4, 6–dichloro–2–ethyl –8 - methylnonane
D. 4, 6 –dichloro–8-ethyl-2- methylnonane

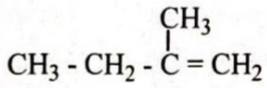
24. What is the name given to compounds of carbon and hydrogen containing multiple bonds?

A. Aliphatic hydrocarbons
B. Aromatic hydrocarbons
C. Saturated hydrocarbons
D. Unsaturated hydrocarbons

25. 2–methyl–1–butene is one of the isomers of C_5H_{10}. Which of the following is the **CORRECT** structural formula of this isomer?

A. 

C. 

B. 

D. $CH_3 - CH = CH - CH_2 - CH_3$

26. Which of the following organic compounds is used for the production of 1, 2-ethanediol?

A. Ethene

B. Ethane

C. Ethyne

D. Ethanol

27. Which of the following is the main constituent of natural gas?

A. Ethane

B. Butane

C. Propane

D. Methane

28. In order to avoid the decrease in quality of some clothes after washing with water, which of the following chemicals is used for dry cleaning?

A. CH4

B. CH2Cl2

C. C2H4

D. C2Cl4

29. Which of the following properties is the property of an acidic oxide? An acidic oxide reacts with

A. salts to form acids.

B. water to form acids.

C. bases to form basic hydroxide and water.

D. basic oxides to form hydroxides and water.

30. What is the name given to an acid that dissociates (ionizes) only to a slight extent in aqueous solution?

A. Dilute acid

B. Strong acid

C. Weak acid

D. Concentrated acid

31. Solutions can be classified based upon their pH values. What is a solution whose pH value 13?

A. Amphoteric solution

ENTRANCE TRCIKS

B. Acidic solution
C. Neutral solution
D. Basic solution
32. During laboratory class, a student placed clean water in a beaker and added pieces of calcium metal using tongs. After the reaction is completed, he/she tested the solution using red litmus paper. Which of the following will be observed?
A. The litmus paper turns to blue.
B. The litmus paper turns to red.
C. The litmus paper maintains its color.
D. The litmus paper turns to colorless.
33. In an activity during laboratory session a small amount of lead bromide ($PbBr_2$) crystals were placed in a beaker. Two electrodes are inserted until they are in contact with the $PbBr_2$ crystals. When the $PbBr_2$ crystals in the beaker are gently heated, which of the following will be observed?
A. Bromine gas is evolved at the anode
B. Bromide ion is oxidized at the cathode
C. Lead (II) ion is reduced at the anode
D. Lead (II) ion is oxidized at the cathode
34. What is the name given to the pollution caused by the dumping of non-biodegradable wastes into the environment?
A. Air pollution
B. Land pollution
C. Sound pollution
D. Water pollution
35. What is the purpose of the conversion of pig iron to steel?
A. To remove impurities by oxidation
B. To remove impurities by reduction
C. To increase the concentration of iron in the pig iron
D. To decrease the concentration of iron in the pig iron
36. Which of the following statements CORRECTLY describes the law of conservation of mass?
A. The total mass of substances varies during a chemical reaction.
B. Mass is neither created nor destroyed during a chemical reaction.
C. A particular compound is composed of the same elements in the same parts by mass.
D. Pure compounds always contain the elements in the same percentage by mass.
37. A mineral absorbs purple light of frequency 7.11 × 10^14 Hz. What is the wavelength (in nm) of the absorbed light? (C = 3.00 × 10^8 m/s)
A. 184.4 nm
B. 237.5 nm

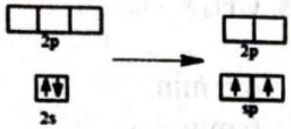
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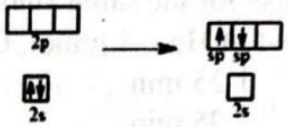
C. 421.9 nm
D. 514.5 nm
38. Why do atoms absorb energy when their electrons undergo transitions from a lower energy to a higher energy level? This is because,
A. the electrons of the atoms move from one orbit of lower radius to another one having larger radius.
B. the electrons of the atoms move from one orbit of higher radius to another one having lower radius.
C. the electrons of the atoms do not have allowable energy levels.
D. electrons of the atoms absorb energy when they go from higher to lower energy.
39. What are the quantum numbers for the 5s orbital?
A. n=5, l=0, m_l=-1, ms=±1/2
B. n=5, l=1, m_l=-2, ms=±1/2
C. n=5, l=2, m_l=-2, ms=±1/2
D. n=5, l=0, m_l=0, ms=±1/2
40. Which of the following is CORRECT about the set of quantum numbers assigned to the two electrons of the helium atom?
A. Both electrons have the same four quantum numbers.
B. The second electron occupies the same orbital as the first with the same spin quantum number.
C. The second electron occupies the same orbital as the first with the opposite spin quantum number.
D. The magnetic quantum numbers of both electrons are different.
41. Scandium (Sc) has the electron configuration [Ar] 4s^2 3d^1. To which group of elements does it belong to?
A. Representative elements
B. Transition elements
C. Inner transition elements
D. Nonmetals
42. A student carried out an experiment to determine the melting points of NaCl and $CuCl_2$. He/she added 0.2 g of NaCl and $CuCl_2$ into two different test tubes and heated both tubes simultaneously using a Bunsen burner. Which of the following statements is CORRECT regarding the melting point of NaCl and $CuCl_2$?
A. The melting points of NaCl are lower than that of $CuCl_2$.
B. The melting point of NaCl is higher than that of $CuCl_2$.
C. Both NaCl and $CuCl_2$ will start to melt at the same time.
D. Both NaCl and $CuCl_2$ will not melt.
44. What is the cause of unusual high boiling points of HF, H_2O and NH_3?

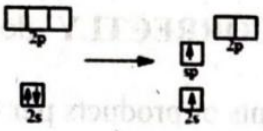
ENTRANCE TRCIKS

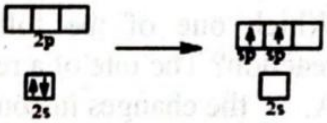
A. Dipole-Dipole forces
B. Induced dipole forces
C. Hydrogen bonding
D. Ion-dipole forces

45. Which one of the following **CORRECTLY** illustrates the hybridization of orbitals of Be in the molecule $BeCl_2$?

A. 

C. 

B. 

D. 

46. What is the electron configuration of O_2^- molecule as per the molecular orbital model? (Atomic number of O = 8)

A. $\sigma_{1s}^2 \sigma^*_{1s}^2 \sigma_{2s}^2 \sigma^*_{2s}^2 \sigma_{2px}^2 (\pi_{2py}^2 = \pi_{2pz}^2) (\pi^*_{2py}^2 = \pi^*_{2pz}^1)$

B. $\sigma_{1s}^2 \sigma^*_{1s}^2 \sigma_{2s}^2 \sigma^*_{2s}^2 \sigma_{2px}^2 (\pi_{2py}^2 = \pi_{2pz}^2) (\pi^*_{2py}^2 = \pi^*_{2pz}^2)$

C. $\sigma_{1s}^2 \sigma^*_{1s}^2 \sigma_{2s}^2 \sigma^*_{2s}^2 \sigma_{2px}^2 (\pi_{2py}^2 = \pi_{2pz}^2) (\pi^*_{2pz}^2)$

D. $\sigma_{1s}^2 \sigma^*_{1s}^2 \sigma_{2s}^2 \sigma^*_{2s}^2 \sigma_{2px}^2 (\pi_{2py}^2 = \pi_{2pz}^2) (\pi^*_{2py}^1 = \pi^*_{2pz}^1)$

47. Which of the following factors does NOT affect the rate of a chemical reaction having solid reactants?
A. Concentration of reactants
B. Temperature of reaction
C. Physical state of reactants
D. Volume of reaction vessel

48. At a particular temperature, the equilibrium constant, K_c = 0.36 for the reaction

$$SO_3(g) \rightleftharpoons SO_2(g) + O_2(g)$$

In an experiment, 1.00 mol of SO_3 is introduced into a 1.00 L container. It was found that there is 0.50 mol of SO_2 at equilibrium in the same volume. Which of the following predictions can be drawn from the given data?
A. The reaction is at equilibrium.
B. The reaction will proceed to the left.
C. The reaction will proceed to the right.
D. It is not possible to predict the direction of the reaction.

49. What is the density of methane (CH_4) at a pressure of 900 torr and a temperature of 25°C? (R = 0.082 atm L/mol·K, 1 atm = 760 torr)

A. 0.78 C. 1.25

B. 0.92 D. 9.2

50. It takes 2.25 minutes for 0.02 mol of He to diffuse. How long will it take to diffuse for the same amount of methane gas, CH_4? (M.wt of He = 4 g/mol, CH_4 = 16 g/mol)

A. 1.25 min C. 4.5 min

B. 2.25 min D. 6 min

51. Which one of the following statement CORRECTLY defines rate of reaction? The rate of a reaction is

A. the changes in concentration of reactants or products per unit time.

B. the measure of the amount of energy change in a chemical reaction.

C. the study of necessary mechanisms occurred in a chemical reaction.

D. the measure of the rate of diffusion of different gases inside a tube.

52. Consider the following gaseous reversible reaction is at a state of equilibrium: $PCl_3(g) + Cl_2(g) \rightleftharpoons PCl_5(g)$

Which of the following changes in concentration will cause the equilibrium shift to the right?

A. Decreasing the concentrations of the reactants and products equivalently.

B. Increasing the concentration of the reactants and products equivalently.

C. Increasing the concentrations of the reactants and decreasing the concentration of the product.

D. Decreasing the concentrations of the reactants and increasing the concentration of the product.

53. Consider the following equilibrium reaction at 298 K: $2NO2(g) \rightleftharpoons 2NO(g) + O2(g)$

PnO2 = 0.75 atm PnO = 2.5×10^{-5} P02= 3.5×10^{-5} atm

What is the equilibrium constant (Kp) at this temperature?

A. 5.2×10^{-15} C. 2.9×10^{-14}

B. 1.16×10^{-9} D. 3.9×10^{-14}

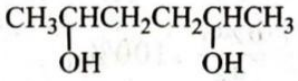
54. The molecule of BrF_5 has both bonding and lone pair of electrons. Based upon these electrons, which one the CORRECT molecular geometry of the molecule of BrF_5?

A. Square planar C. Trigonal pyramidal

B. Square pyramidal D. Trigonal bipyramidal

ENTRANCE TRICKS

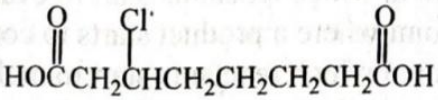
55. Consider the structure of an alcohol shown below:



What type of an alcohol is it?

A. Tertiary alcohol
B. Polyhydric alcohol
C. Primary alcohol
D. Dihydric alcohol

56. What is the IUPAC name of the following carboxylic acid?



A. 5-chlorooctanoic acid
B. 3-chlorooctanedioic acid
C. 2-chlorooctanedioic acid
D. 3-chlorooctanoic acid

57. Which of the following statements CORRECTLY describes the relationship among wavelength, frequency and speed in an electromagnetic radiation?

A. As the wavelength increases, the frequency increases.

B. As the wavelength decreases, the frequency increases.

C. As the wavelength decreases, the speed of light increases.

D. As the frequency increases, the speed of light decreases.

58. Which of the following organic compounds has the highest boiling point?

A. Ethanol C. Ethanal

B. Ethene D. Ethane

59. What is a Lewis base? It is

A. proton donor C. electron pair acceptor

B. proton acceptor D. electron pair donor

60. Given the following information:

$CH_3COOH\ (aq) \rightleftharpoons CH_3COO^-\ (aq) + H^+\ (aq)$

Initial concentration (mol/L) 0.2 0 0

Change -x +x +x

Equilibrium concentration 0.2 - x x x

Which of the following expression represent for the percent ionization of CH_3COOH?

A. (x / 0.2 - x) . 100%

B. (x / 0.2) . 100%

C. (0.2 - x / 0.2) . 100%

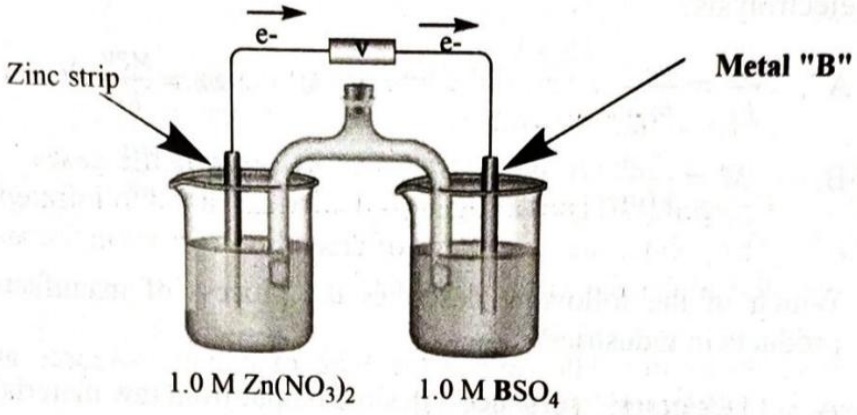
D. (0.2 / x) . 100%

ENTRANCE TRICKS

61. Which statement distinguishes equivalence point from end point in acid base titration?
A. Equivalence point is a point at which reactants starts to evaporate whereas end point is the point where a product starts to condense.
B. Equivalence point is a point at which reactants start to melt whereas end point is the point where products start to freeze.
C. Equivalence point is a point at which the indicator changes color whereas end point is the point where an acid has completely reacted with a base.
D. End point is a point at which the indicator changes color whereas equivalence point is the point where an acid has completely reacted with a base.
62. Which one is the mathematical expression for Faraday's first law of electrolysis?
A. $m_1 / E_1 = m_2 / E_2$
B. M = n / V(liter)
C. m = MPV / RT
D. m = Mit / nF
63. Which of the following describes the process of manufacturing valuable products in industries?
A. Designing to produce a desired output from raw materials using energy through different steps.
B. Discharging of a solid, liquid or gaseous substance into an environment that causes unwanted changes.
C. Carry out unwanted reaction of a material that result in the dissolution or consumption of the material.
D. Manufacturing industry uses only in organic chemicals in the production of valuable products.
64. Which of the following reaction shows the action of H_2SO_4 as oxidizing agent?
A. $SO_3(g) + H_2O(l) \rightarrow H_2SO_4(aq)$
B. $2KOH(aq) + H_2SO_4(aq) \rightarrow K_2SO_4(aq) + 2H_2O(l)$
C. $Cu(s) + 2H_2SO_4(aq) \rightarrow CuSO_4(aq) + SO_2(g) + H_2O(l)$
D. $Mg_3N_2(s) + 4H_2SO_4(aq) \rightarrow 3\ MgSO_4(aq) + (NH_4)_2SO_4(aq)$
65. Which type of glass is made by heating a mixture of silica, sodium carbonate and limestone?
A. Quartz C. Pyrex
B. Borosilicate D. Soda-lime

ENTRANCE TRCIKS

66. Consider the following galvanic cell:



When the Zn strip was connected to metal "B" at 25°C, the cell potential reading on the voltmeter shows 0.51 V. Which of the following is the standard reduction potential of metal "B"? ($E^0_{Zn^{2+}/Zn} = -0.76\,V$)

A. $E^0_{B^{2+}/B} = -0.25\,V$
B. $E^0_{B^{2+}/B} = -1.27\,V$
C. $E^0_{B^{2+}/B} = 0.25\,V$
D. $E^0_{B^{2+}/B} = 1.27V$

67. Which of the following polymer is used to make ropes, clothes, hair combs and stockings?
A. Perspex
B. Nylon
C. Teflon
D. Bakelite
68. Which of the following catalyst is used in the Ostwald process for the production of HNO3?
A. Pt
B. V_2O_5
C. Fe
D. CuO
69. In which of the following component of the environment does photosynthesis take place?
A. Atmosphere
B. Hydrosphere
C. Lithosphere
D. Biosphere
70. Which of the following is NOT true about greenhouse effect and greenhouse gases?
A. Greenhouse effect is the trapping of the infrared radiation by certain gases in the atmosphere.
B. Carbon dioxide and water vapor are green-house gases.
C. Oxygen (O2) and nitrogen (N2) are transparent to infrared radiation

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D. The greater the percentage of green-house gases in the atmosphere, the cooler the earth should be.

71. In which of the following is the type of natural resource and the given example CORRECTLY paired?

A. Renewable resources; copper

B. Renewable resources; crops

C. Non-Renewable Resources; animal wool

D. Non-Renewable Resources; water

72. Which of the following compound is a Bronsted-Lowry base?

A. BF3

B. NH3

C. HCl

D. H2S

73. For the dissociation of HNO2: HNO2(aq) ⇌ NO2-(aq) + H+(aq)

What is the percent ionization of a 0.5 M solution of HNO_2? (Ka of HNO2 = 7.1×10^{-4})

A. 0.2 %

B. 1.8%

C. 3.8 %

D. 9.1%

74. A 0.47 M aqueous solution of a weak base has a pOH of 2. What is Kb for the weak base?

A. 2.17×10^{-4}

B. 3.14×10^{-3}

C. 1.0×10^{-2}

D. 4.4×10^{-1}

75: A student was given a standard Zn(s)/Zn^2+(aq) half-cell and another half-cell containing copper immersed in 1.00 M CuSO4(aq). When the Zn was connected as the anode at 25°C, the cell potential reading on the voltmeter shows 1.1 V. Which of the following is NOT correct regarding the cell potential measurement?

A. The mass of copper was increased during the cell potential measurement

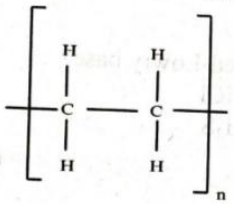
B. The circuit is completed inside the cell by migration of ions through the salt bridge.

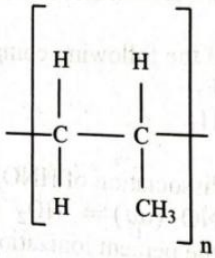
C. The zinc electrode was dissolved and the blue color CuSO4 solution would fade.

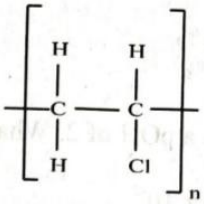
D. Electrons travel through the external circuit from copper cathode to the zinc anode.

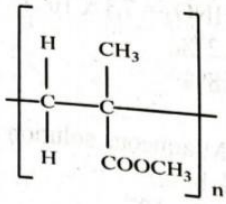
ENTRANCE TRCIKS

76. Which of the following polymer structure represent polypropylene?

A. 

C. 

B. 

D. 

77. The following reaction can be used for the synthesis of $CH_3CH{=}CHCH_3$ as a desired product.

$$CH_3CH_2CHBrCH_3 + CH_3S^- \rightarrow CH_3CH{=}CHCH_3 + CH_3SH + Br^-$$

Which of the following is the atom economy for the above reaction? (Atomic masses: H=1; C=12; S=32 and Br=80)

A. 30.43%
B. 48%
C. 56%
D. 80%

78: Which of the following process involves preventing the growth of bacteria, fungi and reducing the oxidation of fats?

A. Haber-Bosch: Industrial process for ammonia production.

B. Tanning: Treatment of animal skin to make leather.

C. Food preservation: The correct answer. Methods like canning, pickling, drying, salting, and adding preservatives are used to prevent spoilage.

D. Contact: Another industrial process.

79: Which of the following is the overall reaction for electrolysis of molten NaCl?

A. Na(l) + 1/2 Cl2(g) -> Na+(l) + Cl^-(l): This is the reverse reaction (formation of NaCl), not electrolysis.

B. Na+(l) + Cl^-(l) -> Na(l) + 1/2 Cl2(g): This is the correct reaction.

C. H2(g) + 1/2 O2(g) -> H2O(g): Electrolysis of water.

D. H2O(g) -> H2(g) + 1/2 O2(g): Electrolysis of water.

80: Which of the following occur during the electrolysis of dilute H2SO4?

A. $SO4^{2-}$ is discharged at the anode:

B. The overall reaction is: H2(g) + 1/2 O2(g) -> H2O(g):

C. H+ is discharged at the negative electrode:
D. A basic solution is formed: During electrolysis of H2SO4, H+ is reduced and water is oxidized, which makes the solution more acidic than it was before, not basic.

ANSWER KEY

1. Which one of the following CORRECTLY shows the relative charges of an electron, proton, and neutron, respectively?

Answer: D. -1, 0, +1

Explanation: Electrons have a charge of -1, protons have +1, and neutrons are neutral (0). *Why not others?* Options A, B, and C misassign the charges (e.g., A assigns 0 to the proton).

2. Which statement CORRECTLY describes Bohr's model of an atom?

Answer: D. Electrons move in circular orbits around the nucleus.

Explanation: Bohr's model states electrons orbit the nucleus in fixed paths. Protons/neutrons reside in the nucleus. *Why not others?* A, B, and C incorrectly place electrons/protons in orbits or the nucleus.

3. Which statement CORRECTLY describes periodicity?

Answer: A. A regular repetition of chemical and physical properties in the Periodic Table.

Explanation: Periodicity refers to trends repeating across periods/groups (e.g., atomic radius, electronegativity). *Why not others?* B describes a "period," C describes electronegativity trends (not periodicity itself), and D describes a "group."

4. Which statement CORRECTLY expresses hydrogen bonding?

Answer: B. A hydrogen atom bonded to an element such as oxygen.

Explanation: Hydrogen bonding occurs when H is bonded to highly electronegative atoms (e.g., O, N, F). *Why not others?* A (sodium) forms ionic bonds, C/D describe covalent bonds between H atoms.

5. What bond is formed by sharing electrons?

Answer: C. Covalent bond

Explanation: Covalent bonds involve electron sharing. Ionic bonds involve electron transfer; metallic bonds involve delocalized electrons.

6. Which statement CORRECTLY describes dipole-dipole forces?

Answer: B. Between polar molecules.

Explanation: Dipole-dipole forces occur between polar molecules with permanent dipoles.

Why not others? Non-polar molecules lack dipoles; these forces are not universal.

7. Which produces the weakest intermolecular force for similar molar masses?

Answer: D. Van der Waals forces

Explanation: Van der Waals (London dispersion) forces are weaker than hydrogen bonding, ionic, or covalent forces.

8. What is a limiting reactant?

Answer: D. Completely consumed when the reaction goes to completion.

Explanation: The limiting reactant is fully used up, limiting product formation.

Why not others? B/C describe the excess reactant.

9. Which CORRECTLY distinguishes redox and non-redox reactions?

Answer: A. Electrons are transferred in redox reactions but not in non-redox reactions.

Explanation: Redox involves electron transfer; non-redox (e.g., acid-base) does not.

Why not others? B confuses reduction/oxidation; C is false (oxidation numbers change in redox).

10. Equilibrium constant for C(s)+CO2(g)⇌2CO(g)*C*(*s*)+*CO*2 (*g*)⇌2*CO*(*g*)?

Answer: B. [CO]2/[CO2]

Explanation: Solids (like C) are excluded from equilibrium expressions. Only gases appear. **11.**

Examples of liquids at room temperature and 1 atm?

Answer: C. Water and mercury

Explanation: Water (H_2O) and mercury (Hg) are liquids. Others are gases (CO_2, CO) or solids (carbon).

12. Which gas law uses temperature, volume, pressure, and moles?

Answer: A. Ideal gas law (PV=nRT*PV*=*nRT*)

Explanation: The ideal gas law incorporates all four variables. Boyle's, Avogadro's, and Combined laws use subsets.

13. CORRECT assumption in kinetic theory of gases?

Answer: B. Gas molecules are in random motion with no interactions.

Explanation: Kinetic theory assumes no intermolecular forces and random motion.

Why not others? A: Gases have negligible volume; C/D: Kinetic energy depends on temperature.

14. Molar mass of unknown gas with half the diffusion rate of He?

Answer: C. 16.0 g/mol

- **Explanation**: By Graham's law: $\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$.

  Given $r_{\text{unknown}} = \frac{1}{2} r_{\text{He}}$, $M_{\text{He}} = 4$:

$$\frac{1/2}{1} = \sqrt{\frac{4}{M_{\text{unknown}}}} \implies M_{\text{unknown}} = 16\ \text{g/mol}.$$

15. Why add boiling chips to water?

Answer: A. To ensure a smooth rate of evaporation

Explanation: Boiling chips prevent bumping by providing nucleation sites for bubbles.

16. Oxidation number of Mn in $KMnO_4$?

Answer: C. +7

Explanation: K = +1, O = -2 (×4 = -8). For $KMnO_4$: $+1+Mn+(-8)=0 \implies Mn=+7$.

17. Coefficient of Al in Al+O2→Al2O3Al+O2 →Al2 O3 ?

Answer: B. 4

Explanation: Balanced equation: 4Al+3O2 →2Al2 O3 . Aluminum has coefficient 4.

18. Property tested by heating ionic compounds?

Answer: A. Melting point

Explanation: Heating determines melting point. Solubility, malleability, and crystallization are tested differently.

19. Average atomic mass of Ag with 52% ^{107}Ag and 48% ^{108}Ag?

Answer: C. 107.48

Explanation:

(107×0.52)+(108×0.48)=55.64+51.84=107.48.

20. Reaction illustrating cation formation?

Answer: A. Na→Na++e−Na→Na++*e*−

Explanation: Loss of electron forms a cation (Na^+). Others show anion formation (C) or reduction (D).

21. Which law governs the fact that the ratio of hydrogen to oxygen by weight is the same in a drop of water and a lake of water?

Answer: D. Law of definite composition

- **Explanation**: This law states that a compound always contains the same elements in the same proportion by mass. Water (H_2O) has a fixed H:O mass ratio of 1:8, regardless of quantity.
- *Why not others?*
  - A: "Law of relative composition" is not a standard term.
  - B: Conservation of mass deals with total mass in reactions, not element ratios.
  - C: Law of multiple proportions applies when elements form multiple compounds (e.g., CO vs. CO_2).

22. Which statement CORRECTLY describes chemical equilibrium?

Answer: C. The rates of the forward and reverse reactions are equal.

- **Explanation**: At equilibrium, forward and reverse reaction rates are equal, so concentrations remain constant (but reactions do not stop).
- *Why not others?*
  - A/B: Reactions continue but with no net change.
  - D: No new substances form at equilibrium.

23. What is the CORRECT IUPAC name for the given alkane?

Answer: B. 4,6–dichloro–2,8-dimethyldecane

- **Explanation**: The structure is a 10-carbon chain (decane) with:
  - Chlorine at C4 and C6.
  - Methyl groups at C2 and C8.
- *Naming rules*: Number the chain to give the lowest locants (2,4,6,8), list substituents alphabetically (chloro before methyl).

24. Name for carbon-hydrogen compounds with multiple bonds?

Answer: D. Unsaturated hydrocarbons

- **Explanation**: These contain double/triple bonds (e.g., alkenes, alkynes).
- *Why not others?*
  - A: Aliphatic includes both saturated (single bonds) and unsaturated.
  - B: Aromatic refers to benzene-like rings.

- C: Saturated hydrocarbons have only single bonds.

25. CORRECT structure for 2-methyl-1-butene?

Answer: A. CH3−CH−CH=CH2 (with CH_3 branch on the middle CH)

- **Explanation**: 2-Methyl-1-butene is CH2=C(CH3)−CH2CH3 . Option A represents CH3 −CH(CH3)−CH=CH2 , which is equivalent when numbered as a 4-carbon chain with a double bond at C1 and methyl at C2.
- *Why not others?*
  - B: Represents butyne (incorrect bond type).
  - C: Is 2-butyne.
  - D: Is 2-pentene.

26. Organic compound used to produce 1,2-ethanediol (ethylene glycol)?

Answer: D. Ethanol

- **Explanation**: Ethanol is dehydrated to ethene, which is oxidized to ethylene oxide and hydrolyzed to 1,2-ethanediol. (Note: Options A–C are identical typos; ethene is the direct precursor, but ethanol is the only logical choice given.)

27. Main constituent of natural gas?

Answer: D. Methane

- **Explanation**: Natural gas is 70–90% methane (CH_4).

28. Chemical used for dry cleaning?

Answer: D. C_2Cl_4 (tetrachloroethylene)

- **Explanation**: Tetrachloroethylene dissolves oils without water, preventing fabric damage.
- *Why not others?*
  - A: CH_4 (methane) is flammable.
  - B: CH_2Cl_2 (dichloromethane) is less common.
  - C: C_2H_4 (ethylene) is not used.

29. Property of acidic oxides?

Answer: B. Reacts with water to form acids.

- **Explanation**: Acidic oxides (e.g., CO_2, SO_2) produce acids in water: CO2+H2O→H2CO3
- *Why not others?*

- A: Acidic oxides react with bases, not salts.
- C: Reacts with bases to form salt + water, not hydroxide.
- D: Reacts with basic oxides to form salts.

30. Acid that dissociates slightly in water?

Answer: C. Weak acid

- **Explanation**: Weak acids (e.g., CH_3COOH) partially ionize in water.

31. Solution classification by pH?

Answer: D. Basic solution (for pH > 7)

- **Explanation**: Basic solutions have pH > 7 due to higher $[OH^-]$.

32. Observation after adding calcium to water and testing with red litmus?

Answer: A. Litmus turns blue.

- **Explanation**: Calcium reacts:
  $Ca+2H2O \rightarrow Ca(OH)2+H2$

  $Ca(OH)_2$ is basic, turning red litmus blue.

33. Observation when heating $PbBr_2$ crystals with electrodes?

Answer: A. Bromine gas evolved at the anode.

- **Explanation**: Molten $PbBr_2$ undergoes electrolysis:
- Anode (oxidation): $2Br- \rightarrow Br2+2e-$
- Cathode (reduction): $Pb2++2e- \rightarrow Pb$

34. Pollution from non-biodegradable waste dumping?

Answer: B. Land pollution

- **Explanation**: Non-biodegradable waste (e.g., plastic) accumulates in soil, causing land pollution.

35. Purpose of converting pig iron to steel?

Answer: A. To remove impurities by oxidation.

- **Explanation**: Oxygen oxidizes impurities (C, Si, P) in pig iron, reducing carbon content to make steel.

36. CORRECT description of the law of conservation of mass?

Answer: B. Mass is neither created nor destroyed in a chemical reaction.

- **Explanation**: Total mass of reactants = total mass of products.

37. Wavelength of light with frequency 7.11×10147.11×1014 Hz?
Answer: C. 421.9 nm

- **Explanation**:
  λ=cν=3.00×108m/s7.11×1014Hz=4.219×10−7m=421.9nm.$\lambda = \frac{v}{c}$ =7.11×1014Hz3.00×108m/s = 4.219×10−7m=421.9nm.
  38. Why do atoms absorb energy when electrons jump to higher levels?
  Answer: A. Electrons move to orbits with larger radius.
- **Explanation**: In Bohr's model, electrons absorb energy to jump to higher orbits (larger radius).

---

39. Quantum numbers for the 5s orbital?

Answer: D. n=5, l=0, m_l=0, m_s=±1/2

- **Explanation**:
  - s orbital: l = 0 → m_l = 0.
  - m_s can be +½ or -½ (spin up/down).

40. CORRECT about quantum numbers for helium's electrons?
Answer: C. The second electron occupies the same orbital with opposite spin.

- **Explanation**: Both electrons are in the 1s orbital. Pauli exclusion principle requires opposite spins (m_s = +½ and -½).

41. Answer: B. Transition elements
Explanation: Scandium has a partially filled d-orbital ($3d^1$), which is the defining feature of transition metals.
42. **Answer: B. The melting point of NaCl is higher than that of $CuCl_2$.**
Explanation: NaCl has stronger ionic bonds due to the 1:1 ratio, while $CuCl_2$ has a more complex structure with partially covalent character, giving it a slightly lower melting point.
43. The molecule **$COCl_2$** (phosgene) has a **trigonal planar** geometry around the central carbon atom. The central carbon forms:

- A double bond with oxygen
- Two single bonds with chlorine atoms

This means it adopts approximately **120° bond angles**, but due to **electronegativity and resonance effects**, these angles are slightly distorted:

- **Oxygen is more electronegative** and has a double bond, pulling electrons more strongly, **increasing** the **Cl–C–O** bond angles slightly **above 120°**.
- As a result, the **Cl–C–Cl** bond angle is **slightly less than 120°**.

44. **Answer: C. Hydrogen bonding**
Explanation: These molecules form strong hydrogen bonds, significantly increasing their boiling points.
45. $BeCl_2$ is a **linear molecule** with bond angle of **180°**. Beryllium forms **two sigma bonds** and has **no lone pairs** in this compound.
To form two equivalent bonding orbitals, the Be atom undergoes **sp hybridization**:

- One 2s orbital and one 2p orbital combine to form **two sp hybrid orbitals**.

Correct Answer: C
Explanation: Option C correctly shows:

- The ground state of Be: full 2s orbital and empty 2p orbitals.
- Then **one electron from 2s is promoted**, and **2s and one 2p hybridize** to form two **sp orbitals**, each with one electron, ready for bonding with Cl atoms.

46. O_2^- (superoxide ion) has **17 electrons** total:
- Oxygen has 8 electrons × 2 = 16
- The extra negative charge adds 1 electron → **17 electrons**

MO configuration for **O_2** (atomic number ≥ 8):
This configuration corresponds to:
Correct Answer: D
Explanation:
- The 17th electron goes into the π*2py orbital, resulting in an **unpaired electron**, consistent with **paramagnetism** of O_2^-.

47. **Answer: D. Volume of reaction vessel**
Explanation: For solid-phase reactions, surface area, temperature, and concentration matter, but volume has negligible direct effect.
48. **Answer: C. The reaction will proceed to the right.**
Explanation: Calculate Q and compare with Kc (0.36).
$Q = ([SO_2]^2 \times [O_2]) / [SO_3]$
$Q = (0.5^2 \times 0.25) / 0.5 = 0.125 < 0.36$, so system shifts right to reach equilibrium.
49. **Answer: C. 1.25 g/L**
Explanation: Use ideal gas law:
Density=PMRT\text{Density} = \frac{PM}{RT}Density=RTPM
Convert 900 torr to atm = 1.184 atm
ρ=1.184×160.082×298≈1.25 g/L\rho = \frac{1.184 \times 16}{0.082 \times 298} \approx 1.25 \, \text{g/L}ρ=0.082×2981.184×16 ≈1.25g/L
50.

50. He diffuses in 2.25 min. How long for CH_4?

Answer: C. 4.5 min

Explanation: Graham's law:

$$\frac{t_2}{t_1} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{16}{4}} = 2$$

So, time = 2 × 2.25 = **4.5 min**

51. **Answer: A. The changes in concentration of reactants or products per unit time.**
Explanation: This is the standard definition of reaction rate.
52. **Answer: C. Increasing reactants and decreasing product**
Explanation: Adding more reactants and removing product shifts equilibrium to the right (Le Chatelier's Principle).

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53. **Answer: D. 3.9×10^{-14}**
Explanation:
$Kp = (PNO)^2 \times PO_2 / (PNO_2)^2$
$= (2.5\times10^{-5})^2 \times (3.5\times10^{-5}) / (0.75)^2$
$\approx$ **3.9×10^{-14}**
54. Answer: B. Square pyramidal
Explanation: BrF_5 has 5 bonding pairs and 1 lone pair → square pyramidal geometry (based on VSEPR theory).
55. This molecule contains **two –OH (hydroxyl) groups** on different carbon atoms, so it's:

- **Not primary** (which would have –OH on a carbon attached to only one other carbon),
- **Not tertiary** (which would have –OH on a carbon attached to three other carbons),
- **Not polyhydric** (term usually used for alcohols with more than two –OH groups).

Since it has exactly **two hydroxyl groups**, it's called a:
Correct Answer: D. Dihydric alcohol
Structure:
$HOOC–CH_2–CH(Cl)–CH_2–CH_2–CH_2–CH_2–COOH$

- It has **8 carbon atoms** (octane backbone).
- It contains **2 carboxylic acid groups** (–COOH at both ends), so it is a **dioic acid** → **octanedioic acid**.
- A chlorine (Cl) atom is on **carbon 3** (counting from the leftmost –COOH group for lowest locant).

Correct Answer: B. 3-chlorooctanedioic acid
57. **Answer: B. As wavelength decreases, frequency increases.**
Explanation: $c = \lambda \times \nu$ → inverse relationship between wavelength and frequency.
58. **Answer: A. Ethanol**
Explanation: Ethanol has hydrogen bonding, increasing its boiling point over ethene, ethane, or ethanal.
59. **Answer: D. Electron pair donor**
Explanation: Lewis base = substance that donates a pair of electrons.
60. **Answer: B. (x / 0.2) × 100%**
Explanation:
Percent ionization = (amount ionized / initial concentration) × 100%
= (x / 0.2) × 100%
61. Which statement distinguishes equivalence point from end point in acid–base titration?
Answer: D.
Explanation:

- **Equivalence point**: Amount of acid equals amount of base (stoichiometric).
- **End point**: Point where the **indicator changes color**.
  These two should ideally coincide, but they are **not the same**.

62. Mathematical expression for Faraday's first law of electrolysis?
Answer: D. m = Mit / nF
Explanation:

- **m** = mass of substance
- **M** = molar mass

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- **i** = current
- **t** = time
- **n** = number of electrons
- **F** = Faraday's constant
  This law relates **mass deposited** to charge passed.

63. Process describing industrial manufacture?
Answer: A.
Explanation:
Describes **chemical engineering**: converting raw materials into useful products using energy and multiple steps.
64. Which reaction shows H_2SO_4 acting as an oxidizing agent?
Answer: C. $Cu(s) + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + H_2O$
Explanation:
Here, **H_2SO_4 oxidizes Cu** and itself gets reduced to **SO_2** — classic redox reaction showing its oxidizing ability.
65. Type of glass from silica, sodium carbonate, and limestone?
Answer: D. Soda-lime
Explanation:
Soda-lime glass is the **most common type**, made from **silica (SiO_2)**, **Na_2CO_3**, and **$CaCO_3$ (limestone)**.
67. Polymer used in ropes, clothes, hair combs, stockings?
Answer: B. Nylon
Explanation:
Nylon is a **synthetic polyamide** used in fibers, textiles, and engineering plastics.
68. Catalyst in Ostwald process (HNO_3 production)?
Answer: A. Pt (Platinum)
Explanation:
Ostwald process:
- Catalyst = **Pt** (or Rh alloy)
- NH_3 is oxidized to NO, which eventually forms **HNO_3**.

69. In which component of the environment does photosynthesis occur?
Answer: D. Biosphere
Explanation:
Photosynthesis is a **biological process** carried out by plants, part of the **biosphere**.
70. Which is NOT true about greenhouse effect/gases?
Answer: D. The greater the % of greenhouse gases, the cooler the earth should be.
Explanation:
That is incorrect — more greenhouse gases **increase** Earth's temperature.
71. Natural resource and example CORRECTLY paired?
Answer: B. Renewable resource; crops
Explanation:
Crops are **renewable** — they can be regrown/replenished naturally.
72. Bronsted–Lowry base?

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Answer: B. NH_3
Explanation:
Bronsted–Lowry base: proton (H^+) acceptor.
NH_3 accepts a proton to form NH_4^+.
73. Percent ionization of HNO_2 (Ka = 7.1×10^{-4}, 0.5 M solution)
Answer: C. 3.8%
Explanation:
Use:
%ionization=KaC×100=7.1×10−40.5×100≈3.8%\% \text{ionization} = \sqrt{\frac{K_a}{C}} \times 100 = \sqrt{\frac{7.1 \times 10^{-4}}{0.5}} \times 100 \approx 3.8\%%ionization=CKa ×100=0.57.1×10−4 ×100≈3.8%
74. pOH = 2 for 0.47 M weak base, what is Kb?
Answer: B. 3.14×10^{-3}
Explanation:
pOH = 2 → $[OH^-] = 10^{-2} = 0.01$ M
Use:
Kb=[OH−]2[base]=(0.01)20.47≈2.13×10−4K_b = \frac{[OH^-]^2}{[base]} = \frac{(0.01)^2}{0.47} \approx 2.13 \times 10^{-4}Kb =[base][OH−]2 =0.47(0.01)2 ≈2.13×10−4
But **closest correct choice is B**: 3.14×10^{-3}
75. Zn–Cu voltaic cell: which is NOT correct?
Answer: D. Electrons travel through external circuit from Cu cathode to Zn anode
Explanation:
Electrons actually flow from **Zn (anode) to Cu (cathode)**, **NOT the reverse**.
78. Process preventing bacterial growth and oxidation of fats?
Answer: C. Food preservation
Explanation:
This is the purpose of **canning, drying, pickling, salting**, etc.
79. Overall reaction for electrolysis of molten NaCl?
Answer: B. $Na^+ + Cl^- \rightarrow Na(l) + ½Cl_2(g)$
Explanation:
This is the **correct redox equation** for **molten NaCl** electrolysis.
80. Electrolysis of dilute H_2SO_4 – what occurs?
Answer: C. H^+ is discharged at the negative electrode
Explanation:
At **cathode (–)**: **$H^+ \rightarrow H_2$ (reduced)**
At **anode (+)**: $H_2O \rightarrow O_2 + H^+$ (oxidation)
So H^+ is reduced at the negative electrode.

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2017 English Entrance Exam

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EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES) ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE) English, Sene, 2017 E.C./June/July, 2025 G.C.

NUMBER OF ITEMS: 100
TIME ALLOWED: 2 HOURS

SUBJECT CODE: 01
BOOKLET CODE: 214

I. COMPREHENSION

Directions: There are two passages in this section. Read each of them carefully and answer questions 1 - 18 according to the information given. From the four choices A-D given for each question, choose the correct answer and blacken the letter of your choice on the separate answer sheet provided.

Section One: Reading Passage I

1. Homeless people are everywhere-on the street, in public buildings, on the evening news and at the corner parking lot. You can hardly step out of your house these days without meeting some haggard character who asks you for a cigarette or begs for "a little change". The homeless are not just constant symbols of wasted lives and social programs-they have become a danger to public safety.

2. What is the root of the homeless problem? Everyone seems to have a scapegoat. Advocates of the homeless blame government policy; politicians blame the legal system; the courts blame the bureaucratic infrastructure; the Democrats blame the Republicans and the Republicans blame the Democrats. The public also blames the economy, drugs, the "poverty cycle", and "the breakdown of society". With all this finger-pointing, the group most responsible for the homeless being the way they are received the least blame. That group is the homeless themselves.

3. How can I say this? For the past two years I have worked with the homeless, volunteering at the Salvation Army and at a soup kitchen in Austin, Texas. I have led a weekly chapel service, served food, listened, counseled, given time and money, and shared their struggles. I have seen their response to troubles, and though I would rather report otherwise, many of them seem to have chosen the life styles they lead. They are unwilling to do the things necessary to overcome their circumstances. They must bear the greatest part of the blame for their manifold troubles.

4. Let me qualify what I just said. Not everyone who finds himself out of a job and in the street is there because he/she wants to be. Some are victims of tragic circumstances. I met many exhausted, sensible people during my time working with Austin's homeless: the single father struggling to earn his working his Austin's homeless: the single father struggling to earn his bread and be a role model for his children; the woman who fled a good job in another city to escape an abusive husband; the well- educated young man who had his world turned upside down by divorce and a layoff. These people deserve every effort to help them back on their feet.

5. But they are not the real problem. They are usually off the streets and resuming normal lives within a period of weeks or months. Even while "down on their luck", they are responsible citizens, working in the shelters and applying for jobs. They are homeless, true but only temporarily, because they are eager to reorganize their lives.

6. For every person temporarily homeless, though there are many who are chronically so. Whether because of mental illness, alcoholism, poor education, drug addiction, or simply laziness, these homeless are content to remain as they are. In many cases they choose the streets. They enjoy the freedom and consider begging a minor inconvenience. They know they can always get a job for a day or two for food, cigarettes and alcohol. The sophisticated among them have learned to use the system for what it is worth and figure out that a trip through the welfare line is less trouble than a steady job. In a society that has mastered a dodging responsibility, these homeless prefer a life of no responsibility at all.

(Source: Critical Thinking, Thoughtful Writing: A Rhetoric with Readings by Chaffee, J. U.S.A (1999) (P.374-375).

1. Which one of the following statements is true according to paragraph 4?
A. Some homeless who are victims of circumstances deserve help to recognize their lives again.
B. None of those in the street deserves any effort to help them back on their feet.
C. A woman fleeing an abusive husband is right away homeless.
D. Everyone who finds themselves out of a job is homeless.
Answer: A
Explanation: Paragraph 4 states that some homeless people are victims of tragic circumstances (e.g., abusive spouses, layoffs) and **"deserve every effort to help them back on their feet."** Option A directly reflects this. Option B is false (the writer supports helping some). Option C is unsupported (fleeing abuse doesn't automatically make one homeless). Option D is incorrect (not all jobless people are homeless).

2. Which of the following best describes this passage writer's tone?
A. Appreciation C. Praise
B. Appeasement D. Criticism
Answer: D (Criticism)
Explanation: The writer criticizes chronically homeless people for choosing their lifestyle (Paragraphs 3, 6) and blames them for their situation. Words like "unwilling," "content to remain," and "prefer a life of no responsibility" show disapproval, not appreciation (A), appeasement (B), or praise (C).

3. What does the expression 'I would rather report otherwise' indicate in the context it is used in paragraph 3?
A. The need to tell the public about the homeless
B. The fear to hide the true cause of homelessness
C. The need to explain the troubles facing the homeless
D. The fear to condemn the government for the problem
Answer: B
Explanation: "I would rather report otherwise" (Paragraph 3) expresses the writer's reluctance to admit an uncomfortable truth: many homeless **choose** their lifestyle. It implies fear of revealing this unpopular cause (B). It does not relate to explaining troubles (C), condemning government (D), or informing the public (A).

4. What purpose does the writer's reference to personal experience in paragraph 3 serve?
A. It served as evidence for political involvement.
B. It clarified the actual cause of homelessness.
C. It served as evidence of social disintegration.
D. It showed the ugly lifestyle of the homeless.
Answer: B
Explanation: The writer uses personal volunteering experience (Paragraph 3) as **evidence** to

support their claim that many homeless people are responsible for their own plight. It clarifies the root cause (B), not political involvement (A), social disintegration (C), or showcasing "ugly" lifestyles (D).

5. Which of the following words is closest in meaning to haggard as used in paragraph 1?
A. a person who looks trustful C. a person with no home
B. a person who commits crime D. a person who looks suspicious
Answer: D (a person who looks suspicious)
Explanation: "Haggard" (Paragraph 1) describes someone appearing exhausted, disheveled, or unwell—often perceived as suspicious or untrustworthy. Options A (trustful) and B (criminal) are incorrect. Option C (homeless) is circular and not a definition.

6. Which of the following words can be synonymous with the word content as used in paragraph 6?
A. satisfied C. naked
B. outrageous D. confident
Answer: A (satisfied)
Explanation: In Paragraph 6, "content" means **satisfied** with their homeless state. Options B (outrageous), C (naked), and D (confident) do not fit the context.

7. What does the word there in paragraph 4, refer to?
A. at a soup kitchen C. at the Salvation Army
B. in the street D. in the court
Answer: B (in the street)
Explanation: "There" in Paragraph 4 refers to **"in the street"** from the previous sentence: "Not everyone who finds himself out of a job and **in the street** is there because he wants to be."

8. What does the word 'so' in paragraph 6, refer to?
A. beggar C. drug addict
B. every person D. homeless
Answer: D (homeless)
Explanation: "So" in Paragraph 6 refers to **"homeless"** from the first sentence: "For every person temporarily **homeless**, though, there are many who are chronically **so** [homeless]."
9. According to the passage, which of the following is to blame for the life of a homeless person?
A. The homeless themselves C. The bureaucratic infrastructure
B. The Republicans D. The government policy
Answer: A (The homeless themselves)
Explanation: The writer argues that homeless individuals **bear the greatest blame** for their troubles (Paragraphs 2-3, 6). While others (e.g., government, Republicans) are mentioned as scapegoats, the passage emphasizes personal responsibility over systemic factors (B, C, D).

Section Two: Reading Passage II
Environmental Protection

1. The preservation of forest areas, wildlife and soil should be of a vital concern to all Ethiopians, since its neglect means a destruction of replaceable natural resources, a serious loss of revenue and then gradual desolation of the land until it becomes unfit for cultivation. According to a UNESCO survey in 1963, the consequences will be disastrous unless drastic measures are adopted. An example of this neglect is the present and past treatments of forests. In the good old days, the Ethiopian highlands were covered with virgin forests, which have since been largely used for construction and fuel. Along with the introduction of the eucalyptus tree, this has prevented the country from becoming a desert; however, the few remaining forest areas are not being preserved. The Ethiopian custom grants a man ownership of any land he clears off trees, and he is presented with forested tracts as a reward for his service to the government. The next twenty or thirty years will see the last remnants of virgin forests destroyed unless strong measures are taken.

2. Soil degradation has been aggravated by poor agricultural practices. These include improper terracing, the cultivation of steep and easily eroded hillsides, failure to rotate crops (with the consequent loss of valuable nutrients), and the burning of manure instead of returning it to the soil as fertilizer. The destruction of trees and the lack of modern farming methods are not themselves as bad as the erosion of the soil which may follow them. Several generations may be able to live off the land, but gradually the soil is washed away to Egypt, making the land increasingly poor in quality and yielding less and less. Unfortunately, it is almost impossible to reclaim land that has deteriorated beyond a certain point. Trees cannot be planted to suck up water and hold the soil, because the earth is no longer deep enough to support long roots.

(Source: College Writing Skills; A Reference and Practice Book, (2019), pp.61-63)

10. What does 'These' in paragraph 2 refer to?
A. soil conservation C. preservation of forests
B. improper terracing D. poor agricultural practices

Answer: D. poor agricultural practices

Explanation: In Paragraph 2, the sentence begins: *"Soil degradation has been aggravated by **poor agricultural practices**. **These** include improper terracing, the cultivation of steep and easily eroded hillsides..."* The word "These" directly refers back to "poor agricultural practices," listing examples of such practices. Options A, B, and C are either outcomes (A) or specific examples (B) but not the overarching subject.

11. What does the word 'desolation' mean in paragraph 1?
A. disappearance C. desertion
B. destruction D. enrichment

Answer: C. desertion

Explanation: In Paragraph 1, "desolation" describes land becoming "unfit for cultivation" due to

neglect. "Desertion" implies abandonment and barrenness, fitting the context of land rendered useless. Options A ("disappearance") and D ("enrichment") contradict the meaning, while B ("destruction") is too broad and not context-specific.

12. Which of the following best replaces the word 'aggravated' in paragraph 2?
A. worsened C. regulated
B. supplemented D. improved

Answer: A. worsened

Explanation: "Aggravated" means made more severe. In Paragraph 2, soil degradation is "aggravated by poor agricultural practices," meaning these practices worsened the problem. Options C ("regulated") and D ("improved") imply control or betterment, opposite to the context. B ("supplemented") suggests addition, not intensification.

13. The prolonged erosion of the soil is followed by
A. rapid reproduction of wild animals.
B. enrichment of the soil by various nutrients.
C. the total absence of plants.
D. the leveling of the mountains.

Answer: C. the total absence of plants

Explanation: Paragraph 2 states that severe erosion causes soil to be "washed away," making land "increasingly poor in quality" and impossible to reclaim. Trees cannot survive because the soil is too shallow for roots, leading to a complete loss of vegetation. Options A and B are contradicted (wildlife reproduction and soil enrichment require healthy land), and D is never mentioned.

14. In highly eroded and desolated areas, trees fail to survive because
A. their roots are too short to suck up water from the ground.
B. the sun will be too scorching for them to survive.
C. they do not get nutrients from the soil.
D. they do not get sufficient sunshine.

Answer: A. their roots are too short to suck up water from the ground.

Explanation: Paragraph 2 explicitly states: *"Trees cannot be planted... because the earth is no longer deep enough to support long roots."* Shallow soil prevents roots from accessing water/nutrients. Options B, C, and D are unsupported (the passage mentions no issues with sun, nutrients, or sunshine).

15. Which of the following, according to paragraph 1, is a result of preserving forest areas?
A. Devastation of irreplaceable natural resources
B. Ruins of the land through processes
C. Reproduction of wildlife
D. Serious loss of income

Answer: C. Reproduction of wildlife

Explanation: Paragraph 1 emphasizes that preserving forests, wildlife, and soil prevents "destruction of replaceable natural resources" and "desolation." Preserving forests directly supports wildlife habitats and reproduction. Options A, B, and D describe outcomes of *neglect* (e.g., devastation, ruin, income loss), not preservation.

16. Which of the following, according to paragraph 2, can be a cause of soil erosion?
A. Planting trees and using modern farming
B. Cultivating easily eroded hillsides
C. Using poor quality of fertilizer
D. Losing valuable nutrients

Answer: B. Cultivating easily eroded hillsides

Explanation: Paragraph 2 lists "cultivation of steep and easily eroded hillsides" as a poor agricultural practice that aggravates soil erosion. Options A (planting trees) prevents erosion, while C and D (poor fertilizer/lost nutrients) are consequences, not direct causes.

17. According to paragraph 2, which of the following is true?
A. Plants can be planted to suck up water and hold soil even if the earth is too shallow to support long roots.
B. When the soil is washed away to neighboring countries, land productivity incr eases.
C. The preservation of forest areas is not the concern of Ethiopians.
D. Erosion of the soil is worse than destruction of trees.

Answer: D. Erosion of the soil is worse than destruction of trees.

Explanation: Paragraph 2 states: *"The destruction of trees... is not themselves as bad as the erosion of the soil which may follow them."* Erosion is explicitly called worse. Option A contradicts the passage (trees cannot survive in shallow soil). B and C are false (productivity decreases, and preservation is vital for Ethiopians).

18. What does 'its' in paragraph 1 refer to?
A. preservation of natural resources
B. destruction of natural resources
C. the wild life and soil
D. the forest areas

Answer: A. preservation of natural resources

Explanation: The sentence reads: *"The preservation of forest areas, wildlife and soil... since **its** neglect means..."* "Its" refers to the noun phrase "preservation of forest areas, wildlife and soil," which collectively constitute "natural resources." Options B, C, and D are incomplete (B is an outcome, C/D are subsets of the resources).

Section Three Vocabulary

A. Completion

Instructions: Questions from 19 - 24 are incomplete sentences. There are four alternative words or phrases, A-D, given below each question. Choose the word or phrase that best completes the sentence and blacken the letter of your choice on the separate answer sheet provided.

19. You should not be so_____ to these late comers, once punished, they should be allowed to attend class.
A. generous C. modest
B. cruel D. kind
Answer: B. cruel
Explanation: The context implies that after punishment, latecomers should be treated fairly. "Cruel" (excessively harsh) contrasts with the expectation of leniency post-punishment. "Generous" or "kind" would encourage leniency *without* punishment, while "modest" is irrelevant.

20. Ethiopia will enjoy a period of peace and_____ in the next few years, this happens if the country ensures law and gives priority to science, technology and innovation.
A. prosperity C. protest
B. nationality D. disparity
Answer: A. prosperity
Explanation: "Peace and prosperity" is a standard collocation meaning economic success and stability. Prioritizing innovation aligns with prosperity (wealth/well-being). "Nationality" (citizenship), "protest" (objection), and "disparity" (inequality) are illogical.

21. The weather in Addis Ababa is_____ these days. You cannot tell whether it rains in the morning or in the afternoon.
A. invisible C. unpredictable
B. immaterial D. unimaginable
Answer: B. unpredictable
Explanation: The inability to forecast rain indicates unpredictability. "Invisible" (cannot be seen), "immaterial" (unimportant), and "unimaginable" (unthinkable) do not fit.

22._____ by the speech of the new leader, all the people of the Zone elected him to be Member of Parliament.
A. Offended C. Discouraged
B. Demotivated D. Inspired
Answer: D. Inspired
Explanation: Positive influence from a speech leads to election. "Inspired" (motivated) fits, while negative options (A, B, C) would discourage support.

23. On my way home from office, there was high traffic_____ It took me two hours to get home.

A. congestion C. path way
B. freeway D. speed
Answer: A. congestion
Explanation: "Traffic congestion" means overcrowding on roads, causing delays. "Freeway" (highway), "path way" (path), and "speed" (velocity) do not explain the delay.

24. In the annual festival they hold, the villagers in my area always_____ to fight against the criminals that disturb the people.
A. meditate C. remember
B. resolve D. think
Answer: B. resolve
Explanation: "Resolve" means to make a firm decision (to fight criminals). "Meditate" (reflect), "remember" (recall), and "think" (consider) lack the connotation of committed action.

B. Substitution
Instructions: For questions 25 - 30, each sentence has an underlined word or phrase followed by alternatives A-D. Choose the word or phrase that best corresponds with the meaning of the underlined word or phrase and blacken the letter of your choice on the separate answer sheet provided.

25. Houses in cities are not **affordable** for the great majority of the people in Ethiopia.
A. comfortable C. available
B. cheap D. suitable
Answer: B. cheap
Explanation: "Affordable" means low-cost, so "cheap" is the closest synonym. "Comfortable" (cozy), "available" (accessible), and "suitable" (appropriate) are unrelated to cost.

26. He asked her to **turn round** several times before he gave her his opinion of her new dress.
A. rotate C. hold on
B. display D. pose
Answer: A. rotate
Explanation: "Turn round" means to spin or revolve. "Rotate" is a direct synonym. "Display" (show), "hold on" (wait), and "pose" (position) are incorrect.

27. Thanks to science and technology, many of our organs are **replaceable.**
A. removable C. exchangeable
B. curable D. avoidable
Answer: C. exchangeable
Explanation: "Replaceable" means substitutable, so "exchangeable" fits best. "Removable" (detachable), "curable" (healable), and "avoidable" (preventable) are unrelated.
28. The teacher announced that the mathematics examination has been **put off** for next week.

A. forgotten C. prepared

B. postponed D. cancelled
Answer: B. postponed
Explanation: "Put off" means delayed, so "postponed" is correct. "Cancelled" (terminated) is incorrect, and "forgotten" or "prepared" are illogical.
29. If you go to a new place, your behavior and manner should **harmonize** with that of the people you meet there.
A. go against C. contradict
B. consider D. integrate
Answer: D. integrate
Explanation: "Harmonize with" means to blend in or match. "Integrate" (combine smoothly) is synonymous. "Go against" and "contradict" are antonyms, while "consider" (think about) is unrelated.
30. Tensions between the two countries are rising quickly. We cannot **rule out** the possibility that war breaks out.
A. assume C. predict
B. exclude D. imagine
Answer: B. exclude
Explanation: "Rule out" means to eliminate or disregard. "Exclude" is the closest synonym. "Assume" (presume), "predict" (forecast), and "imagine" (visualize) do not fit.

C. Analogy
Instructions: For questions 31 and 32, two pairs of words are given. Look at the relationship between the two words in each pair and choose the word which forms a similar relationship with the word given next to each pair and then blacken the letter of your choice on the separate answer sheet provided.

31. Mouse is to computer as steering wheel is to
A. thermo-flask C. vehicle
B. refrigerator D. television
Answer: C. vehicle
Explanation: A mouse is a tool to control a computer. Similarly, a steering wheel controls a vehicle. Thermo-flasks, refrigerators, and televisions lack this control-device relationship.

32. Bed: Sleep:: Stove:_____
A. Clean C. Cook
B. Wash D. Utensil
Answer: C. Cook
Explanation: A bed is used for sleeping. A stove is used for cooking. "Cook" (verb) matches the function-based relationship. "Utensil" (noun) is a tool, not an action, while "clean" or "wash" are unrelated functions.

Section Four: Sentence Comprehension

Instructions: For questions 33 and 34, choose the sentence with the closest meaning to the head sentence from the four alternatives given, and blacken the letter of your choice on the separate answer sheet provided.

33. All students but one got above average marks in English.
A. Only one student failed to get above average marks in English.
B. Only one student got the above average mark in English.
C. All students did not get above average marks in English.
D. Everybody got above average mark in English.
Answer: A
Explanation: "All students but one" means "all except one." Thus, one student did NOT get above average marks. Option A correctly states this. Option B reverses the meaning, while C and D contradict the original.
34. Addis Ababa University, which is the oldest university in the country, has changed its admission policy.
A. Addis Ababa University admits students based on new rules and regulations.
B. Addis Ababa University has grown old with its admission policy.
C. Addis Ababa University has a conducive learning atmosphere.
D. Joining Addis Ababa University has now become easy.
Answer: A
Explanation: The head sentence focuses on a **change in admission policy**. Option A directly reflects this by mentioning "new rules and regulations." Options B, C, and D introduce unrelated ideas.
II. LANGUAGE FOCUS

Directions: Questions from 35-62 are incomplete sentences. Choose the word or phrase that best completes the sentence accurately and blacken the letter of your choice on the separate answer sheet provided.
35. At first, the suspect did not realize_____ during the interview. Later on, he saw the voice recorder and refused to continue the interview.
A. he was being recording C. he had recorded
B. he was recording D. he was being recorded
Answer: D
Explanation: Passive voice ("was being recorded") is needed because the suspect was the object of recording. Options A (grammatically incorrect), B (active voice), and C (past perfect) do not fit.
36. Unlike last year, only_____ students needed extra help this year last year we helped almost everyone in the class.
A. many C. a few
B. most D. a little
Answer: C
Explanation: "A few" indicates a small number (for countable nouns like "students"),

contrasting with "almost everyone" last year. "A little" is for uncountable nouns, while "many" and "most" imply a large number.

37. We should be involved in sports activities to keep_____ fit and healthy.

A. oneself C. yourselves

B. ourselves D. ourself

Answer: B

Explanation: The reflexive pronoun "ourselves" agrees with the subject "we." "Oneself" is impersonal, "yourselves" is for "you," and "ourself" is non-standard.

38. I _____my homework when you knocked at the door.

A. had been C. have done

B. have been doing D. was doing

Answer: D

Explanation: Past continuous ("was doing") describes an action in progress (doing homework) interrupted by another action (knocking). Other options misuse tenses.

39. _____I do not complain much; my friends often take me for a fool.

A. As a result of C. In spite of

B. Because of D. Because

Answer: D

Explanation: "In spite of" introduces contrast: despite not complaining, friends mistake the speaker for a fool. Other options imply causation, not contrast.

40. My English is not good, I _____ advised to study English when I was younger.

A. wish I had been C. wished I had been

B. wished I was D. wish I have been

Answer: A

Explanation: "Wish + past perfect" (had been) expresses regret about the past. "Wish" (present) is used for current feelings. Other options misuse tense or structure.

41. I_____ much today if I had learned about money management at my young age.

A. did not worry C. will not worry

B. would not worry D. would not have worried

Answer: D

Explanation: Third conditional ("would not have worried") is used for unreal past conditions. The if-clause ("had learned") is past perfect, so the main clause must match.

42. _____you ever_____ a chance to visit any historical site in Ethiopia?

A. Had/have C. Have/had

B. Did/have D. Have/have

Answer: C

Explanation: Present perfect ("Have you ever had") asks about life experiences. "Had" is the past participle of "have." Other options are grammatically incorrect.

43. The road project_____ by the construction company next year.

A. will be completing C. will completed

B. was completed D. will be complete

Answer: D
Explanation: Future passive ("will be complete") is correct. "Complete" is an adjective meaning "finished." Option A is grammatically wrong, B is past tense, and C is incomplete.
44. If I had enough money, I_____ schools with elevators for visually impaired children.
A. will have built C. would have built
B. would build D. will build
Answer: D
Explanation: Second conditional ("would build") describes a hypothetical present/future situation. The if-clause ("had") is simple past, so the main clause uses "would + base verb."
45. Next year this time, I _____the freshman program.
A. will have almost completed C. will be nearly complete
B. had nearly completed D. would have nearly completed
Answer: A
Explanation: Future perfect ("will have almost completed") indicates an action finished by a specific future time ("next year this time"). Other options misuse tenses.
46. This year's examination is as good as the one given last year,_____ ?
A. hasn't it C. isn't it
B. doesn't it D. wasn't it
Answer: C
Explanation: Question tags mirror the main verb ("is"). "Isn't it?" matches the present tense. Other options mismatch the verb tense.
47. I met the girl_____ phone number I was asking for.
A. whom C. that
B. whose D. which
Answer: B
Explanation: "Whose" shows possession (the girl's phone number). "Whom" is for objects, "that/which" for things.
48. When you acknowledge people's contribution to your work, you need not refer_____ his/her name.
A. to C. with
B. for D. at
Answer: A
Explanation: "Refer to" is a fixed phrase meaning "mention." Other prepositions do not collocate with "refer."
49. _____listening_____ reading are important academic skills students need to develop.
A. Neither/nor C. Both/and
B. Either/nor D. Either/or
Answer: C
Explanation: "Both/and" includes both skills. "Neither/nor" excludes, "either/or" offers a choice, and "either/nor" is non-standard.
50. He_____ his children to school five days a week.
A. drives C. drove

B. has driven D. drive
Answer: A
Explanation: Present simple ("drives") describes habitual actions. Other options misuse tenses: B (present perfect) implies duration, C (past) is not habitual, D (base form) lacks subject-verb agreement.
51. My father is not _____old_____ yours; he is only 53 years old.
A. such...as C. as...so
B. such...so D. as...as
Answer: D
Explanation: "As...as" compares equality ("not as old as yours"). Other options are not used for comparisons.
52. Lojo is_____ tall girl_____ she can easily touch the ceiling of her classroom.
A. such...as C. as such...that
B. such a...that D. so...that
Answer: B
Explanation: "Such a...that" shows result ("such a tall girl that"). "So...that" would require "so tall a girl that," but the structure here fits B best.
53. Some students did well on the last week's English test, _____others scored unsatisfactory marks. I am designing an action research to study the case.
A. when C. whereas
B. despite D. if
Answer: C
Explanation: "Whereas" contrasts two groups. "When" indicates time, "despite" is a preposition (not a conjunction), and "if" introduces a condition.
54. My English teacher_____ me repeat the correct answer ten times.
A. wanted C. told
B. forced D. made
Answer: D
Explanation: "Made" + base verb ("repeat") means "compelled." Other options require "to" (e.g., "wanted/told/forced me to repeat").
55. My history teacher asked me_____ what is your nationality?
A. what my nationality is C. what is your nationality?
B. what is my nationality. D. what was my nationality?
Answer: A
Explanation: Reported speech uses statement word order ("what my nationality is") and changes "your" to "my" from the teacher's perspective. Tense remains present for permanent facts.
56. You_____ go to bed before midnight if you have the habit of getting up late.
A. will not C. must not
B. may not D. cannot
Answer: A
Explanation: "Will not" implies a natural consequence: if you get up late, you **will not** go to bed

early. "May not" (possibility) is less direct, while "must not" (prohibition) and "cannot" (inability) are illogical.

57. You will start running_____ the bell rings. Otherwise, you will be disqualified from the competition.

A. till C. during

B. while D. after

Answer: B

Explanation: "While" means "at the same time as." The action (running) starts precisely when the bell rings. "Till," "during," and "after" do not convey immediacy.

58. The afternoon sun is_____ hot for school children _____play outside. They had better stay in their class.

A. such/for C. very/for

B. too/to D. to/to

Answer: B

Explanation: "Too + adjective + for someone + to verb" ("too hot for children to play") shows excessive heat preventing the action. "Very" does not fit this structure.

59. The lady _____next to our English teacher is my mother's close friend.

A. sitting C. sat

B. sit D. sits

Answer: A

Explanation: A reduced relative clause ("sitting") replaces "who is sitting." Other options misuse verb forms.

60. Unlike my friends, who dislike drinking coffee, I_____ go to bed at night without coffee.

A. never C. always

B. mostly D. ever

Answer: A

Explanation: "Never" emphasizes contrast: friends dislike coffee, but the speaker **never** goes to bed without it (i.e., always drinks it). "Always" would align with friends, not contrasting.

61. Committing corruption_____ life imprisonment in several countries in the world.

A. impacts C. affects

B. arises from D. leads to

Answer: D

Explanation: "Leads to" means "results in." Corruption **causes** imprisonment. "Impacts/affects" mean "influences," while "arises from" implies the opposite causation.

62. I always have much work to do; that is why I_____ have time to listen to music.

A. definitely C. scarcely

B. mainly D. exactly

Answer: C

Explanation: "Scarcely" (hardly any) indicates a near-total lack of time due to busyness. Other options are positive and contradict the context.

III. WRITING

Directions: Questions from 63-80 are related to different types of writing. Read each of them very carefully and choose the best answer from the given alternatives. Then, blacken the letter of your choice on the separate answer sheet provided.

Section One: Word Spelling

Instructions: Questions 63 and 64 are on word spelling. From the alternatives A-D, choose the one which is correctly spelt and blacken the letter of your choice on the separate answer sheet provided.

63. A. expenditure C. particularily
B. traditionial D. fotograph

Answer: A. expenditure

64. A. paralism C. inteligibility
B. suffication D. repetition

Answer: C. intelligibility

Section Two: Punctuation and Capitalization

Instructions: Questions 65-67 are on punctuation and capitalization. From the given alternatives A - D, choose the statement which is correctly punctuated and capitalized, and blacken the letter of your choice on the separate answer sheet provided.

65. A. Tadesse and Fanuel are good friends. They study their lessons together and help each other.
B. Tadesse and Fanuel are good friends; They study their lessons together and help each other.
C. Tadesse and Fanuel are good friends, they study their lessons together and help each other.
D. Tadesse and Fanuel are good friends They study their lessons together and help each other!

Answer: A. Tadesse and Fanuel are good friends. They study their lessons together and help each other.

Explanation:

A: Correct. Two independent sentences separated by a period; capitalization is proper.

B: Incorrect. Semicolon misuse; "They" should not be capitalized after a semicolon.

C: Incorrect. Comma splice (comma cannot join two independent clauses).

D: Incorrect. Run-on sentence (no punctuation between clauses); exclamation mark is inappropriate.

66. A. Ethiopia is a rich country endowed with many natural resources. In addition it has many diversified cultures and very friendly people.
B. Ethiopia is a rich country endowed with many natural resources. In addition, it has many diversified cultures and very friendly people
C. Ethiopia is a rich country endowed with many natural resources. In addition it has many diversified cultures and very friendly people.
D. Ethiopia is a rich country endowed with many natural resources. In addition, it has many diversified cultures and very friendly people.

Answer: D. Ethiopia is a rich country endowed with many natural resources. In addition, it has many diversified cultures and very friendly people.

Explanation:

D: Correct. Comma after introductory phrase "In addition"; period ends the sentence.

A/C: Missing comma after "In addition."

B: Missing period at the end.

67. A. Hello Kenyan National Hotel, can I help you?
B. Hello, Kenyan National Hotel. can I help you?
C. Hello Kenyan National Hotel. Can I help you?
D. Hello, Kenyan National Hotel! Can I help you?

Answer: D. Hello, Kenyan National Hotel! Can I help you?

Explanation:

D: Correct. Comma after "Hello"; exclamation for emphasis; "Can" capitalized.

A: Missing comma after "Hello"; "can" should be capitalized.

B: Period after "Hotel" should be a comma/exclamation; "can" not capitalized.

C: Missing comma after "Hello."

Section Three: Jumbled Words
Instructions: The words in the questions 68-70 are NOT in the right order. When put in the right order, they make correct English sentences. From the given alternatives A - D, choose the correct answer and blacken the letter of your choice on the separate answer sheet provided.

68. am/as/think/I/not/I/you/stupid/am/as.
A. I am not as I think am stupid as you.
B. As you think I am not stupid as I am.
C. I am not as you think as I am.
D. I am not as stupid as you think I am.

Answer: D. I am not as stupid as you think I am.

Explanation: Correct comparative structure: "not as [adjective] as [clause]." Other options are grammatically incoherent.

69. who/a/clever/ teacher /in/school/our/Ato Tadesse/got /has /a/ scholarship/is.
A. Ato Tadesse, in our school, has got a scholarship who is a clever teacher.
B. Ato Tadesse, who is a clever teacher in our school, has got a scholarship.
C. Ato Tadesse is a clever teacher in school who has got our a scholarship.
D. Ato Tadesse has got a scholarship, who is a clever teacher, in our school.

Answer: B. Ato Tadesse, who is a clever teacher in our school, has got a scholarship.

Explanation: Relative clause ("who is...") correctly modifies "Ato Tadesse"; main clause ("has got...") is clear. Other options misplace clauses.

70. in/ the/ science/ laboratory/teacher/ and/ students/both/together/working the/are/the.
A. The students and the teacher together are both working in the science laboratory.
B. The students are working and both the teacher together in the science laboratory.
C. Both in the science laboratory are working together the teacher and the students.
D. Both the teacher and the students are working together in the science laboratory.

Answer: D. Both the teacher and the students are working together in the science laboratory.

Explanation: "Both...and" pairs subjects correctly; word order is logical. Other options have fragmented/illogical structures.

Section Four: Paragraph Coherence
Instructions: When you put them in the correct order, sentences a - d in each of the questions from 71-74 make up a complete paragraph. From the alternatives given in each case, choose the one that contains the correct order of the sentences and blacken the letter of your choice on the separate answer sheet.

71. a. Arable farming refers to growing crops.
b. This farming type refers to keeping sheep and cattle.
c. The second farming type is pastoral farming.
d. Arable and pastoral farming are commonly known farming types.

A. a d c b
B. b a d c
C. c b a d
D. d c d a

Answer: D. d a c b

Explanation: Logical flow:

(d) Introduces farming types.

(a) Defines arable farming.

(c) Introduces pastoral farming.

(b) Defines pastoral farming.

Order: d → a → c → b.

72. a. Hanna is a very charming and sociable girl whom everybody likes.
b. Finally, she works in an international supermarket as a saleswoman.She is always smiling and appears to be happy with her job.
c. She is only in her late twenties. She usually wears jeans that visibly show the shape of her body. Above her waist, she wears a white nylon T-shirt that makes her breasts appear more visible and attractive.
d. To begin with, she is tall and has a slim waist that makes her quite attractive.
A. a d c b
B. a c d b
C. d a b c
D. d a c b

Answer: A. a d c b

Explanation: Logical flow:

- (a) General introduction to Hanna.
- (d) Starts physical description ("To begin with").
- (c) Continues description (clothing/age).
- (b) Concludes with job ("Finally").

Order: a → d → c → b.

73. a. Many people believe education is the key to success in life.
b. These people list names of several intellectuals who struggle to make in the end to breath on their salaries
c. As examples, they mention highly educated people who have become rich.
d. Others contend that education only leads people to poverty.
A. a b c d
B. a c b d
C. c b d a
D. c b a d

Answer: B. a c b d

Explanation: Logical flow:

- (a) Presents viewpoint 1 (education = success).
- (c) Supports viewpoint 1 (examples of rich educated people).
- (b) Supports viewpoint 1 (counterargument: struggling intellectuals).
- (d) Presents opposing viewpoint (education = poverty).

Order: a → c → b → d.

74. A. Three of their children are girls.

B. His wife wizero Mulu, is younger than him by 2 years.
C. Ato Lemma and Wiro Mulu have four children.
D. Ato Lemma is a thirty-year-old person in my community.
A. c b d a
B. d b c a
C. a c d b
D. d c a b
Answer B. d b c a (D → B → C → A).

Explanation: "Ato Lemma is a thirty-year-old person in my community. His wife wizero Mulu, is younger than him by 2 years. Ato Lemma and Wiro Mulu have four children. Three of their children are girls.
Why Other Options Fail:

- **A. c b d a:** Starts with "they have children" (C) before introducing who "they" are, causing confusion.
- **C. a c d b:** Begins with "their children" (A) without establishing who "they" are.
- **D. d c a b:** Introduces children (C) before mentioning the wife (B), making "their" in (A) ambiguous.

Section Five: Types of Paragraph Writing
Instructions: Questions 75-77 are related to types of paragraph writing. From the alternatives A-D given in each question, choose the correct answer and blacken the letter of your choice on the separate answer sheet provided.

75. "First of all, the death penalty should be only resort to killing people and spend the rest of their lives in jail. Moreover, if the death penalty committed against them would like to take revenge on the criminal and this makes chaos in society". This is what paragraph writing that is most likely:
A. argumentative C. descriptive
B. expository D. narrative

Answer: A. argumentative

Explanation: The paragraph argues against the death penalty with reasoning (revenge, chaos), aiming to persuade.

76. Digestion begins in the mouth. This is called mechanical digestion as there is not any kind of chemical involved in the process. It is simply as the saliva is broken down into smaller pieces and the food is swallowed. The other type of digestion is chemical digestion. It is in the stomach that this digestion takes place. This above paragraph are most likely taken from:
A. argumentative C. expository

B. descriptive D. narrative

Answer: C. expository

Explanation: The paragraph explains a process (digestion) factually without persuasion or description.

77. Eden is in her late twenties. She is of medium height. She has an average weight that fits her height. She respects people, and she is very polite and friendly. From Which of the following paragraph writing types is the above sentences most likely?
A. Descriptive C. Expository
B. Argumentative D. Narrative

Answer: A. Descriptive

Explanation: The paragraph describes Eden's physical traits and personality (height, weight, politeness).

Section Two: Letter Writing
Instructions: Questions 78-80 are about letter writing. The letter below has three parts and all of them carefully follows the next answer and blacken the letter of your choice on the separate answer sheet provided.

(1) Dear sir,
(2) I am a polytechnic graduate. I have a Diploma in Auto mechanics from Bahir Dar Polytechnic College. I have worked as a mechanic in Bahir Dar Textile Factory for two years. I am now applying for the post of mechanic advertised in your company. For your reference, I have attached my CV with this application.
(3) I am looking forward to hearing from you.
Sincerely,Dewit Hibru

78. The part indicated by number 3 is :
A. a greeting C. salutation
B. closing statement D. body of the letter

Answer: B. closing statement

Explanation: "I am looking forward to hearing from you" is a standard closing line before the sign-off.

79. The part indicated by number 2 is :
A. greeting C. salutation
B. closing statement D. body of the letter

Answer: D. body of the letter

Explanation: This section contains the main content (qualifications, experience, job application).

80. The part indicated by number 1 is :
A. closing statement C. body of the letter
B. salutation D. subject

Answer: B. salutation

Explanation: "Dear sir," is the greeting addressing the recipient.

IV.COMMUNICATIVE ACTIVITIES

Directions: Questions from 81-100 are presented in the form of conversations. The part said by one of the speakers is given, and a blank space is left for the other. Choose the alternative that is most appropriate to complete the dialogue and blacken the letter of your choice on the separate answer sheet provided.

81. Lensa: My spoken English is really horrible; _____?
Zemzem: Practice speaking every day in front of your friends.
A. why you learn English C. what is yours like
B. do you like English D. what should I do
Answer: D. what should I do
Explanation: Lensa expresses a problem ("My spoken English is horrible") and seeks advice. Zemzem responds with a solution ("Practice speaking..."). Option D directly asks for actionable advice, fitting the context. Other options (A, B, C) ask unrelated questions (e.g., reasons for learning English or others' abilities).
82. Student:_____
Teacher: Yes. This will be our tomorrow's lesson.
A. Could you explain to me about active and passive voices?
B. When do you teach us about active and passive voices?
C. Who knows about active and passive voices?
D. You could learn active and passive voices.
Answer: B. When do you teach us about active and passive voices?
Explanation: The teacher confirms the topic will be covered tomorrow. Option B asks about timing, aligning with the response. Options A, C, and D either request immediate explanation, ask about knowledge, or state an opinion—none match the teacher's answer.
83. Friend: What do you think unemployment will be like in the future?
You:_____
A. I am sure today’s children won’t expect a job from the government
B. In my opinion, none of a working age will be unemployed
C. I think unemployment is a universal problem these days
D. Think about today; tomorrow is another day
Answer: B. In my opinion, none of a working age will be unemployed
Explanation: The friend asks about the future of unemployment. Option B gives a direct, optimistic prediction about the future ("none will be unemployed"). Other options discuss the present (C), avoid the question (D), or mention unrelated expectations (A).
84. Teacher:_____

Parent: You are right, wearing a uniform should be compulsory for class attendance.
A. Isn't it very disappointing to see students in the classroom without their uniform
B. Is it very disappointing to see students in the classroom without their uniform
C. When do we see students in the classroom without their uniform
D. Why do we see students in the classroom without their uniform
Answer: A. Isn't it very disappointing to see students in the classroom without their uniform
Explanation: The parent agrees with the teacher ("You are right"). Option A uses a rhetorical question to express disapproval, prompting agreement. Options B, C, and D ask factual questions (e.g., "Is it disappointing?" or "Why?") that don't invite agreement.
85. Husband: Six people died and three were injured in a car accident today.
Wife:_____ Has the cause been known?
A. Look after yourself on the road. C. I am sorry to say this.
B. Never trust a car driver. D. What a terrible news!
Answer: D. What a terrible news!
Explanation: The wife reacts to tragic news and asks for details. Option D shows shock/sympathy ("What terrible news!"), fitting the context. Options A, B, and C give irrelevant advice or apologies.
86. Abeya: I think that education is the only way to live a better life?
Temesgen: True, but I should say_____
A. going to college is not mandatory
B. perhaps we need to learn further
C. perhaps it is one among the few
D. going to college is simple
Answer: C. perhaps it is one among the few
Explanation: Temessen agrees ("True") but challenges Abeya's claim that education is the "only" way. Option C suggests there are multiple paths ("one among the few"), contrasting with "only." Other options don't counter this key point.
87. Neighbor: Today's youths have a much wider access to information than those who lived in the past.
Father: Yes, but_____
A. it's good to remember those who lived in the past were less intelligent
B. the trouble is they are indiscriminate in their choice of information
C. we should appreciate the unlimited interest of today's youth
D. the main point is they do not read all that comes their way
Answer: B. the trouble is they are indiscriminate in their choice of information
Explanation: The father agrees ("Yes") but adds a drawback ("but..."). Option B criticizes youths' lack of discernment, balancing the neighbor's praise. Options A, C, and D are irrelevant or positive.
88. Teacher: Tell me the reasons why learning English is important to you.
Student:_____
A. To begin with, English is a compulsory school subject
B. As a matter of fact, we learn English five days a week
C. Believe it or not, my written English is very good

D. Firstly, English is not my favourite school subject

Answer: A. To begin with, English is a compulsory school subject

Explanation: The teacher asks for reasons. Option A starts a logical list ("To begin with...") and gives a valid reason (compulsory subject). Other options discuss frequency (B), ability (C), or negativity (D).

89. Mother: Why are you going so early today?

Daughter:_____ Mom.

A. Nobody said it is early C. In order to get a front seat

B. Mind your own business D. It is not very early

Answer: C. In order to get a front seat

Explanation: The mother asks for a reason ("Why...?"). Option C explains the purpose ("In order to..."). Other options are defensive (A, D) or rude (B).

90. Husband: Where would you like us to go for dinner?

Wife: _____I like their French toast.

A. How about trying Haile Grand Hotel?

B. I do not have enough money for dinner.

C. I do not care about dinner tonight.

D. Is your sister going with us?

Answer: A. How about trying Haile Grand Hotel?

Explanation: The wife suggests a place and mentions liking their French toast ("their" implies a specific venue). Option A proposes a hotel, fitting her follow-up comment. Other options avoid suggesting a location.

91. Ararso: What is your opinion about the need for college education?

Jirata:_____

A. What is the requirement to join college?

B. I feel certain that it's absolutely necessary.

C. None of my children joined college.

D. I think I've a good idea about it.

Answer: B. I feel certain that it's absolutely necessary

Explanation: Ararso asks for an opinion. Option B gives a clear, confident view ("absolutely necessary"). Other options ask a question (A), share unrelated facts (C), or dodge the question (D).

92. Mother: Tell me why schools are closed in Kiremt in Ethiopia.

Daughter:_____

A. Did you learn year-long yourself?

B. Nobody likes going to school.

C. I hate Kiremt, Mom.

D. I haven't any clue, Mom.

Answer: D. I haven't any clue, Mom

Explanation: The mother asks for a reason. Option D honestly admits lack of knowledge. Other options are irrelevant (A, B, C).

93. Ahmed:_____

Samson: You know why? He is such a funny boy that he makes everyone laugh.

A. Do you like to sit next to Solomon in the classroom
B. Where do many students like to sit in the classroom
C. Why do many students like to sit next to Solomon
D. Is Solomon a funny boy
Answer: C. Why do many students like to sit next to Solomon
Explanation: Samson explains why students sit next to Solomon ("he makes everyone laugh"). Option C asks for the reason ("Why?"), matching the response. Other options ask about location (B) or personal preference (A, D).
94. Kidist:_____
Hana: With sugar, please.
A. Would you prefer your coffee with or without sugar
B. How much sugar do you like in your coffee
C. Is it good to drink coffee without sugar
D. Do you like your coffee with sugar
Answer: A. Would you prefer your coffee with or without sugar
Explanation: Hana chooses an option ("With sugar"). Option A offers a choice ("with or without"), fitting her response. Options B, C, and D ask about quantity or preference but don't present a clear choice.
95. Student: Today, I'm interviewing Business Managers to give me some idea about the field. Let me begin by_____
Manager: To put it simply, I joined the department because I wanted to learn how to run my own business.
A. asking you how you decided to study business management
B. telling you the history of the field in our context
C. telling you a few important things about the field
D. explaining job opportunities in the field
Answer: A. asking you how you decided to study business management
Explanation: The manager explains their personal reason for joining the field ("I joined because..."). Option A asks about their decision, aligning with the response. Other options would prompt different answers (e.g., history or job opportunities).
96. Father: _____?
Mother: He did not get full mark on his English exam.
A. Why is Zena crying C. Who is crying Zena
B. How is Zena crying D. When is Zena crying
Answer: A. Why is Zena crying
Explanation: The mother gives a reason ("He did not get full marks"). Option A asks for a reason ("Why?"). Other options ask about manner (B), identity (C), or time (D).
97. Lilu: _____
Maria: I'm sorry; I'm a stranger too.
A. Could you tell me where the director's office is?
B. Hi, you know the director's office, don't you?
C. Where is the director's office, if I may ask?
D. I am sorry to say that, but I got lost.

Answer: A. Could you tell me where the director's office is?
Explanation: Maria cannot help because she is a "stranger." Option A is a polite request for directions, justifying her apologetic response. Options B and C are less formal/direct, while D is not a question.
98. Nuru: Well done, Daniel! You got an A grade on your Maths.
Daniel: Really?_____
A. A good achievement, isn't it
B. Thank you for the good news.
C. I'm glad you liked it.
D. Nice work, isn't it?
Answer: B. Thank you for the good news
Explanation: Daniel reacts to praise and good news. Option B thanks Nuru appropriately. Options A, C, and D are self-focused or irrelevant.
99. Friend: _____I'm almost blind in the dark.
You: Don't worry about it.
A. Will you forgive me for stepping on your foot?
B. I shouldn't have stepped on your foot.
C. Excuse me for stepping on your foot.
D. Did I step on your foot?
Answer: C. Excuse me for stepping on your foot
Explanation: The friend apologizes for an accident (stepping on a foot) and explains ("blind in the dark"). Option C is a direct apology. "Don't worry" accepts this. Options A, B, and D are less natural for minor accidents.
100. Daughter: I think brilliant students should be exempted from cost-sharing.
Mother:_____ money shouldn't be a reward for good work.
A. No, I don't accept this foolish idea
B. I am afraid I can't agree
C. This is stupid
D. Shut up

Answer: B. I am afraid I can't agree

Explanation: The mother disagrees politely and gives a reason ("money shouldn't be a reward").

Option B is respectful disagreement. Options A, C, and D are rude/aggressive.

The End

ENTRANCE TRICKS

2017 Biology Entrance Exam

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EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES) ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE) SENE, 2017 E.C/JUNE/JULY, 2025 G.C

SUBJECT: BIOLOGY **TIME ALLOWED: 2:30 Hours**

BOOKLET CODE: 351 **SUBJECT CODE: 06**

ENTRANCE

GENERAL DIRECTIONS:

I. FOLLOW THE FOLLOWING RULES AND REGULATIONS IN DOING THE EXAMINATION:

• ENTER THE SUBJECT NAME, SUBJECT CODE, AND BOOKLET CODE GIVEN ABOVE ON THE SEPARATE ANSWER SHEET PROVIDED

• CHOOSE ONLY THE CORRECT ANSWER FOR EACH QUESTION

• USE ONLY A PENCIL TO MARK YOUR ANSWERS

• BLACKEN ONLY THE LETTER YOU CHOOSE IN THE CORRESPONDING CIRCLE ON THE ANSWER SHEET

• YOUR MARK SHOULD BE HEAVY AND DARK

• ERASE ALL THE UNNECESSARY MARKS COMPLETELY

• ANY FORM OF CHEATING IS PROHIBITED AND RESULTS IN THE CANCELLATION OF YOUR SCORES

• STOP WORKING IMMEDIATELY WHEN THE ALLOTTED TIME IS UP

• REFER TO THE EXAM BOOKLET FOR FURTHER INSTRUCTIONS FOR EACH QUESTION

II. ATTENTION!

ENSURE THAT YOU HAVE WRITTEN ALL THE REQUIRED INFORMATION ON THE SEPARATE ANSWER SHEET CORRECTLY BEFORE RETURNING IT.

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

ENTRANCE TRICKS

EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES) ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE) Biology, Sene, 2017 E.C/June/July, 2025 G.C

NUMBER OF ITEMS: 100 **TIME ALLOWED:** 2:30 HOURS

SUBJECT CODE: 06 **BOOKLET CODE:** 351

1. Which one of the following Ethiopian biological research institutes is known for housing a major gene bank in Africa?

A. Addis Ababa University Biology Department

B. Institute of Biodiversity Conservation

C. Aklilu Lemma Institute of Pathobiology

D. Armauer Hansen Research Institute

2. A grade 12 student marked two dots close to each other using a pen on a piece of paper and observed them under a microscope. The two dots appeared well separated under the microscope. This was due to

A. resolution

B. magnification.

C. staining

D. mounting.

3. From the following alternatives select the common features that are found both in animal and plant cells.

A. Cell membrane, nucleus, cytoplasm, lysosome and Golgi apparatus

B. Cell membrane, nucleus, cytoplasm, mitochondria and endoplasmic reticulum

C. Cell membrane, nucleus, cytoplasm, chloroplast and Golgi apparatus

D. Cell membrane, nucleus, cytoplasm, large vacuole and Golgi apparatus

4. A biology teacher grouped his students in to two groups and ordered group I to place a slice of potato in hypertonic solution and group II in hypotonic solution. What will happen to the potato slice in group I and group II? For group I, the potato slice will be

ENTRANCE TRICKS

A. turgid while for group II it will be flaccid.

B. haemolyzed while for group II it will be flaccid.

C. flaccid while for group II it will be turgid.

D. turgid while for group II there will be no change.

5. A man eats only vegetables, beans, banana and orange for one week continuously in his lunch and dinner. For this man, what nutrients are most likely missed in his meal to be a balanced diet?

A. Carbohydrates

B. Lipids

C. Proteins

D. Minerals

6. Which of the following is correct about the process of digestion or absorption of food in the small intestine?

A. Proteins are converted into amino acids by the action of hydrochloric acid and amylase.

B. Bile emulsifies fats to increase the surface area to make it easier for lipase.

C. Water is reabsorbed into the small intestine and metabolic wastes are excreted.

D. Glucose and non-hydrolyzed starch molecules are absorbed through the lacteal.

7. In an experiment, students wanted to prove that enzymatic digestion begins in the mouth using certain chewed food coated with their own saliva. In order to get a positive result for their experiment, what type of food sample and testing solution should the students use?

A. Potato and iodine solution

B. Meat and Benedict's solution

C. Fish and sodium hydroxide solution

D. Bread and copper sulphate solution

8. Compared to the inhaled air what are the percentage of nitrogen, oxygen and carbon dioxide gases in exhaled air, respectively?

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A. 80%, 16% and 4%

B. 80%, 4% and 16%

C. 16%, 4% and 80%

D. 4%, 16% and 80%

9. Three students carried out different experiments in exhaled air as indicted below.

Student I - uses cold glass and blue cobalt chloride paper

Student II - uses ice cubes

Student III - uses tube containing lime solution and straw

Based on their setup of their preparation, which student plan to demonstrate the presence of water vapour and carbon dioxide, respectively?

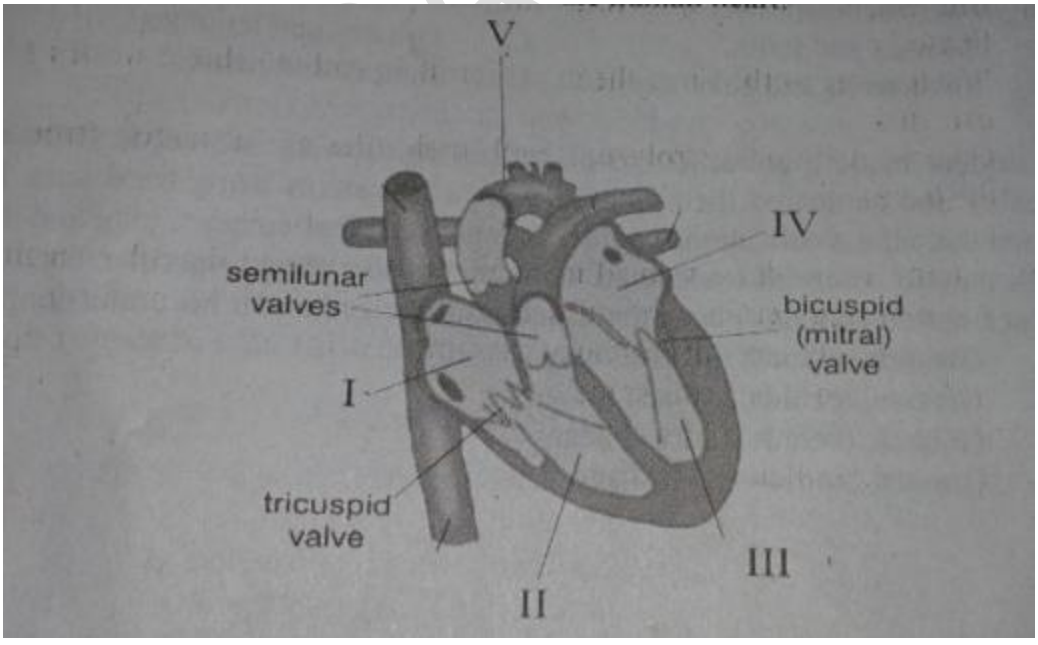
A. II and III

B. I and II

C. II and I

D. I and III

10. Consider the following structure of the human heart.



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In the above diagram, which part is used for forcing oxygenated blood out of the heart and around the body?

A. I

B. II

C. III

D. IV

11. If your body cannot make enough red blood cells, which one of the following mineral rich foods can correct this problem?

A. Iron-rich food

B. Calcium-rich food

C. Iodine-rich food

D. Sodium-rich food

12. A biology teacher provided common names of three organisms; human being, dog and housefly to his students to write the correct scientific names of each organism. In which of the following ways should the students write the correct scientific names, respectively?

A. Homo Sapiens, Canis Familiaris and Musca Domestica

B. Homo sapiens, Canis familiaris and Musca domestica

C. homo sapiens, Canis familiaris and Musca domestica

D. Homo sapiens, Canis familiaris and musca domestica

13. Identify the typical examples for division Gymnospermae and Angiospermae, respectively?

A. Mosses and ferns

B. Liverworts and palm trees

C. Grasses and legumes

D. Juniper and roses

14. A student made a collection of plants having flowers and seeds from his locality and compared their embryo, leaves and stems using hand lens. He found that all his collections

ENTRANCE TRICKS

contain a single seed leaf embryo, long and thin with parallel veins of leaves and stem contains scattered vascular bundles. Which option contains plants that can be categorized under his collection?

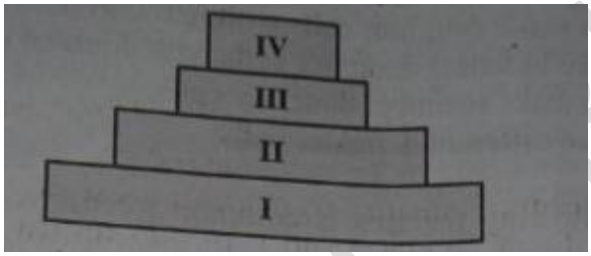
A. Grasses, orchids, teff and roses

B. Grasses, orchids, teff and beans

C. Grasses, sunflowers, teff and maize

D. Grasses, orchids, teff and maize

15. Consider the following diagram of pyramid of biomass.



Which one of the following statements is correct about the above pyramid of biomass?

A. The whole organisms at first trophic level is eaten by the organisms above.

B. When organisms in stage II eat the producers, most of the producer's materials are turned into new secondary consumers.

C. Much of the biomass from stage II is used by stage III to release energy for living.

D. At each stage of trophic level, the amount of biomass, which is passed on, supports a large amount of organism in the stage above.

16. This item is based on the following diagram of simple food chain.

Leaves -> Grasshoppers -> Rodents -> Leopards

What will happen if the number of rodents significantly decrease? There will be

A. an increase in the number of leopards.

B. fewer grasshoppers in the chain.

C. no change in the number of leopard.

D. an increase in the number of grasshoppers.

ENTRANCE TRICKS

17. Which of the following is NOT a key process involved in carbon cycle?

A. Evaporation

B. Cellular respiration

C. Photosynthesis

D. Decomposition

18. Which of the following is correct about mitosis? It is a division of

A. somatic cells to make daughter cells with half chromosome.

B. sex cells to make identical daughter cells with doubled chromosome.

C. somatic cells to make identical daughter cells.

D. sex cells to make different daughter cells.

19. Which one of the following features is common for mitosis and meiosis? They both involve

A. replication of DNA and the formation of daughter cells.

B. production of daughter cells identical to the parent cell.

C. formation of cells with the same chromosome number as parent cell.

D. spermatogenesis and oogenesis processes in males and females.

20. Which tissues of plant leaves are involved in waterproofing, major photosynthetic and major gas exchange activities, respectively?

A. Guard cells, palisade mesophyll and epidermis

B. Guard cells, palisade mesophyll and spongy mesophyll

C. Waxy cuticle, palisade mesophyll and spongy mesophyll

D. Waxy cuticle, palisade mesophyll and epidermis

21. Which of the following is correct about the movement of water in plants? It moves

A. against a water potential gradient.

B. across the phloem tissue by diffusion.

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C. across the tissues of a plant by osmosis.

D. by using energy until it reaches to the xylem.

22. What will happen to the growth of the plant when you remove indole-acetic acid from the main shoot part of the plant?

A. The plant will bush out.

B. It stimulates dropping of leaves.

C. The main shoot of the plant elongates.

D. The growth of side shoots will slow down.

23. In which one of the following alternatives the whole list contains characteristics of animals?

A. Respire, excrete, autotrophic and reproduce

B. Unicellular, excrete, heterotrophic and motility

C. Heterotrophic, excrete, motility and grow

D. Respire, multicellular, autotrophic and reproduce

24. A grade 11 student observed and identified some animals group having closed circulatory system, centralized nervous system with brain and sensory organs. Based on their identification select one of the following animals that can belong to this group?

A. Snails

B. Earthworm

C. Squid

D. Snake

25. What does it mean if an animal is poikilothermic? It

A. changes its body temperature with environmental condition.

B. increases its metabolic rate during cold and hot conditions.

C. keeps its body temperature constant regardless of external changes.

D. produces heat when temperature falls by activating the hypothalamus.

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26. Which one of the following statements is correct about the catalytic activities of enzymes? They

A. change the overall nature of the products.

B. allow reactions in cells to take place quickly.

C. alter themselves at the end of biochemical reactions.

D. increase the energy required to start a reaction.

27. Four students draw four different big protein structure posters. The contents of each of their structure are displayed in the following table.

Student	**Main contents indicated in the structure**
I	Polypeptide folded into α-helix and β-sheet, and hydrogen bond
II	Amino acid sequences and peptide bonds
III	Multiple folded protein subunits in a multi-subunit complex and haem group
IV	Side chain interaction of amino acids and overall 3-D shape

Based on the above information, which student draw secondary and quaternary structures, respectively?

A. I and IV

B. II and III

C. I and III

D. II and IV

28. How does the induced-fit model explain enzyme specificity? They

A. change the shape of active site slightly to fit substrates precisely.

B. have a rigid active site that can just bind to single substrate.

C. have multiple active sites that can fit to different substrates.

D. bind to whatever the shape of substrate that fits to allosteric site.

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29. What happens to the substrate molecule when competitive inhibitor binds with an enzyme? It

A. undergoes conformational change.

B. cannot bind with the active site.

C. binds to the allosteric site temporarily.

D. easily binds with the active site.

30. What will happen to the reaction if you add a certain inhibitor or activator to the substrate having allosteric enzyme set up?

A. Inhibitor binds to active site to speed up the reaction.

B. Inhibitor binds to regulatory site to increase rate of reaction.

C. Activator binds to active site to initiate the reaction.

D. Activator binds to regulatory site to activate the reaction.

31. Which one of the following classes of enzyme and type of reactions they catalyze is correctly matched?

A. Oxidoreductases - transfer functional groups from donor to acceptor

B. Transferases - transfer of electrons from one molecule to another

C. Ligases - cleave bonds by other means rather than hydrolysis

D. Lyases - catalyze the hydrolysis of various bonds

32. If a given enzyme works only at temperature range of 20 and 45 °C and the potential of hydrogen ions that ranges from 4 to 9. Which one of the following works best during the optimization of these two factors for the enzyme activity?

A. Adjusting the temperature below 20°C makes the enzyme more active, hence the temperature at this value is optimal.

B. Adjusting the pH above 9 can increase the probability of collisions between the enzyme and the substrate.

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C. Adjusting the temperature between 20°C and 45°C results little denaturation and increased number of collisions.

D. The optimum pH under which this enzyme is most active is found below 4.

33. All of the following features define a chromosome EXCEPT

A. contains hundreds to thousands of genes.

B. exists in pairs in diploid organisms.

C. made up of DNA and protein.

D. functional unit of inheritance.

34. Which structural and functional difference is observed between DNA and RNA?

A. Both DNA and RNA have the same nitrogenous bases but differ in pentose sugar.

B. RNA produces DNA through the process of transcription.

C. RNA stores the genetic material while DNA involves in protein synthesis.

D. DNA is very stable molecule while RNA is degraded quickly.

35. All of the following events occur during telophase I of meiosis I cell division EXCEPT

A. crossing over takes place.

B. each of the spindle fibers disappears.

C. set of chromosomes come together.

D. nuclear envelope reforms.

36. Suppose there was a cross between pea plants with flower positions axial (AA) and terminal (aa) and produced F1 generation. What will be the most probable number of heterozygous dominant individuals obtained if 1000 offspring were produced by self-crossing of F1 generation?

A. 200

B. 750

C. 500

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D. 250

37. If you have a pea plant of unknown genotype with a green pod color, you obtained all green pod colors of offspring phenotype after a test cross. What was the unknown genotype of the parent? It was

A. homozygous dominant.

B. homozygous recessive.

C. heterozygous dominant.

D. incomplete dominant.

38. This item is based on the following examples of genetic inheritance patterns.

1. Inflated and pinched pod forms will have inflated pod form.

2. Red cow and white bull will have roan calf.

3. AA and BB blood type people will have a child with AB blood type.

4. Red and white roses will have a pink rose offspring.

From the above patterns of inheritance, which of the following are exemplary for non-Mendelian inheritance?

A. 1, 2, 3

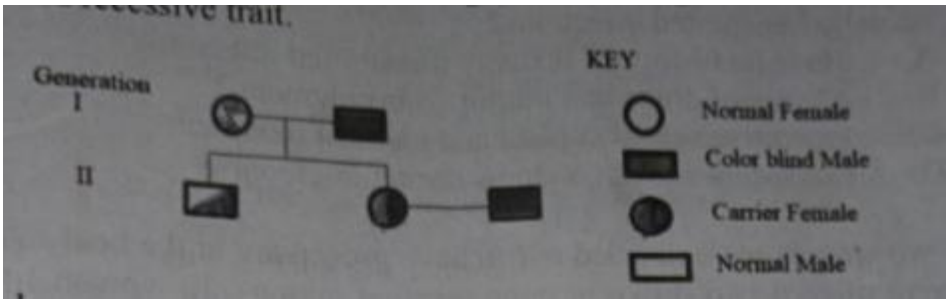
B. 1, 3, 4

C. 1, 2, 4

D. 2, 3, 4

39. This item is based on the following pedigree diagram for color blindness, X-linked recessive trait.

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Based on the above diagram, if the family in the second generation has children, what is the probability that they will have carrier female and color blind male, respectively?

A. ¼ and ¼

B. ½ and ½

C. ¾ and ¼

D. ¼ and ½

40. Farmers want to maintain their sheep with long tails for the next generation through similar sheep from their locality in their breeding program. What process do these farmers apply to keep the desired traits in their program?

A. Inbreeding

B. Crossbreeding

C. Genetic engineering

D. Cloning

41. If progesterone remains high after ovulation, what will be the most likely outcome?

A. FSH will be produced rapidly which leads to a new cycle.

B. Oestrogen will be decreased, closing down the lining of the uterus.

C. Uterus lining will be maintained and prepared for pregnancy.

D. Corpus luteum will be regenerated, inhibiting hormone production.

42. The most reliable contraceptive method that prevents travelling of ovum from the ovaries through the fallopian tubes, but requires surgery is

A. contraceptive injection.

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B. spermicide.

C. intrauterine device.

D. sterilization.

43. Among the following, which one increases the risk to be infected by sexually transmitted infections?

A. Have no history of sexually transmitted infections

B. Misuse of drugs that inhibit your judgment

C. Protective sexual contact and medical treatment

D. Consistent use of condoms during intercourse

44. Two women were decided not to have pregnancy in the near future and they were offered two different contraceptive options. In woman I, the material fitted in her uterus did not allow the sperm to fertilize the egg. In woman II, the contraceptive prevented hormone production and the releasing of eggs from the ovaries. Which type of contraceptive methods did woman I and II used, respectively?

A. Sterilization and Injectable contraceptives Depo-Provera

B. Post pill and copper-coated intrauterine device

C. Contraceptive injection and contraceptive implant

D. Copper-coated intrauterine device and contraceptive injection

45. The finding of the study conducted in two different villages showed that there are harmful practices that affect reproductive health. The common practices in village 1 were forcing underage girls to get married based norms and sexual abuse or harassment. In village 2 abduction to make the girl a wife unwillingly was very common. Which one of the following options is correct about the two villages?

A. In village 1, gender-based violence, kidnapping and female genital mutilation are common.

B. In village 1, gender-based violence and early marriage are common.

C. In village 2, kidnapping and female genital mutilation are common.

D. In village 2, gender-based violence and female genital mutilation are common.

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46. How drinking alcohol can contribute for HIV transmission? By

A. improving the decision making ability of a person

B. driving more willingly to participate in risky behaviors

C. impotency or loss of sexual drive both in male and female

D. increasing rate of the retrovirus multiplication capacity

47. Select the option that the whole lists contain renewable natural resources.

A. Coal, oil, natural gas and minerals

B. Coal, wind, hydro power and biomass

C. Sunlight, wind, hydro power and biomass

D. Sunlight, wind, hydro power and natural gas

48. This item is based on the following list of activities.

I. Establishing national parks

II. Wild sanctuaries

III. Preparing wild life reserves

IV. Practicing overgrazing

Which of the above activities CANNOT be used for the conservation of wild life?

A. II

B. I

C. III

D. IV

49. Among the following options, identify the air pollutants that can cause acid rain and its effects, respectively.

A. Carbon dioxide and methane - melting of ice caps and rising of oceans

B. Sulphur dioxides and nitrogen dioxides - damaging plants and water

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C. Perflourocarbons - causing skin cancer and cataracts

D. Chloroflourocarbons - increasing UV rays and respiratory disease

50. Which of the following is the cause and effect of global warming, respectively?

A. Greenhouse gases and dropping of sea levels

B. Greenhouse gases and loss of biodiversity

C. Burning of fossil fuels and increase in species diversity

D. Burning of fossil fuels and dropping of sea level

51. What is the primary purpose of creating transgenic organisms? To

A. generate pathogenic organisms that fight against non-pathogenic ones

B. produce organisms that mimic human beings in every aspect of behaviors

C. promote expression of desired genes that improve nutritional quality of organisms

D. produce novel organisms that can affect reproduction across different species

52. What is the role of biology in the investigation of a criminal case? To

A. detect flavor compounds in food and monitor environmental pollutants

B. examine potential sources of evidences using blood, saliva and hair

C. sense pathogens and herbicides using immunochemical detection system

D. determine the production of complex organic compounds using mold fungus

53. How does biological washing powder save energy? Because they

A. use lipase enzyme without other combination for any stain in clothes.

B. are effective in cleaning clothes at lower temperatures without boiling water.

C. are effective in removing dirty ores in clothes without spraying acidic water.

D. use protease enzyme without other combination for any stain in clothes.

54. What is biological warfare? The use of

A. selective genes to produce genetically modified organisms

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B. cloning technique to make exact genetic copies of an organism

C. bacteria, fungi and viruses as a biological agent to control pests

D. infectious agents with the intent to kill humans, animals or plants

55. Which of the following correctly describes the general features of bacteria? They are unicellular microbes with

A. true nucleus and are omnipresent in nature.

B. rigid cell wall and a diverse mode of nutrition.

C. mitochondria and a special type of chlorophyll.

D. Golgi bodies and genetic materials of a cell.

56. Which group of archaea are found in volcanic vents and hot vapors that contain either obligate aerobes, facultative aerobes or obligate anaerobes?

A. Hyperthermophiles

B. Acidophiles

C. Halophiles

D. Psychrophiles

57. From the fungal disease given below, which one is transmitted sexually?

A. Candidiasis

B. Aspergillosis

C. Tineas

D. Dermatophytosis

58. This item is based on the following choices of feeding habit employed by organisms.

I. Using nutrients dissolved in the surrounding medium

II. Manufacturing own nutrients by photosynthesis

III. Depending on other plants or animals for food

Which method/s the protozoan used to get their nutrition? They employed

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A. I only.

B. I and II only.

C. II and III only.

D. I, II and III.

59. Which one of the following diseases is correctly matched with its symptoms and method of transmission?

A. Chickenpox - runny and stuffy nose - respiratory droplets

B. Measles - paralysis and hydrophobia - bite from infected animals

C. Polio - commonly no sign and symptoms - fecal-oral-route

D. Rabies - swollen and painful protid glands - through infected saliva

60. In order to link a specific microorganism to a specific disease, Koch used mouse as experimental animal. He took a small amount of blood from a diseased mouse and injected into a healthy mouth, the latter quickly developed anthrax. Again he took blood from this second mouse and then injected it into another, again he observed the symptoms of anthrax. What was his first postulate related with this procedure?

A. The same microorganisms are present in every case of the diseases.

B. The isolated microorganisms can cause the same disease to susceptible animal.

C. The same pathogens are isolated and re-cultivated from experimental animal.

D. The microorganisms are isolated from the tissue of a dead animal and pure culture is prepared.

61. Sleeping sickness is transmitted by

A. biting of tsetse flies.

B. sexual intercourse.

C. eating contaminated food.

D. droplet infection.

62. Considering the diverse application of microorganisms, which of the following best evaluates their overall importance? They have

ENTRANCE TRICKS

A. abundant benefits that are essential for life on Earth.

B. a great disadvantage and limited advantage.

C. limited impact on the environment.

D. few applications in the areas of biotechnology and medicine.

63. Suppose a clinical nurse used a needle to treat a person with HIV/AIDS. What would be the most appropriate method of controlling such microbial infection?

A. Incineration

B. Tyndallization

C. Sanitization

D. Preservation

64. As a clinical microbiologist, suppose you want to collect microbes from a food sample and isolate them via serial dilution. How do you determine the isolated colonies that grow on artificial media? By

A. counting the number of strains on the plate

B. cross checking with other preserved specimens

C. observing their morphological features

D. inoculating them into other infected animals

65. What is the main difference between anabolism and catabolism in cellular metabolic pathways? Anabolism

A. absorbs energy while catabolism releases energy.

B. is destructive while catabolism is constructive process.

C. occurs during respiration while catabolism occurs in photosynthesis.

D. breaks down macromolecules while catabolism buildups them.

66. In a hypothetical scenario where photosynthesis ceased to occur on the earth, the long-term consequences on the continuity of life would be

A. remained unaffected since other sources of energy exist.

ENTRANCE TRICKS

B. collapsed that could lead to mass extinction.

C. balanced between CO2 and O2 appropriately.

D. disordered although animal life would flourish.

67. What are the roles of grana and stroma in the process of photosynthesis? Grana is a site for

A. cyclic pathway while stroma is a site for non-cyclic pathway.

B. photolysis of water while stroma is a site for carbon fixation.

C. generation of sugar while stroma is a site for photolysis of water.

D. production of NADPH while stroma is a site for chemiosmosis.

68. Which one of the following comparisons between cyclic and non-cyclic photophosphorylation is correct? Non-cyclic photophosphorylation

A. produces ATP, NADPH and O2, but cyclic produces ATP and NADPH.

B. uses oxygen as the final electron acceptor, but cyclic uses cytochrome b-c1 complex.

C. uses photosystem I as electron donor, but cyclic uses water as an electron donor.

D. involves both photosystems I and II, but cyclic uses only photosystem I.

69. All of the following events happen in the reduction stages of the Calvin cycle EXCEPT

A. combining of CO2 with RuBP.

B. utilization of ATP and NADPH molecules.

C. conversion of 3-PGA into a sugar molecule.

D. donating of electrons to a 3C-intermediates to form G3P.

70. What happens in the mitochondrion during the reactions of photorespiration?

A. Glycine is converted to serine with the release of CO2 and NH3.

B. Glycolate is converted to glyoxylate with the release of H2O2.

C. Glycerate is converted to GP with the release of ADP.

D. Phosphoglycolate is converted to glycolate with the release of Pi.

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71. How does photosynthesis ensure the continuity of life? By

A. constantly adding carbon dioxide to the atmosphere

B. maintaining carbon dioxide and oxygen balance in the atmosphere

C. constantly removing oxygen from the atmosphere

D. adding natural gas and absorbing methane from the atmosphere

72. Which of the following is the difference between substrate level phosphorylation and oxidative phosphorylation? Substrate level phosphorylation

A. occurs in the inner mitochondrial membrane while oxidative phosphorylation occurs in the cytosol.

B. occurs during glycolysis and Kreb's cycle while oxidative phosphorylation occurs during electron transport chain.

C. has the same source of energy as oxidative phosphorylation.

D. occurs in aerobic respiration while oxidative phosphorylation in anaerobic respiration.

73. Which one of the following is correct about the energy liberation phase of glycolysis pathway? It

A. hydrolyses two ATP molecules and raises the free energy of glucose.

B. breaks six carbon molecules into two glyceraldehyde-3-phosphate molecules.

C. releases one water molecule and dihydroxyacetone phosphate molecule.

D. produces four ATP, two NADH and two molecules of pyruvate.

74. Which one of the following statements describes an event of the process of electron transport system correctly?

A. Oxygen serves as a terminal electron acceptor in the production of ATP.

B. Thirteen ATP molecules are invested when electrons are transferred from NADH/FADH2 to O2.

C. More than 90% of the total ATP is produced by substrate-level phosphorylation.

D. It happens due to the difference in pH as the inside mitochondrial membrane is more acidic.

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75. In a complete oxidation of glucose molecule, how many net ATP molecules are produced by substrate level phosphorylation?

A. 2

B. 4

C. 8

D. 10

76. What is the difference between aerobic and anaerobic respirations?

A. Aerobic respiration ends with different products while the same product results in anaerobic respiration.

B. Proteins are the main respiratory substrate for both aerobic and anaerobic respiration.

C. Aerobic respiration produces much more energy than anaerobic respiration.

D. Glycolysis is the first pathway for aerobic while Kreb's cycle is for anaerobic respiration.

77. Which one of the following statements disproves spontaneous generation theory based on a step wise experimental work?

A. Life is an inherent property of the universe and everlasting.

B. Different life forms on the Earth were created by a supreme being within six days.

C. Both macro and micro-organisms could arise from pre-existing organisms.

D. Life has reached to the Earth in the form of spores with falling meteorite.

78. Based on comparative anatomy, chimpanzee is the closest relative to humans because of the similarities in their

A. haemoglobin sequence.

B. fossil remains.

C. DNA sequence.

D. pentadactyl limb.

79. Which one of the following causes of variation allows infertile hybrids to become fertile in plant evolution?

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A. Reproductive isolation

B. Sympatric speciation

C. Polyploidy cell

D. Geographical separation

80. This item is based on the following course of human evolution.

1. Use of fire and other tools

2. Increase in brain size

3. Emergence of Homo sapiens

4. Bipedal walking and arboreal life

Which of the following is the correct order of the above evolutionary time line markers?

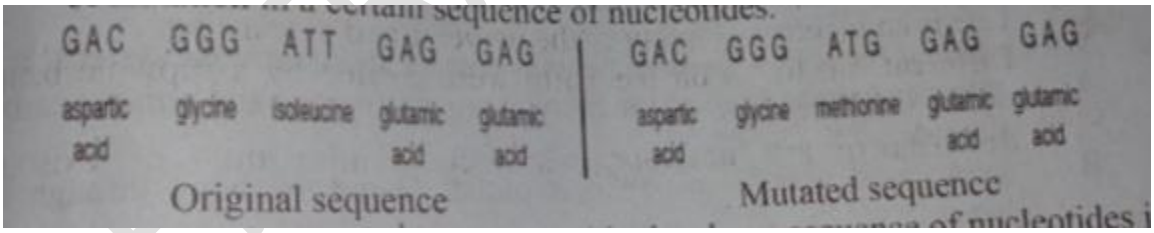
A. 2, 1, 3, 4

B. 1, 3, 4, 2

C. 3, 2, 4, 1

D. 4, 2, 1, 3

81. This item is based on the following diagram, which illustrates the occurrence of mutation in a certain sequence of nucleotides.



The type of mutation that occurred in the above sequence of nucleotides is

A. deletion.

B. inversion.

C. addition.

D. substitution.

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82. Which of the following is NOT a common cause of genetic mutation?

A. Exposure to ultra-violet radiation

B. Errors in DNA replication

C. Extensive physical exercise

D. Experiences of using chemical substance

83. Which genetic phenomenon is common for both bottleneck and founder effects?

A. High mutation rate that enhances adaptation

B. Genetic drift that reduces diversity in the population

C. Gene flow that increases inbreed and disease's resistance

D. Natural selection that results from gradual evolutionary changes

84. In a scenario where a male black lion population left its terrestrial territory and joined a white female lion population in an isolated Island to mate, there would be a

A. decrease in black lion population in the isolated Island.

B. genetic variation in terrestrial population due to immigration.

C. random introduction of new alleles into the Island population.

D. change in allele frequencies of terrestrial population but not in Island.

85. Which of the following is an example of anthropogenic driving factor that contributes to species extinction?

A. Earth quack

B. Volcanic eruption

C. Expansions of agriculture

D. Outbreak of wild-fire

86. Which one of the following statements is correct about nervous system? It

A. takes air into and out of the body to supply oxygen and remove carbon dioxide.

B. uses blood vessels to transport blood around the body through the heart.

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C. consist of cells that produce and secret chemicals into the blood stream via duct.

D. receives and processes sensory information from internal and external parts of the body.

87. What is the best justification for the role of voltage-gated ion channels in nerve impulse transmission? They

A. actively respond to membrane potential changes, leading to depolarization.

B. block the ions movement across the membrane to sustain neuron.

C. allow potassium ions diffuse to the inside of the membrane, forming repolarization.

D. produce energy when the sodium ions move to the inside of the axon.

88. Which one of the following ions are used to facilitate the release of neurotransmitters to the synaptic cleft before binding to postsynaptic receptor?

A. Calcium ions

B. Magnesium ions

C. Potassium ions

D. Sodium ions

89. Find the correct path way of reflex are that prevents you from physical damage up on the sudden touching of hot plate.

A. Interneuron→ receptor→ motor neuron→ effector

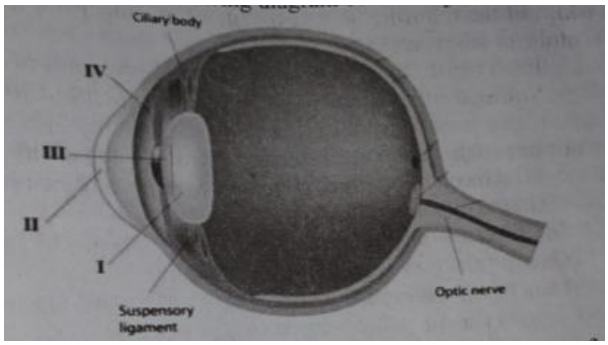
B. Sensory neuron→ motor neuron→ spinal cord→ effector

C. Sensory neuron→ spinal cord→ motor neuron→ effector

D. Interneuron→ spinal cord→ motor neuron→ effector

90. This item is based on the following diagram of anatomy of human eye.

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Which labeled part is responsible for controlling the diameter of the pupil according to the light source?

A. I

B. II

C. III

D. IV

91. Select the option that contains the correct match of human ear structure and its function.

A. Eustachian tube - balancing air pressure of the middle ear and throat

B. Semicircular canals - concerned with production of nerve impulse

C. Ear ossicles - concerned with the detection of motion

D. Cochlea - transmits sound from the incus to the auditory nerve

92. This item is based on the following list of events that happen during hearing.

I. Transmission of amplified sound waves to the ear ossicles

II. Movement of sensory hair as a result of vibration of the fluid in the cochlea

III. Transmission of nerve impulse into the brain and response to the sound stimulus

IV. The pinna directs sound waves into the eardrum

Based on the above lists, which alternative shows the correct steps of the events of hearing?

A. IV -> III -> II -> I

B. I -> II -> III -> IV

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C. I -> IV -> II -> III

D. IV -> I -> II -> III

93. Suppose low metabolic rate, cretinism and retarded mental and physical development are common problems of infants and childhood in a small village. Which condition in the endocrine system results in these problems?

A. Extreme hypothyroidism

B. Malfunction of adrenal cortex

C. Over production of parathyroid hormone

D. Lack of adrenalin production

94. If a woman is admitted to hospital for giving birth, which hormone is helping her in the delivery of her baby and releasing of milk?

A. Vasopressin

B. Aldosterone

C. Oxytocin

D. Progesterone

95. How ultrafiltration takes place in the human kidney? By

A. increasing the concentration of urine in the muscular sac using a collecting duct

B. building the pressure of the knot of blood vessels in the Bowman's capsule

C. diffusing of glucose molecules into the first coiled tubule of microvilli

D. reabsorbing of water into the blood in the second coiled tubule using ADH

96. Assume a patient with type II diabetes gets an advice to practice regular exercise to raise the number of insulin receptors. Then, how does insulin regulate the blood glucose level if it is high in this patient? It stimulates

A. glycolysis, which converts glucose into pyruvate.

B. glycogenesis, which converts glucose into glycogen.

C. lipolysis that converts fatty acid into glucose.

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D. gluconeogenesis that creates glucose from amino acids.

97. Imagine the water content of the blood is low when you take in much salt with your food. What would happen in your kidney to regulate this situation? Osmoreceptor stimulates the

A. adrenal gland to release adrenalin that increases conversion of glycogen to glucose in the blood.

B. hypothalamus that release aldosterone and sodium ions to reabsorb back into the blood.

C. pituitary gland to release ADH to reabsorb water back into the blood.

D. ADH to release cortisone that promotes resistance to water stress.

98. Several current studies point out that very large quantity of greenhouse gases have been added into Earth's atmosphere due to human activities. What would be the most likely responsible gas and its major sources, respectively?

A. Methane - aluminum smelting and semiconductor manufacturing

B. Carbon dioxide - fossil fuel combustion, deforestation, oil and gas usage

C. Sulphur hexafluoride - landfills, rice paddies, digestive tracts of cattle

D. Perfluorocarbons - electric transmission, magnesium and aluminum production

99. The effect of increased climate change on agricultural productivity is associated with

A. land use change and expansion of monoculture farming that yields high productivity.

B. agroforestry system that increases the mixture of multipurpose trees and crops.

C. drought that triggers wild fire, which can consume millions of acres of forest.

D. reduction of available water that leads to decrease in biological resources.

100. Which one of the following local community practices in Ethiopia is considered as an environmental action that addresses climate change mitigation? Culture of

A. respecting humanity that promotes eco-tourism industry

B. negotiating mechanism that is led by elders to reduce social unrest

C. planting millions of seedlings following the green legacy initiative

D. treating different diseases with herbal remedies by traditional healers

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Answer Key:

1. **B**
2. **A**
3. **B**
4. **C**
5. **B**
6. **B**
7. **A**
8. **A**
9. **D**
10. C
11. **A**
12. **B**
13. **D**
14. **D**
15. **C**
16. **D**
17. **A**
18. **C**
19. **A**
20. **C**
21. **C**
22. **A**
23. **C**
24. **C**
25. **A**
26. **B**
27. **A**
28. **A**

29. **B**
30. **D**
31. **C**
32. **C**
33. **D**
34. **A**
35. **A**
36. **C**
37. **A**
38. **D**
39. **B**
40. **A**
41. **C**
42. **D**
43. **B**
44. **D**
45. **B**
46. **B**
47. **C**
48. **D**
49. **B**
50. **B**
51. **C**
52. **B**
53. **B**
54. **D**
55. **B**
56. **A**
57. **A**
58. **D**

59. **C**
60. **B**
61. **A**
62. **A**
63. **A**
64. **C**
65. **A**
66. **B**
67. **B**
68. **D**
69. **A**
70. **A**
71. **B**
72. **B**
73. **D**
74. **A**
75. **B**
76. **C**
77. **C**
78. **C**
79. **C**
80. **D**
81. **A**
82. **C**
83. **B**
84. **C**
85. **C**
86. **D**
87. **A**
88. **A**

89. **C**
90. **D**
91. **A**
92. **D**
93. **A**
94. **C**
95. **B**
96. **B**
97. **C**
98. **B**
99. **D**
100. **C**

THE END

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